

Project I — Master Design Plan

Project I

Master Design Plan

Pre-production planning — June 2026

Unity desktop idle gacha turn-based hero collector RPG

Reading guide. Yellow highlighted blocks (like this) mark every change, addition, or clarification vs. the source PDF.

For a one-page index of every change, see the Changelog section.

Project I — Changelog vs. Source PDF

Purpose: Single-page index of every change, addition, and clarification this master plan makes vs. the original *Project I.pdf* (208 pages, 6,795 lines of text).

How to read: every entry in *the master plan* that came from this list is wrapped in `<mark>...</mark>` tags so it renders as a yellow highlight in the rendered PDF. If you see a yellow block, it's described here.

Decisions locked by user (from planning session)

#	Topic	Decision	Source
D-01	Combat style	Auto turn-based by default; manual mode is an opt-in toggle in expanded view. Speed determines turn order. Cooldowns in turns.	Q&A
D-02	Elements	3-element triangle locked:  Fire /  Water /  Wind. Glacier reclassified from Ice → Water. Frost/Freeze remain as Water-flavored status effects.	Q&A
D-03	Endgame	Tower of Trials + World Map / Expedition + Hero Trials + Boss Rush. Character merge & gear grinding are the core power-growth loops. Events post-launch only.	Q&A
D-04	Narrative	No story for v1. Bare flavor only on hero sheets.	Q&A
D-05	Equipment depth	Mid-depth: main stat + 2-3 random sub-stats + 2/4-piece set bonuses.	Q&A
D-06	Monetization	Standard gacha: Battle Pass + Monthly Card + Beginner Pack + Limited Banner bundles + Dupe Shop. Pity 50/210.	Q&A
D-07	PvP / Guilds	Out of scope. Single-player design locked.	Q&A
D-08	KO behavior	If the entire party dies, the stage auto-repeats with rewards reduced (no first-clear bonus, ~50% loot). Mid-stage hero KO: dead until end of stage; revived between battles.	Q&A
D-09	Auto vs. Manual	Auto is the default for all combat. Manual is a per-stage toggle. Compact mode = always auto.	Q&A
D-10	Sweep / Skip tickets	NOT in v1. Idle is the only “skip” mechanism.	Q&A
D-11	Steam Market trading	NOT in scope. Project I is gacha-monetized, not loot-trade.	Q&A
D-12	Pets / Companions	NOT in v1. Kept in BACKLOG.md for post-launch consideration.	Q&A
D-13	Launch roster	15 5★ heroes total: 9 Standard pool + 1 launch Limited banner (Auron) + 5	Q&A

#	Topic	Decision	Source
		future Limited banners (Akari, Ragnar, Seraphina, Mei Lin, Ayame). Plus 15 4★ heroes (designs forthcoming).	
D-14	Render format	Master plan rendered to PDF via pandoc + weasyprint; all new content highlighted via HTML <code><mark></code> tags.	Q&A

Inconsistencies in source PDF — fixes applied or pending user answer

#	Issue	Status	Resolution
F-01	“Ice” element on Glacier — only 3 elements exist (Fire/Water/Wind)	✓ Fixed	Glacier →  Water. Frostbite / Freeze become Water-flavored status effects.
F-02	Source-PDF draft-name confusion in the Fire Assassin sheet	✓ Resolved	Fire Assassin canonical name is Kaida . All draft-name traces removed.
F-03	Source-PDF spelling typo for Water Mage	✓ Resolved	Water Mage canonical name is Neria (the alt spelling was a draft typo, not a rename).
F-04	Class weapon list contradicts actual hero weapons (Tank=Shield is broken; many hero weapons don’t match class spec)	✓ Fixed	Reframed as “weapon archetypes” — each class has a <i>family</i> of allowed weapons. Existing hero designs preserved.
F-05	“No Mana, No Energy” but Energy is referenced as a stat (necklace) and as an effect (Marina’s Ultimate)	✓ Fixed	All Energy references deleted. Marina’s Ultimate “restores Energy” effect simply removed (no replacement) — her ult is now pure team heal + cleanse + Healing Received buff.
F-06	Cooldown Reduction as an equipment stat conflicts with integer turn cooldowns	✓ Fixed	CDR stat removed. Replaced on Magic Armor + Necklaces with Effect Hit % (Mages) / Healing Done % (Supports).
F-07	Secondary stats (Magic Power, Healing Power, Damage Reduction, Shield Strength) not in core stat list	✓ Fixed	Folded into core 7 stats. Magic Power → ATK; Healing Power → ATK × per-skill heal multiplier; Damage Reduction → DEF; Shield Strength → HP-derived.
F-08	9-hero roster vs. 15-hero detailed sheets in PDF	✓ Fixed	Launch roster locked at 15 5★ heroes total: 9 Standard pool + 1 launch Limited (Auron) + 5 future Limited banners (Akari, Ragnar, Seraphina, Mei Lin, Ayame). Plus 15 4★ heroes (designs TBD).
F-09	(???) and (BANNER 2) markers on certain heroes (Ayame, Akari, Ragnar, Seraphina, Mei Lin, Auron)	✓ Fixed	Lock confirmed: Auron = launch Limited banner; Akari/Ragnar/Seraphina/Mei Lin/Ayame = future Limited banners (designs preserved as WIP placeholders).
F-10		✓ Standard	APPEARANCE section becomes part of standard character sheet template; will be

#	Issue	Status	Resolution
	APPEARANCE sections only present on 4 of 15 character sheets		backfilled for the missing 11 when sheets are next revised.
F-11	Chibi skill animation count ambiguity — is Skill 1 + Skill 2 = 1 sheet or 2?	 Fixed	One chibi animation sheet per skill: Idle / Attack / Skill 1 / Skill 2 / Ultimate / Death = 6 sheets per hero.
F-12	“Guild content” referenced as source of Sacred Stones, but Guilds are out of scope	 Fixed	Sacred Stones source moved to Weekly Boss Rush (NEW endgame mode) + Tower of Trials milestone rewards.
F-13	Auron + Glacier are visually “ice” but element-tagged as Water	 Acknowledged	Kept as Water with ice-themed visual identity. Status effects use Frostbite/Freeze under the Water element banner.

Referenced but undefined in source PDF — defined here

#	Item	Defined In
U-01	Skill upgrade system	<code>gdd/03-stats-and-progression.md</code>
U-02	Hero Essence currency + Essence Shop	<code>gdd/05-gacha-and-currencies.md</code> (Dupe Shop)
U-03	Duplicate Tokens (folded into Hero Essence — single dupe currency)	<code>gdd/05-gacha-and-currencies.md</code>
U-04	Sacred Stones (Shrine upgrade material) — tier ladder	<code>gdd/06-idle-and-shrine.md</code>
U-05	Promotion Materials — full ladder	<code>gdd/03-stats-and-progression.md</code>
U-06	Element Crystals (Fire / Water / Wind)	<code>gdd/03-stats-and-progression.md</code>
U-07	Achievements list + reward tiers	<code>gdd/12-daily-weekly-loops.md</code>
U-08	Battle Pass full design	<code>gdd/13-battle-pass.md</code>
U-09	Daily / Weekly missions	<code>gdd/12-daily-weekly-loops.md</code>
U-10	Events architecture (for post-launch)	<code>gdd/21-liveops-and-telemetry.md</code>
U-11	Damage formula (full math)	<code>gdd/01-combat.md</code>
U-12	Enemy / boss roster + design rules	<code>gdd/09-campaign-structure.md</code>
U-13	Campaign chapter/act/world structure	<code>gdd/09-campaign-structure.md</code>
U-14	Tutorial / onboarding flow	<code>gdd/16-tutorial-onboarding.md</code>
U-15	Save system spec	<code>gdd/17-save-system.md</code>

● **New systems added (each gets its own GDD sub-doc, all highlighted)**

#	System	GDD Sub-doc
N-01	Combat speed multiplier (1x/2x/3x/4x)	gdd/01-combat.md
N-02	Auto-AI spec (skill priority, targeting)	gdd/01-combat.md
N-03	Idle stage selection UI + drop preview	gdd/06-idle-and-shrine.md
N-04	First-clear vs. repeat rewards	gdd/09-campaign-structure.md
N-05	3-star rating system per stage	gdd/09-campaign-structure.md
N-06	Formation / positioning (Front Row / Back Row)	gdd/01-combat.md
N-07	KO + party-wipe behavior	gdd/01-combat.md
N-08	Buff / debuff stacking rules	gdd/01-combat.md
N-09	Turn order timeline UI	gdd/01-combat.md , gdd/20-ui-menus-tabs.md
N-10	Boss preview / scout	gdd/01-combat.md
N-11	Manual retry + auto-repeat on wipe	gdd/01-combat.md
N-12	Auto-loot pickup in compact mode	gdd/15-desktop-window-ux.md
N-13	Counter-attack mechanic	gdd/01-combat.md
N-14	Speed tie-breaker rules	gdd/01-combat.md
N-15	Boss enrage timer (30 turns default)	gdd/01-combat.md
N-16	Difficulty modifier rules (Normal/Hard/Nightmare/Hell)	gdd/09-campaign-structure.md
N-17	Damage formula	gdd/01-combat.md
N-18	Skill upgrade system (skill stones from stages)	gdd/03-stats-and-progression.md
N-19	Equipment sub-stats (2-3 rolled subs)	gdd/04-equipment.md
N-20	Equipment set bonuses (6 sets: Berserker / Bastion / Swiftwind / Mystic / Vampire / Lifegiver)	gdd/04-equipment.md
N-21	Equipment salvage + sub-stat reroll	gdd/04-equipment.md
N-22		gdd/10-tower-of-trials.md

#	System	GDD Sub-doc
	Tower of Trials (100 floors, monthly reset)	
N-23	World Map / Expedition (4 parties, 4 timer tiers)	<code>gdd/11-world-map-expedition.md</code>
N-24	Daily / Weekly missions + 7-day login bonus	<code>gdd/12-daily-weekly-loops.md</code>
N-25	Hero Trials (weekly rotating single-hero stages)	<code>gdd/12-daily-weekly-loops.md</code>
N-26	Boss Rush (weekly chapter-bosses gauntlet)	<code>gdd/12-daily-weekly-loops.md</code>
N-27	Battle Pass (50 tiers, 60-day cycle, free + premium + premium-plus)	<code>gdd/13-battle-pass.md</code>
N-28	Monetization plan + ARPU targets	<code>gdd/14-monetization.md</code>
N-29	Tutorial / onboarding (9-stage flow, reroll-friendly)	<code>gdd/16-tutorial-onboarding.md</code>
N-30	Save system (local JSON + cloud sync hook, tamper-resistant)	<code>gdd/17-save-system.md</code>
N-31	Audio direction (mixing bus, SFX/ music taxonomy)	<code>gdd/18-audio-direction.md</code>
N-32	Localization plan (string tables, EN v1, CJK/EU expansion)	<code>gdd/19-localization-plan.md</code>
N-33	Wishlist system (Standard 5★ guaranteed rotation)	<code>gdd/05-gacha-and-currencies.md</code>
N-34	Free daily pull on at least one banner	<code>gdd/05-gacha-and-currencies.md</code>
N-35	Pity-tracker UI in gacha screen	<code>gdd/05-gacha-and-currencies.md</code>
N-36	Beginner step-up banner (replaces flat 20% discount)	<code>gdd/05-gacha-and-currencies.md</code>
N-37	Reroll-friendly tutorial pacing	<code>gdd/16-tutorial-onboarding.md</code>
N-38	Mail / Inbox system	<code>gdd/20-ui-menus-tabs.md</code>
N-39	Achievements + Codex + Bestiary + Hero Gallery	<code>gdd/12-daily-weekly-loops.md</code> , <code>gdd/20-ui-menus-tabs.md</code>
N-40	Full settings menu (40+ options, sub-tabbed)	<code>gdd/15-desktop-window-ux.md</code> , <code>gdd/20-ui-menus-tabs.md</code>

#	System	GDD Sub-doc
N-41	Accessibility suite (color blind, high contrast, reduce shake, text scaling)	gdd/15-desktop-window-ux.md
N-42	Audio sliders (Master/Music/SFX/Voice/UI)	gdd/15-desktop-window-ux.md
N-43	Hotkey rebinding	gdd/15-desktop-window-ux.md
N-44	Pause-when-unfocused, battery-saver, display scale, etc.	gdd/15-desktop-window-ux.md
N-45	Confirm-on-spend toggle for premium currency	gdd/15-desktop-window-ux.md
N-46	Hot-config (server-driven balance)	gdd/21-liveops-and-telemetry.md
N-47	Telemetry / analytics events	gdd/21-liveops-and-telemetry.md
N-48	Anti-cheat basics + IAP receipt validation	gdd/21-liveops-and-telemetry.md
N-49	Event calendar + maintenance broadcast plumbing	gdd/21-liveops-and-telemetry.md
N-50	Team presets (3-5 saved team configs)	gdd/01-combat.md , gdd/20-ui-menus-tabs.md
N-51	Recommended team builder	gdd/20-ui-menus-tabs.md
N-52	Quick-equip / bulk-select / lock-favorite	gdd/04-equipment.md , gdd/08-inventory-and-storage.md
N-53	Hero shard / fragment system (duplicates give shards toward chosen hero)	gdd/05-gacha-and-currencies.md
N-54	Replay last battle / damage log	gdd/01-combat.md
N-55	Auto-pause on PC sleep + cloud save conflict resolution	gdd/17-save-system.md



Removed / explicitly deferred to backlog

#	Item	Reason
R-01	Pets / companion system	User decision — not v1
R-02	Steam Market trading	User decision — not in scope
R-03	PvP / Arena	User decision — not in scope
R-04	Guilds	User decision — not in scope
R-05	Sweep / Skip tickets	User decision — idle is the only skip
R-06	Hero skins / costumes	Deferred to post-launch (BACKLOG)
R-07	Hero affinity / friendship	Deferred to post-launch (BACKLOG)
R-08	Hero synergy team-comp bonuses	Deferred to post-launch (BACKLOG)
R-09	DPS meter / damage history toggle	Deferred to post-launch (BACKLOG)
R-10	Story / narrative campaign	User decision — no story for v1
R-11	Friends list / referral	Single-player locked
R-12	Full screen reader support	Deferred to v2 accessibility pass



Document index

After this changelog, all design content lives in:

- **design/project-i-master-plan.md** — top-level overview and links
- **design/gdd/** — 21 GDD sub-docs
- **design/heroes/** — 16 hero sheets (9 ready + 7 placeholders)
- **design/BACKLOG.md** — deferred features for post-launch consideration
- **design/Project-I-Master-Plan.pdf** — rendered PDF with highlights

Project I — Master Design Plan

Status: Pre-production planning, June 2026 **Engine:** Unity (target Windows PC; macOS post-launch)
Genre: Idle Gacha Turn-Based Hero Collector RPG **Window paradigm:** Desktop / Taskbar (TBH-inspired) **Source PDF:** /Users/tata/Downloads/Project I.pdf (208 pages)

Yellow <mark> highlights mark every addition, change, or clarification vs. the source PDF. See `CHANGELOG-from-source-PDF.md` for the full index.

1. Vision

Project I is a **desktop idle gacha** that sits in the corner of your screen while you work, fighting battles, hatching eggs, collecting loot — and that you can open whenever you want to deep-dive into team-building, gear crafting, and gacha pulls. The combat is **auto-paced turn-based**: heroes act in Speed order using cooldown-gated skills. Auto mode is the default; manual mode is an opt-in toggle for players who want to lean into hard content.

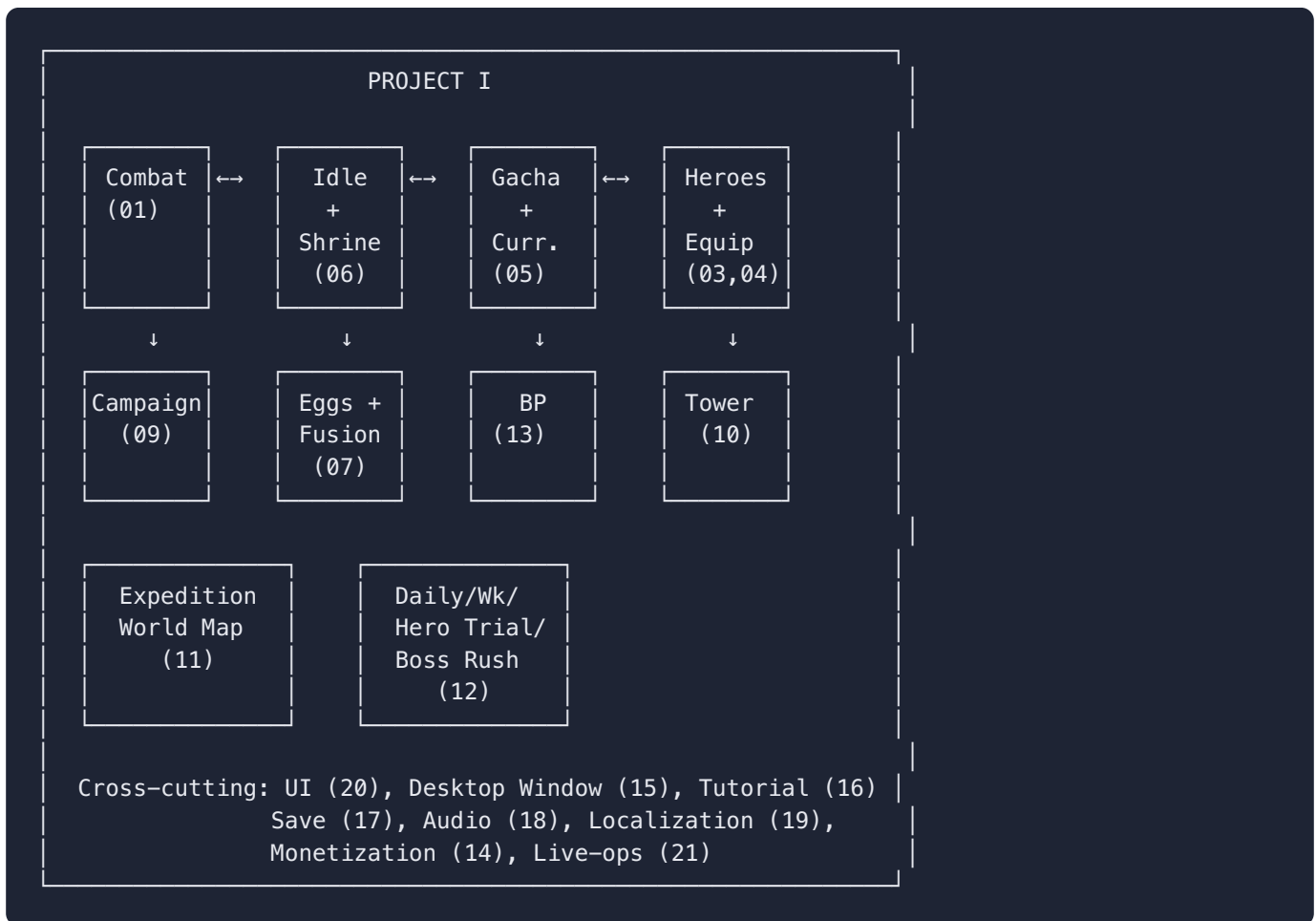
The closest reference is TBH: Task Bar Hero (desktop format) crossed with Epic Seven (combat depth) crossed with AFK Arena (idle progression). It is **single-player only** — no PvP, no guilds, no co-op.

Free-to-play with standard gacha monetization: Battle Pass, Monthly Card, Limited Banner bundles, Beginner Pack, Crystal packs, Dupe Shop. No pay-to-win, no power-creep treadmills, no energy gates.

2. Core Pillars

Pillar	Description
Idle First	The game makes meaningful progress while you're not playing. Active play accelerates, never replaces, idle.
Desktop-Native	Compact, peripheral, always-visible window is the identity. Expanded mode is only for management.
Deep Collection	15 5★ launch heroes + 15 4★ planned × 5 progression axes (Level, Star, Awakening, Skills, Gear) = years of growth.
Fair Gacha	Visible pity, Wishlist, free daily pull, generous tutorial rewards, Hero Shards as anti-bricked-pull insurance.
No Predatory Hooks	No stamina, no power creep, no FOMO-laced cash purchases. Real value, transparent pricing.
Accessibility-Mindful	Color blind, high contrast, text scaling, reduce motion shipped at launch. Screen reader phase 2.

3. Top-Level System Map



4. Documents Index

Core gameplay systems

#	Doc	Topic
01	Combat	Turn order, damage formula, auto-AI, formation, KO, buffs, retry, difficulty, enrage
02	Elements & Classes	3-element triangle, 5 classes, weapon archetypes
03	Stats & Progression	7 core stats, Level, Star, Awakening, Skill Level, Promotion materials
04	Equipment	Slots, sub-stats, set bonuses, enhancement, salvage, reroll
05	Gacha & Currencies	Banners, pity, currencies, wishlist, free daily pull, Step-Up
06	Idle & Shrine	Idle rewards, offline simulation, Shrine, idle stage picker, ROI
07	Eggs & Fusion	Egg tiers, Hatchery, fusion ladder
08	Inventory & Storage	Character + item storage, lock/favorite, filters, bulk select

Content & endgame

#	Doc	Topic
09	Campaign Structure	3 Acts × 4 Chapters × 10 Stages × 4 Difficulties, 3-star, first-clear rewards, featured chapter
10	Tower of Trials	100-floor monthly mode with hero locks and modifiers
11	World Map / Expedition	4 dispatch slots, mission rotation, crew bonuses
12	Daily / Weekly / Hero Trials / Boss Rush / Achievements	Recurring content, login bonus, ~150 achievements

Monetization & meta

#	Doc	Topic
13	Battle Pass	50 tiers, 60-day cycle, Free + Premium + Premium-Plus, BP-exclusive hero
14	Monetization	Crystal packs, Monthly Card, Beginner Pack, Bundles, Dupe Shop, ARPU targets

Platform / production

#	Doc	Topic
15	Desktop Window UX	Compact/Expanded/Tray, hotkeys, 9 settings sub-tabs, accessibility, multi-monitor
16	Tutorial / Onboarding	9-stage flow, guaranteed Legendary, reroll-friendly, day-by-day onboarding
17	Save System	Local + Steam Cloud, deterministic idle, tamper-resistance
18	Audio Direction	SFX taxonomy, music cues, mixing bus, no VO v1
19	Localization Plan	String tables, EN v1, CJK + EU expansion architecture
20	UI / Menus / Tabs	14 top-level tabs, Mail/Inbox, Codex, Achievements
21	Live-ops / Telemetry / Security	Hot-config, event calendar, telemetry, anti-cheat, IAP receipts
22	Boss Design	PLACEHOLDER — 12 chapter boss stubs, design constraints, common patterns
23	Art Direction	PLACEHOLDER — areas needing definition, asset budget, open questions
24	Day-1 Launch Content	Complete lockdown of everything shipping Day 1
25	Enemy AI	5-tier AI scaling system (Mindless → Optimal) by enemy level + boss override

Production & business

Doc	Topic
Production & Business Plan	Engineering ADRs, legal, marketing, QA, support, analytics, team, timeline, risks, budget
Unity Setup Guide	Step-by-step setup for Unity + Git + cross-machine workflow

Supporting documents

- `CHANGELOG-from-source-PDF.md` — All 50+ changes vs the source PDF
- `BACKLOG.md` — 29 deferred features parked for future consideration
- `heroes/` — Per-hero detailed sheets (pending audit answers — see § 9)

5. Roster Summary

Total launch 5★ roster: 15 heroes. Plus 15 4★ heroes (designs TBD — user is producing these separately).

Element	Tank	Assassin	Archer	Mage	Support
 Fire	Valdrik 	Kaida 	Ragnar 	Emberia 	Akari 
 Water	Glacier 	Seraphina 	Auron 	Neria 	Marina 
 Wind	Mei Lin 	Kael 	Lyria 	Ayame 	Sylphie 

Legend:  Standard banner (in pool from day 1);  Launch Limited Banner (Auron — first paid banner, live at launch);  Future Limited Banners (released over post-launch banner schedule).

Available at launch (10 heroes): - **Standard banner pool (9):** Valdrik, Emberia, Kaida, Glacier, Marina, Neria, Sylphie, Lyria, Kael - **Launch Limited Banner (1):** Auron

Future Limited Banner releases (5): Akari, Ragnar, Seraphina, Mei Lin, Ayame (release schedule defined post-launch)

This means F2P players via the 210-pull Standard Selector have 9 heroes available, Wishlist works on those 9, and Auron is the chase target for the first banner window. Subsequent Limited banners introduce one new hero at a time.

4★ roster: 15 heroes planned but not yet designed. They will likely form the bulk of the Hatchery / Egg / promotion-fodder gameplay AND be obtainable as Epic-tier gacha drops. See `design/heroes/4-stars-TBD.md`.

6. Core Gameplay Loops

Daily Loop (~30 min)

```

Login
↓
Claim Shrine
↓
Claim idle rewards
↓
Run daily missions (8 of them)
↓
Dispatch 4 expeditions
↓
1 Tower floor run
↓
1-2 Hero Trial / Boss Rush attempts (if weekly modes active)
↓
Set idle target, tab away
  
```

Weekly Loop (~3 hours total across the week)

```

Monday: weekly missions reset, login bonus continues
↓
Mid-week: featured chapter of the week (event-like 50% gold boost)
↓
Across week: complete weekly missions, climb tower, advance BP tiers
↓
End of week: claim weekly mission rewards, plan next week
  
```

Seasonal Loop (~60 days)

```

Day 1: BP season starts
↓
~30 days: BP tier-up via daily/weekly engagement
↓
~45 days: Tower monthly reset (twice per BP season)
↓
Day 60: BP ends, claim grace period, new season begins
  
```

Lifetime Loop (~6+ months)

```
Reach max account level
↓
Awaken favorite heroes to 6
↓
Promote heroes to 6★ via fodder + Apex Catalysts
↓
Max skill levels (10) on multiple heroes
↓
Tower floor 100 climb
↓
Boss Rush full clear on Nightmare
↓
Collect all Hero Shards / build dream roster
```

7. Key Numbers (cheat-sheet)

Concept	Value
Launch 5★ heroes	15 (10 available at launch + 5 future limited banners)
Launch 4★ heroes	15 (designs TBD)
Launch elements	3 (Fire / Water / Wind)
Launch classes	5 (Tank / Assassin / Archer / Mage / Support)
Stats per hero	7 (HP / ATK / DEF / SPD / CR / CD / ACC)
Equipment slots per hero	7 (Weapon, Helmet, Chest, Gloves, Boots, Ring, Necklace)
Equipment sets	6 launch (Berserker, Bastion, Swiftwind, Mystic, Vampire, Lifegiver)
Max hero level	60
Max star tier	6★ (5★ → 6★ requires Apex Catalyst)
Awakening levels	0-6
Skill ranks per skill	1-10
Campaign chapters	12 (3 Acts × 4 each)
Stages per chapter	10
Difficulty tiers	4 (Normal / Hard / Nightmare / Hell)
Total stage instances	480
Tower of Trials floors	100 (monthly reset)
Expedition slots	up to 4
Battle Pass tiers	50 (60-day season)
Gacha rates	Legendary 0.6% / Epic 7.2% / Rare 92.2%
Pity (all banners)	hard 50 only — guaranteed Legendary at pull 50; no soft pity
Standard Selector unlock	210 pulls
Idle storage cap	12 hours
Offline accumulation rate	75% of active
Wishlist size	3 Standard heroes

Concept	Value
Achievement count at launch	~150

8. Resolved Audit Items (all green)

All source-PDF inconsistencies have been resolved with user input. See [CHANGELOG-from-source-PDF.md](#) for full detail.

#	Issue	Resolution
F-01	Ice element on Glacier	Glacier = Water 
F-02	Fire Assassin draft-name confusion	Canonical: Kaida 
F-03	Water Mage spelling typo	Canonical: Neria 
F-04	Class weapon list	Reframed as “weapon archetypes” 
F-05	Energy stat conflict	Energy deleted entirely 
F-06	CDR vs. turn cooldowns	CDR deleted; replaced by Effect Hit / Healing Done % 
F-07	Secondary stats	Folded into 7 core stats 
F-08	Roster size	15 5★ at launch + 15 4★ TBD 
F-09	(???) markers	Banner roles locked (Section 5) 

9. Hero Sheets

Per-hero detailed sheets live in [design/heroes/*.md](#) . All 15 5★ sheets are authored.

Recommended sheet template (per hero): - Header: name, element, class, role, weapon, rarity, banner pool - Appearance description - Stat grades (HP/ATK/DEF/SPD/CR/CD/ACC) - Passive (name, mechanics, max stacks) - Basic attack (name, cooldown, effects) - Skill 1 (name, cooldown, multiplier, effects) - Skill 2 (name, cooldown, multiplier, effects) - Ultimate (name, cooldown, multiplier, effects) - Character identity (strengths/weaknesses) - Chibi animations (idle / attack / skill 1 / skill 2 / ultimate / death) - Team role - Recommended partners - Auto-AI defaults (recommended Ultimate trigger conditions)

10. Roadmap (high-level only)

Phase	Timing	Goal
Planning (current)	Now	Approve master plan + GDDs + hero sheets
Pre-Production	T+1mo to T+3mo	Engine setup, ADRs, architecture, vertical slice
Production	T+3mo to T+9mo	Implement all 21 GDDs, 10 launch-available heroes (9 Standard + Auron), all 12 chapters
Polish	T+9mo to T+11mo	QA, balance, bugs, performance
Soft Launch	T+11mo	Limited region launch (Steam early access?)
Global Launch	T+12mo	Worldwide launch
Post-Launch	T+12mo+	Events, seasonal BPs, new heroes, backlog items

(Detailed sprint planning lives in `production/sprints/` — not yet written.)

11. Risks & Open Questions

Risk	Mitigation
Desktop window paradigm is unfamiliar	TBH proved the format. Compact mode is the differentiator.
Gacha + idle combination is competitive market	Stand on unique window paradigm + fair monetization.
Cheating on save files	Tamper detection, server validation for cash-touching events.
10 launch-available heroes might feel light	Add the 5 designed-but-deferred heroes (Akari, Ragnar, Seraphina, Mei Lin, Ayame) on a planned Limited Banner cadence. Plus 15 4★ heroes. Plus BP-exclusive heroes per season (every 60 days).
Steam Cloud sync conflicts	Conflict resolution UI in <code>gdd/17-save-system.md</code> .
Launch without Steam Market	Decision made — gacha-only monetization. Communicate clearly.
Tower of Trials hero-lock might frustrate small rosters	Rest Floors every 10 floors mitigate this. Tower Coin shop rewards even casual players.

12. What's NOT in v1

(Per user direction and design decisions. Full list in `BACKLOG.md`.)

- ✗ Story / narrative campaign
- ✗ Voice acting
- ✗ PvP / Arena
- ✗ Guilds
- ✗ Pets / companions
- ✗ Steam Market trading
- ✗ Sweep / skip tickets
- ✗ Hero skins / costumes
- ✗ Mobile / console ports
- ✗ Cross-game crossovers
- ✗ Full screen reader support (v2)
- ✗ Live events (post-launch only)
- ✗ Friends / social features

13. What ARE the differentiators

Differentiator	Why it matters
Desktop window paradigm	One of the very few PC-native idle gachas. TBH proved demand; market has room.
Fair monetization	No energy gates, transparent pity, Wishlist, free daily pull, Hero Shards. Most modern gachas don't go this far.
Auto-AI quality	Per-hero auto-skill overrides + bias slider. Players don't need to babysit combat.
Idle stage selection + ROI display	Most idle games hide the math. Project I surfaces it.
3-star clear + Hero Trials	Replay value per stage, even after the campaign is "done".
Hero Shards	Targeted progression. Even unlucky pulls accumulate toward a specific hero.
Configurable settings (40+ options, 9 sub-tabs)	PC players expect depth. Mobile-port settings menus disappoint.
Accessibility from day 1	Color blind, high contrast, motion reduction, text scaling. Most gacha games skip these.

This master plan is the authoritative source of truth for v1. All implementation work should reference it.

GDD 01 — Combat System

Highlighted (<mark>) blocks are new vs. the source PDF.

1. Overview

Project I's combat is **auto-paced turn-based**. A party of 3 heroes fights enemies (1-5 per wave). Speed determines turn order. Each hero performs one action per turn — Basic Attack, Skill 1, Skill 2, Ultimate, or Defend. Skills have integer turn cooldowns. Compact window mode runs combat fully on auto. Expanded window mode allows the player to toggle manual control on or off per-stage.

2. Player Fantasy

I'm watching my hand-picked dream team out-strategize a fight while I work. When a tough boss appears, I can lean in, tab open, and play it perfectly — but the moment I tab out, the team I trained handles it for me.

The fantasy is **idle commander, not active fighter**. Manual play is a power tool for hard content, not the default expectation.

3. Detailed Rules

3.1 Battle Format

- **Party size:** 3 heroes.
- **Enemies per wave:** 1-5; varies by stage and difficulty.
- **Waves per stage:** 3-5 normal waves, then a boss wave on chapter-end stages.
- **Victory:** all enemies defeated. Stage rewards granted, idle clock advances.
- **Defeat:** all 3 heroes KO'd. Stage auto-repeats with reduced rewards (no first-clear bonus, ~50% loot table roll). No "retry limit," no "stamina charged" — the player simply loses the time spent on that attempt.

3.2 Turn Order

Highest **Speed** stat acts first.

Speed tie-breakers (in order): (1) higher current ATK; (2) lower current HP%; (3) party position (left → right).

Action gauge model: Each combatant has an internal “action gauge” that fills at a rate proportional to Speed. When the gauge reaches 100, that combatant takes a turn and the gauge resets to 0. This produces irregular turn orders when Speed values are uneven (a fast hero can act twice between two slow heroes’ turns) which is far better than strict round-robin.

3.3 Hero Actions

Each turn a hero performs exactly one of:

Action	Cooldown	Notes
Basic Attack	None	Always available
Skill 1	2 turns	Mid-power, usually single-target
Skill 2	3 turns	Higher-power, often AoE / utility
Ultimate	5 turns	Highest-impact ability
Defend	None	-50% damage taken this turn, +30% counter chance (see 3.10)

Defend is automatically chosen by auto-AI when (a) the hero has no off-cooldown skill and (b) basic-attacking would target a debuff-immune enemy. Manual players can use it as a deliberate stall tactic.

3.4 Cooldown Behavior

- Cooldowns tick down at the **start** of the hero’s turn (not end). A 2-turn cooldown means: cast on turn 1, available again on turn 3.
- **Ultimate** cooldown starts only the first time it’s used (does NOT start at battle start). This prevents the boring “wait 5 turns for first ult” pattern.
- **Cooldowns persist between waves within the same stage, but reset between stages.** This is a deliberate choice for pacing — late waves feel earned but each stage starts fresh.

3.5 Damage Formula



```

RawDamage    = ATK × SkillMultiplier
ElementMod    = 1 + ElementBonus           (Fire→Wind / Wind→Water / Water→Fire = +25%; reverse = -25%)
CritRoll      = uniform(0..1) < CritRate / 100
CritMod       = 1 + (CritDamage / 100)     if CritRoll else 1.0
AccRoll       = uniform(0..1) < (Accuracy - EnemyResistance) / 100
DefMit        = EnemyDEF / (EnemyDEF + 1000 + 5 × AttackerLevel)
Damage        = floor(RawDamage × ElementMod × CritMod × (1 - DefMit))
Damage        = max(1, Damage)             (a hit always deals at least 1)

```

Healing uses the same formula sans Element/Crit/DefMit, scaled by `HealingMultiplier` (skill-defined). Healing can crit only if the skill explicitly allows it.

Status effect application rolls separately: `EffectApplied = uniform(0..1) < (Accuracy - EnemyResistance) / 100`. Status effects ignore the attack's hit roll — a status can land on an attack that missed damage if Accuracy is high enough.

SkillMultiplier reference table (skill designer tuning knob):

Action	Default Multiplier	Reason
Basic Attack	1.00×	Baseline
Skill 1	1.50×	Mid-burst
Skill 2	1.80× (AoE) or 2.20× (single-target)	AoE penalty
Ultimate	3.50× (single) or 2.60× (AoE)	Designed around 5-turn investment

DefMit example: Enemy with 800 DEF vs. lvl 40 attacker = $800 / (800 + 1000 + 200) = 40\%$ mitigation. Same enemy vs. lvl 60 attacker = $800 / (800 + 1000 + 300) = 38\%$. Higher-level attackers cut through DEF, but never reduce it to zero.

3.6 Element System

Element	Strong vs.	Weak vs.
 Fire	 Wind (+25%)	 Water (-25%)
 Water	 Fire (+25%)	 Wind (-25%)
 Wind	 Water (+25%)	 Fire (-25%)
Neutral	—	—

Element neutral (enemy of no listed element, e.g. void/stone bosses) applies neither bonus nor penalty.

Off-element penalty escalates with difficulty tier: Normal -25%, Hard -25%, Nightmare -40%, Hell -50%. (See `09-campaign-structure.md`.)

3.7 Class System

Class	Role	Primary Stats	Notes
Tank	Frontline, taunt, shield	HP, DEF, Accuracy	Heavy Armor
Assassin	Single-target burst	ATK, Speed, CritRate, CritDmg	Light Armor
Archer	Sustained ranged DPS	ATK, Accuracy, CritDmg	Light Armor
Mage	AoE & debuff	ATK, Accuracy, Speed	Magic Armor
Support	Heal / buff / cleanse	Accuracy, Speed, HP	Magic Armor

3.8 Formation / Positioning



Two-row formation:

Position	Row	Role
Slot 1	Front	Tank by default
Slot 2	Front	Bruiser / off-tank
Slot 3	Back	DPS / Support / Mage

Damage routing: - **Single-target enemy attacks** target Front Row first unless they have piercing/back-row skills. Back-row heroes take 30% damage from front-row-targeted attacks (splash). - **AoE attacks** hit all 3 slots equally. - **Taunt** redirects all single-target attacks to the taunting hero regardless of row. - **If both Front-Row slots are dead**, the Back-Row hero is treated as Front Row.

Slot count: 3 fixed. Future expansion to 4 or 5 is a backlog item, not v1.

3.9 KO & Wipe Behavior

- A KO'd hero is **dead for the remainder of the current stage**. They cannot be revived mid-stage except by abilities that explicitly say so (e.g., Akari's **Crimson Rebirth**).
- **Between stages**, all heroes revive to 100% HP automatically. There is no “out-of-combat heal” cost.
- **Party wipe** (all 3 KO'd): Stage auto-restarts. Rewards from that attempt are reduced — the player gets 50% of the standard repeat-clear loot table and **no** first-clear bonus or 3-star bonus.

- **Idle-mode wipe loop:** if a party wipes 3 consecutive times on the current idle stage, the idle simulation auto-downgrades to the last stage the party cleared cleanly. A toast notification surfaces this. The player can override and resume the harder stage manually.

3.10 Counter-Attack

Counter-attack is not a base stat. It is granted by: - Specific hero skills (e.g., a hypothetical “Vengeance Stance” passive). - The **Defend** action (+30% chance to counter the next single-target attack against this hero). - Equipment Set bonuses (e.g., the Berserker set — see [04-equipment.md](#)).

A counter triggers on a single-target attack against the counter-eligible hero. It does **not** consume the hero’s next turn. Counter damage uses Basic Attack multiplier (1.00x) and cannot crit unless the source explicitly allows it. Counters cannot themselves be countered (no infinite loops).

3.11 Boss Enrage



Every boss stage has an **enrage timer**. After N turns, the boss applies a permanent buff to itself: +100% ATK, +50% Speed, +200% Skill cooldown reduction .

Difficulty	Boss enrage turn
Normal	30
Hard	25
Nightmare	20
Hell	15

Enrage exists to **prevent infinite-stall builds** (e.g., perma-shield Tank + perma-heal Support) on bosses. Stage bosses without an enrage become trivially farmable, which breaks the difficulty curve.

3.12 Buff / Debuff Stacking Rules



Rule	Behavior
Identical buff from same source	Refreshes duration to its full value. Does NOT increase magnitude.
Identical buff from different sources	Replaces if the new magnitude is higher or duration is longer , otherwise ignored.
Different buffs	Stack freely. A hero can have ATK Up + DEF Up + Speed Up simultaneously.
Identical debuff	Same rules: refresh / replace if stronger or longer.
Stack-based passives (e.g. Lyria's Focus, Kael's Predator)	Cap at 5 stacks. Each new trigger adds 1 stack; old stacks do NOT have individual timers. Stacks persist until the hero dies or the stage ends.
Maximum total buffs/debuffs per combatant	8 buffs + 8 debuffs, hard cap. Excess is rejected silently (oldest stays).
Cleanse priority	Cleansing removes oldest debuff first unless skill specifies otherwise.

3.13 Status Effects (canonical list)

Effect	Type	Behavior	Default duration
Burn	Debuff	Deal 8% of caster's ATK as damage at start of victim's turn	2 turns
Freeze	Debuff	Skip next turn	1 turn
Frostbite	Debuff	-20% Speed, -10% DEF	2 turns
Slow	Debuff	-30% Speed	2 turns
ATK Down	Debuff	-30% ATK	2 turns
DEF Down	Debuff	-30% DEF	2 turns
Accuracy Down	Debuff	-30% Accuracy	2 turns
Speed Down	Debuff	-30% Speed	2 turns
Stun	Debuff	Skip next turn, immune to most other debuffs	1 turn
Haste	Buff	+30% Speed	2 turns
ATK Up	Buff	+30% ATK	2 turns
DEF Up	Buff	+30% DEF	2 turns
Shield	Buff	Absorbs damage equal to value, expires when consumed	2 turns or until depleted
Regen	Buff	Heal 8% of caster's max HP at start of victim's turn	2 turns
Taunt	Debuff (on enemy)	Forces target to single-attack the taunter	1-2 turns
Crowd Control Resistance	Buff	Immune to Freeze, Stun, Taunt	2 turns
Revival Flame	Buff	If KO'd while active, revive at 20% HP, then expire	2 turns

Note: Many character passives also generate unique stacks (Focus, Predator, Crimson Flame, Wind Sigil, Frostguard, Resolve, Tide, Frost Charge, Infernal Momentum, Nature Blessing, Serenity). These follow the stack-based passive rules above (cap 5, no individual timers).

3.14 Auto-AI Specification



When auto mode is active, each hero follows a deterministic decision tree on its turn:

```
function decideAction(hero, allies, enemies):
    if hero.HP < 20% and Defend gives survival benefit:
        return Defend

    if Ultimate is off cooldown:
        if any Ultimate-priority condition is met:
            return Ultimate
        elif 2+ enemies are alive AND Ultimate is AoE:
            return Ultimate
        elif strongest enemy HP > 30% AND Ultimate is single-target:
            return Ultimate

    if Skill 2 is off cooldown:
        return Skill 2 (targeted per Skill 2's type)

    if Skill 1 is off cooldown:
        return Skill 1 (targeted per Skill 1's type)

    return Basic Attack
```

Default targeting by skill type:

Skill type	Auto-target
Single-target damage	Enemy with lowest HP% (finish them off)
AoE damage	N/A — hits all
Single-target heal	Ally with lowest HP%
Self-buff	Self
Team-buff	All allies
Debuff (single)	Enemy with highest ATK (silence them)

Manual override: in the expanded mode Hero Detail screen, the player can toggle each hero's auto behavior per-skill (e.g., “Lyria’s Ultimate: only cast on Boss tag” or “Sylphie’s Skill 1: cast at any ally HP < 60%”). This is a **per-hero preference**, saved to the save file, persistent across battles.

Auto-mode bias slider (advanced setting): Aggressive / Balanced / Conservative. - **Aggressive:** ults released ASAP regardless of enemy count - **Balanced:** as above (default) - **Conservative:** ults held until enemy HP > 70% on at least 2 targets or boss-tagged enemy is alive

3.15 Combat Speed Multiplier



In **expanded mode**, players can set combat speed to 1×, 2×, 3×, or 4×. Default is 1×. Higher speeds shrink animation playback time but **do not** alter game logic (no skipped frames in damage roll). The speed setting is saved as a profile preference.

In **compact mode**, combat runs at 1× always — to keep the desktop CPU footprint predictable and prevent the game from spiking the system while idle. Players who want fast farming should expand to the full window.

Tower / Boss Rush / Hero Trial stages remember per-mode speed presets (e.g., Tower defaults to 3×, Campaign defaults to 1×) — separately saved.

3.16 Boss Preview / Scout



Before entering any boss stage (campaign or Tower), tapping a **Scout** button surfaces:

- Enemy roster (sprites + names + element + class)
- Enemy levels (relative to player average team level)
- Boss skill list (icons only on first scout; full descriptions unlock after first defeat)
- Recommended counter-elements / party comp

Scouting is **free** — no resource cost, no cooldown. It exists to reduce blind-attempt frustration.

3.17 Turn Order Timeline UI



The expanded-mode combat screen shows a **horizontal turn order timeline** at the top, displaying the next 6 turns based on action-gauge fill rates and predicted Speed changes. Used by manual players to plan Skill 2 sequencing. Hidden in compact mode (no room).

Timeline updates each tick — when Speed changes mid-turn (e.g., Haste applied), the timeline reshuffles visibly so the player sees the impact of their buff.

A simpler version of this UI is visible to auto players too — it's purely informational and helps explain *why* the auto-AI picks the moves it does.

3.18 Manual Retry



After party wipe, the player sees a **Retry** modal with three options:

1. **Retry (same team)** — restart immediately. No cost.
2. **Edit team & retry** — opens the team-builder. No cost.
3. **Abandon** — return to stage select. No cost.

Manual retries are unlimited. No daily limit. No stamina cost. The only “cost” is time spent on the failed run.

3.19 Replay / Damage Log



At the end of every battle (win or loss), the player can tap **Replay** to open a damage log showing:

- Total damage dealt per hero (with per-skill breakdown)
- Total damage taken per hero
- Healing done per hero
- Status effects applied (count)
- Highest single-hit damage and which skill caused it

The log is not saved between battles — it’s intended for quick “did I make a mistake?” review during the post-battle screen. It’s also useful for the Hero Trials and Tower of Trials min-max community.

A full replay video / step-by-step playback is **not** in v1 (backlog).

3.20 Team Presets



The Team menu lets the player save up to **5 named team presets** (e.g., “Fire Burn”, “Boss Hunter”, “Tower Climb”, “Wind PvE”). Each preset stores:

- Selected 3 heroes
- Each hero’s current equipment
- Each hero’s manual auto-AI overrides
- Optional preset notes

Switching presets is **one tap** — useful for switching between idle farming team and active boss-fight team.

A new battle’s “Last Used Team” is always auto-saved as preset slot #6 (overwrites). This means the player always has a one-tap restore even if they don’t manually save presets.

4. Formulas (consolidated reference)

See section 3.5 for the full damage equation. Other key formulas:

Concept	Formula
Action gauge fill per tick	<code>gauge += Speed × 0.001</code> (target: full at gauge ≥ 100)
HP scaling per level	$HP_{at_L} = BaseHP \times (1 + 0.04 \times (L-1))$
ATK scaling per level	$ATK_{at_L} = BaseATK \times (1 + 0.035 \times (L-1))$
DEF scaling per level	$DEF_{at_L} = BaseDEF \times (1 + 0.04 \times (L-1))$
Shield damage absorption	Shield absorbs incoming Damage first; excess hits HP
Burn / Regen tick	<code>value = (caster.ATK × 0.08)</code> (Burn) or <code>(caster.maxHP × 0.08)</code> (Regen)
Crit chance hard cap	100% (no overflow)
Accuracy hard cap	200% (overflow grants +damage on certain effects — see Lyria passive)
Status resistance hard cap	95% (5% min hit rate always)

5. Edge Cases

- **Simultaneous KO** (mutual kill, e.g., reflect): defender resolves first. Attacker's death is then resolved.
- **Caster KO'd mid-skill animation**: the skill's effects still apply (animation visually completes). Stat-based status (Burn DoT, Regen) referencing the caster's stats uses the **snapshot at cast time**.
- **Cooldown overflow**: if a buff somehow reduces a cooldown below 0, treat as 0 (next turn the skill is available).
- **0-speed scenario**: if all combatants have Speed=0 (test setup), default to position order, then auto-revive Speed to 1 to prevent infinite loop.
- **All enemies dead mid-Ultimate animation**: Ultimate still completes; loot rolls; victory triggers after animation.
- **Hero counterattack when KO'd by triggering hit**: counter does **not** fire — counter requires the defender to survive the hit.
- **Pre-emptive auto-skip**: if a stage is set to "Auto" and the player tabs away, the game keeps simulating. If they tab back, screen recovers to current state — there is no "rewind" buffer.

6. Dependencies

- **Stats system** (`gdd/03-stats-and-progression.md`) — defines HP/ATK/DEF/Speed/CritRate/CritDmg/Accuracy values.
- **Equipment** (`gdd/04-equipment.md`) — supplies sub-stats and set bonuses (some grant counter, lifesteal, shield boost).
- **Hero abilities** (per-hero sheets in `design/heroes/`) — each hero's Skill 1, Skill 2, Ultimate definitions.
- **Campaign / difficulty** (`gdd/09-campaign-structure.md`) — difficulty modifiers, boss enrage timer values.
- **Idle simulation** (`gdd/06-idle-and-shrine.md`) — the auto-AI runs in idle mode at a tick rate defined there.
- **UI / menus** (`gdd/20-ui-menus-tabs.md`) — turn timeline, scout panel, retry modal, replay log.

7. Tuning Knobs

- Skill multipliers per action type (3.5 table)
- Action gauge fill rate constant (`0.001`)
- Crit / Accuracy / Status hit caps
- Per-stat per-level growth coefficients (3.5%, 4%)
- Boss enrage thresholds per difficulty (3.11)
- Default status effect durations (3.13)
- Off-element penalty per difficulty (3.6)
- Counter chance from Defend action (default 30%)
- AoE penalty (Skill 2 single-target multiplier 2.20× vs AoE 1.80×)
- Auto-mode bias slider behavior thresholds (3.14)
- Repeat-wipe auto-downgrade threshold (3 consecutive wipes in 3.9)

8. Acceptance Criteria

- ✓ A hero with 150 Speed acts before a hero with 130 Speed in a controlled test scenario.
- ✓ A Skill 1 with cooldown 2 cast on turn 1 is unavailable on turn 2 and available on turn 3.
- ✓ Damage formula in 3.5 produces deterministic output for fixed seeds (used in unit tests).
- ✓ Element triangle math: Fire-skill on Wind enemy = 1.25× modifier; Fire-skill on Water enemy = 0.75× modifier; Fire on Neutral = 1.0× modifier.
- ✓ Party wipe results in stage auto-restart, with first-clear bonus suppressed and reward roll reduced to 50% of standard repeat table.
- ✓ Buff stacking — applying the same ATK Up twice from same source extends duration, not magnitude.

7.  Auto-AI never uses Ultimate on a single trash mob when a boss-tagged enemy is alive (Balanced bias mode).
8.  Turn order timeline UI updates within 1 frame when Speed changes mid-turn.
9.  Crit / Accuracy / Status caps are enforced (a hero with 250 CritRate still crits at 100%, not 250%).
10.  Compact mode runs at locked 1× combat speed regardless of expanded-mode preference.
11.  Boss enrage applies the documented buff stack and remains for the rest of the battle.
12.  Counter-attacks only fire if the defender survives the triggering hit.
13.  3 consecutive idle-mode wipes on the same stage auto-downgrade idle target to last-cleanly-cleared stage.

GDD 02 — Elements & Classes

Highlighted (`<mark>`) blocks are new vs. the source PDF.

1. Overview

Project I uses a **3-element triangle** ( Fire /  Water /  Wind) and **5 classes** (Tank / Assassin / Archer / Mage / Support). Every hero is exactly one element and one class.

Launch 5★ roster = 15 heroes — exactly one hero per element × class. Plus 15 4★ heroes (designs TBD).

2. Player Fantasy

I’m building a team where each hero counters something specific. Bringing the wrong element punishes me; bringing the right one feels devastating. Mixing classes is how I balance survivability and damage.

3. Detailed Rules

3.1 Element Triangle

Element	Strong vs.	Weak vs.	Visual identity
 Fire	 Wind	 Water	Red/orange palette; ember particles; phoenix imagery
 Water	 Fire	 Wind	Blue/teal palette; wave/bubble particles; ice-themed for some heroes
 Wind	 Water	 Fire	Green/white palette; leaf/wind-line particles; eastern shrine-maiden imagery

Ice = Water decision (D-02): Glacier (Frostwarden) and Auron (Frostbound Prodigy) are visually ice/frost themed but element-tagged as Water. Their status effects (Frostbite, Freeze) are Water-element effects mechanically. This is a *visual identity* difference, not a separate element.

3.2 Element Bonus / Penalty

See `01-combat.md` § 3.6 for the formula. Quick reference:

Difficulty	Bonus (advantage)	Penalty (disadvantage)
Normal	+25%	-25%
Hard	+25%	-25%
Nightmare	+30%	-40%
Hell	+35%	-50%

Elemental discipline scales with difficulty: Hell mode punishes bringing the wrong element more than Normal does.

3.3 Class System

Class	Role	Primary Stats	Armor	Default Weapon Archetype
Tank	Frontline protector, taunter, shielder	HP, DEF, Accuracy	Heavy	Shield-based (sword+shield, gunlance, etc.)
Assassin	Burst single-target DPS, executioner	ATK, Speed, CritRate, CritDmg	Light	Dual blades, scythes, sickles
Archer	Sustained ranged DPS	ATK, Accuracy, CritDmg, Speed	Light	Bows, crossbows, bracers
Mage	AoE damage, debuff, control	ATK, Accuracy, Speed	Magic	Staves, grimoires, fans
Support	Heal, buff, cleanse, revive	Accuracy, Speed, HP	Magic	Staves, harps, lanterns

Weapon archetype clarification (F-04): The class spec defines a *family* of weapons, not a strict requirement. Glacier's Gunlance is a Tank weapon. Marina's Harp is a Support weapon. Auron's Bracer is an Archer-class projectile weapon. Each hero's weapon must visually communicate the class identity (e.g., Tank weapons feel large/heavy, Assassin weapons feel agile/dual-wielded) but exact form factor is per-character.

3.4 Stat Grades per Class

(S = Exceptional, A = High, B = Above Average, C = Average, D = Below Average)

Class	HP	ATK	DEF	SPD	CR	CD	ACC
Tank	S	D	S	D	D	D	B
Assassin	C	S	D	S	S	S	B
Archer	C	A	D	A	B	A	S
Mage	C	S	D	B	D	D	A
Support	B	D	B	A	D	D	S

These are **class averages** — individual heroes may deviate (e.g., a particular Archer may have S CritRate).
Used for balancing and gear recommendation logic only.

3.5 Launch Roster Map (FINAL)



Launch roster — 15 5★ heroes total:

Element	Tank	Assassin	Archer	Mage	Support
Fire	Valdrik	Kaida	Ragnar	Emberia	Akari
Water	Glacier	Seraphina	Auron	Neria	Marina
Wind	Mei Lin	Kael	Lyria	Ayame	Sylphie

Standard banner pool (9 heroes, available at launch) — Valdrik, Emberia, Kaida, Glacier, Marina, Neria, Sylphie, Lyria, Kael Launch Limited Banner (1 hero, available at launch) — Auron Future Limited Banners (5 heroes, released post-launch over time) — Ragnar, Akari, Seraphina, Mei Lin, Ayame

Banner-pool allocation summary:

Pool	Count	Heroes
Standard	9	Valdrik, Emberia, Kaida, Glacier, Marina, Neria, Sylphie, Lyria, Kael
Limited (launch)	1	Auron
Limited (future, ~1 per banner cycle)	5	Akari, Ragnar, Seraphina, Mei Lin, Ayame

F2P players via the 210-pull Standard Selector have 9 heroes available, and the Wishlist system targets those 9. Limited heroes are time-windowed treats; Standard pool may absorb older Limited heroes after their second rerun ends (TBD post-launch).

4★ roster: 15 heroes planned. Designs are forthcoming (user is producing separately). 4★ heroes drop from Epic gacha tier, from Hatchery (rare egg tier), and from event rewards. 4★ → 5★ promotion requires fodder + Apex Catalyst per `gdd/03-stats-and-progression.md`.

4. Formulas

Element math: see `01-combat.md` § 3.5–3.6.

Class stat priorities feed into auto-equip and recommended-team logic:

```
prioritize(hero, gear) = sum(gear.subStat[i].value × classWeight[hero.class][gear.subStat[i].type])
```

`classWeight[class][stat]` is a table. Example for Assassin:

Stat	Weight
ATK	1.0
CritRate	1.0
CritDmg	0.9
Speed	0.8
Accuracy	0.5
HP	0.2
DEF	0.1

5. Edge Cases

- **Neutral-element enemy:** no bonus / no penalty either way. Used for “void” bosses, story-locked enemies, certain Tower of Trials floors.
- **Element disadvantage on a Support hero:** healing skills are *not* element-modified — Marina heals at full strength even against Wind enemies. Only damage skills feel the element math.
- **Class identity vs. weapon form:** a Mage carrying a “Tome” weapon vs. an “Astral Grimoire” — mechanically identical. Different visual identity only.

6. Dependencies

- `01-combat.md` — uses element multipliers, class stat priorities
- `03-stats-and-progression.md` — defines the 7 core stats
- `04-equipment.md` — Heavy / Light / Magic armor types per class
- `design/heroes/*.md` — each hero's element + class assignment

7. Tuning Knobs

- Element advantage/penalty per difficulty (3.2)
- Class stat-grade matrix (3.4)
- Class weight table (formula in 4)

8. Acceptance Criteria

1.  Each of the 15 5★ launch heroes is exactly one element $\in \{\text{Fire, Water, Wind}\}$ and one class $\in \{\text{Tank, Assassin, Archer, Mage, Support}\}$.
2.  Element triangle math matches `01-combat.md`.
3.  Class stat-grade matrix is referenced by the auto-equip logic.
4.  Standard banner pool contains ≥ 10 heroes (so F2P has meaningful Standard pulls).
5.  The “Standard Selector” guaranteed pick (after 210 Standard pulls) chooses from the Standard-pool heroes only.

GDD 03 — Stats & Progression

Highlighted (`<mark>`) blocks are new vs. the source PDF.

1. Overview

A hero's combat power comes from four independent progression axes:

1. **Level** (1 → 60) — raw stat growth from EXP materials.
2. **Star Promotion** (1★ → 6★) — major power tier; consumes fodder.
3. **Awakening** (0 → 6) — duplicate-driven; new passives unlock at certain awakenings.
4. **Equipment** — see `04-equipment.md`.

Plus a fifth axis added here: **Skill Level** (1 → 10 per skill) — incrementally upgrade Skill 1 / Skill 2 / Ultimate with skill stones farmed from stages.

2. Player Fantasy

Every system I touch makes my favorite hero a little stronger. There's always a next milestone — a star, an awakening, a skill rank, a piece of gear.

3. Detailed Rules

3.1 Core Stats (canonical, 7)

Stat	Symbol	Purpose	Cap
HP	HP	Max health	None (display-only ceiling at 999,999)
ATK	ATK	Outgoing damage scaler	None
DEF	DEF	Damage mitigation (see formula in <code>01-combat.md</code> § 3.5)	None
Speed	SPD	Action gauge fill rate	None (effective ceiling ~300 in practice)
Crit Rate	CR	Critical hit %	100%
Crit Damage	CD	Critical damage multiplier %	None (effective ceiling ~400% in practice)
Accuracy	ACC	Hit & status apply chance	200% (overflow grants special effects on certain heroes)

No secondary stats like Magic Power, Healing Power, Damage Reduction, Shield Strength, Energy Generation, or Cooldown Reduction. Those source-PDF references are deprecated: - “Magic Power” / “Healing Power” → folded into ATK (with healing-multiplier defined per-skill). - “Damage Reduction” → folded into DEF (the DefMit formula handles it). - “Shield Strength” → derived from caster’s max HP at cast time. - “Energy Generation” → deleted; no Energy system exists (cooldowns are turn-based). - “Cooldown Reduction” → deleted; cooldowns are integer turns only.

This decision is pending user confirmation (F-05, F-06, F-07).

3.2 Hero Level System

- **Max Level:** 60
- **Stats per level:** small additive growth to HP / ATK / DEF / Speed
- **No skill unlocks tied to level** (skill unlock is part of Awakening — see 3.4)

Gold + EXP cost curve:

Level Range	Gold per level	EXP per level (cumulative for the range)
1–20	500	~50k total
21–40	2,000	~200k total
41–50	5,000	~500k total
51–60	10,000	~1,000k total

Approx total per hero (1→60): ~200,000 Gold + ~1.7M EXP.

EXP Materials:

Material	EXP
Small EXP Crystal	100
Medium EXP Crystal	500
Large EXP Crystal	2,500
Legendary EXP Crystal	10,000

Sources: campaign stages, Shrine rewards, daily missions, events.

3.3 Star Promotion

Major power tier. Available at level milestones; consumes 5 fodder units of the previous star.

Promotion	Level required	Fodder	Gold cost
1★ → 2★	20	5 × 1★	5,000
2★ → 3★	30	5 × 2★	15,000
3★ → 4★	40	5 × 3★	50,000
4★ → 5★	50	5 × 4★	150,000
5★ → 6★	60	5 × 5★	500,000

Promotion materials (in addition to fodder + gold): - 2★→3★: 10 × Sacred Dust (egg / story drop) - 3★→4★: 20 × Sacred Fragment (chapter completion + Tower 10-floor reward) - 4★→5★: 30 × Sacred Crystal (chapter boss drop, Hard+, Tower 20-floor reward) - 5★→6★: 50 × Radiant Soulstone (Tower 50-floor reward, Boss Rush weekly milestone) - Also required for 6★: **1 × Apex Catalyst** per element (Fire/Water/Wind) — **genuinely rare**. F2P realistic supply: ~1 every 1-2 months. Premium-BP buyers: ~1-2 per month. See § 3.7 Apex Catalyst Supply below for exact sources.

This makes 6★ a long-term aspiration even for daily players.

Important rule (from source PDF, kept): 5★ summoned heroes START at 5★ Level 1 — they do **not** start at 1★. But they still consume 4★ fodder to promote to 5★... wait, they're already 5★. Let me re-read the source. The PDF says "5★ units start at 5★ but still require fodder from lower rarities for promotion." So summoned 5★ heroes are at 5★ Level 1, and progress only via Level (1→60), Awakening, gear, and skills — they cannot go below 5★. To promote any 5★ to 6★, the player consumes 5 × 5★ fodder + the materials above. The 5★ fodder can be either: (a) duplicate 5★ heroes (consumed instead of becoming Awakening copies), or (b) sacrificial 5★ fodder units obtained from special events / shop. This keeps 6★ as the rarest, most aspirational tier.

3.4 Awakening (Duplicate System)

Each hero supports 6 Awakening levels (Awakening 0 = base unlock).

Awakening Level	Total Copies	Effect
0	1	Unlock hero
1	2	+5% all stats, +1 ATK skill cap
2	3	Skill 1 +1 level cap, +5% all stats
3	4	Skill 2 +1 level cap, +5% all stats
4	5	Ultimate +1 level cap, +5% all stats
5	6	Unique passive boost per hero (e.g. Lyria's Focus cap raises from 5 → 7 stacks)
6 (max)	7	Awakened Form — unique visual aura + unlocks Awakening exclusive skill bonus per hero. Capstone reward.

Beyond 6: additional duplicate copies convert to **Hero Essence**. Hero Essence is used in: - The **Dupe Shop** (currently called "Essence Shop") — buy promotion materials, EXP crystals, gold, and select-a-hero fragments. - **Hero Shard exchanges** — see `gdd/05-gacha-and-currencies.md`.

Hero Essence rate: 7th copy = 100 Essence; 8th = 110; 9th = 121; (10% growth per copy, capped at 200 per copy beyond the 16th).

3.5 Skill Level (NEW)



Each of a hero's three skills (Skill 1, Skill 2, Ultimate) can be leveled 1–10 independently. Higher skill levels increase damage / heal / buff magnitude — but do **not** alter cooldowns.

Effect per skill rank (additive to base):

Rank	Effect
1	Base (default for newly unlocked hero)
2–5	+3% effect magnitude per rank (i.e., rank 5 = +12% over base)
6	+5% effect (rank 6 = +17% total)
7	+5% (22%)
8	+7% (29%)
9	+8% (37%)
10	+13% (50% over base)

Skill rank cost:

Rank	Skill Stones	Gold
2	5	1,000
3	10	3,000
4	20	8,000
5	35	20,000
6	60	50,000
7	90	100,000
8	130	200,000
9	180	400,000
10	250	800,000

Skill rank cap by Awakening (Section 3.4):

Skill	Default cap	Cap raised by
Skill 1	5	Awakening 2 → 7, Awakening 4 → 9, Awakening 6 → 10
Skill 2	5	Awakening 3 → 7, Awakening 5 → 9, Awakening 6 → 10
Ultimate	5	Awakening 4 → 7, Awakening 5 → 9, Awakening 6 → 10

Skill Stones — sources: - Campaign repeat-clear (Hard+) — small drop chance per stage clear - Daily Mission “Skill Stones x10” reward - Tower of Trials floor milestones - Hero Trials weekly reward - Dupe Shop purchasable - Hero Essence convertible (50 Essence → 10 Stones)

3.6 Element Crystals



The source PDF references “Wind Crystals” as an example inventory item. Defining the full system:

Element Crystals are tier-3+ ascension materials per element. Used as Promotion Materials for star promotions of heroes of the matching element.

Tier	Drop source
Lesser Element Crystal	Stage repeat-clear, matching-element chapter
Greater Element Crystal	Chapter boss drop, Hard+ matching-element chapter
Pristine Element Crystal	Tower of Trials, Hero Trial weekly, matching-element boss
Apex Catalyst	Cross-element (used for 6★ promotion regardless of source element)

Tier upgrade: 10 Lesser → 1 Greater → 10 Greater → 1 Pristine.

Element Crystals are bound to their element — Fire Crystals cannot promote a Water hero. Apex Catalysts are universal but extremely rare.



3.7 Apex Catalyst Supply (tightened)

Apex Catalyst is the gate item for 5★ → 6★ promotion. Total supply is intentionally tight so 6★ promotion is a long-term aspirational goal.

Source	Frequency	F2P?	Premium-BP?
Tower of Trials floor 70 milestone	1 per Tower monthly reset	✓	✓
Boss Rush — full clear on Nightmare	1 per monthly first-clear (NOT weekly)	✓ (if you can clear)	✓
Dupe Shop	1 per month, 100,000 Hero Essence (doubled from earlier draft)	✓ if accumulated	✓
BP Premium Tier 25 reward	1 per season (60 days)	✗	✓
Limited Banner “Power Bundle” \$29.99	1 per bundle purchase	✗	Paid

Estimated realistic monthly supply:

- **Hardcore F2P** clearing Tower 70 + Boss Rush Nightmare + dumping Essence into Dupe Shop = **3 Apex/month**
- **Casual F2P** clearing only Tower 70 = **1 Apex/month**
- **Average F2P** (Tower 70 + Boss Rush Hard-only) = **1-2 Apex/month**
- **Premium-BP buyer** = **+0.5 Apex/month** from BP
- **Limited Banner big-spender** = **+1 Apex** per banner cycle from Power Bundle

This means 6★ promotions happen ~once every 1-2 months for committed players, ~once a season for casual players. That preserves 6★ as a real aspiration over a year of play.

3.8 Sacred Stones (Shrine Material)



Source-PDF referenced “Sacred Stones” for Shrine upgrades. Tier ladder:

Tier	Used for	Source
Sacred Pebble	Shrine 1→10	Stage clear (any)
Sacred Stone	Shrine 11→20	Chapter boss (Hard+)
Sacred Crystal	Shrine 21→30 (planned post-launch expansion)	Weekly Boss Rush, Tower 50-floor reward
Sacred Relic	Shrine 31+ (post-launch)	Limited event reward only

3.9 Progression Loop Summary

```

Pull a hero → start at 5★ L1 (if 5★) or 1-3★ L1 (if egg/fusion)
↓
Level up via EXP Crystals + Gold
↓
Promote (if not 5★+) by consuming fodder + Promotion Material
↓
Awaken via duplicates → unlocks higher skill rank caps
↓
Level skills via Skill Stones
↓
Equip Sub-Stat / Set-Bonus gear (see 04-equipment.md)
↓
Repeat for new heroes; chase 6★ for favorites

```

4. Formulas

Concept	Formula
HP at level L	$\text{BaseHP} \times (1 + 0.04 \times (L-1))$
ATK at level L	$\text{BaseATK} \times (1 + 0.035 \times (L-1))$
DEF at level L	$\text{BaseDEF} \times (1 + 0.04 \times (L-1))$
Skill magnitude at rank R	$\text{BaseEffect} \times \text{SkillRankMultiplier}[R]$ (see 3.5 table)
Hero Essence from 7+ duplicate	$100 \times 1.1^{(\text{copy} - 7)}$, capped at 200
Stat multiplier from Awakening	$1 + 0.05 \times \text{awakeningLevel}$ (cap at +30% at Awakening 6)
Star promotion stat boost	Variable per star — see GDD 04 for gear sub-stat scaling

5. Edge Cases

- **Hero with 0 dupes** — cannot Awaken at all. Stuck at Awakening 0 until dupe arrives or Hero Shards traded for 1 copy.
- **Hero at 6★ Awakening 6 with Skill Rank 10 on all skills** — fully maxed. Future copies all convert to Essence.
- **Skill rank exceeds the current Awakening cap** — system rejects the upgrade attempt with a clear error.

- **Element Crystal mismatch** — promoting a Fire hero with Wind Crystals is rejected at the confirm step.
- **Apex Catalyst shortage** — block 6★ promotion until catalyst is available; the player can hold all other materials ready.

6. Dependencies

- `01-combat.md` — uses stat values
- `02-elements-and-classes.md` — element + class assignments per hero
- `04-equipment.md` — sub-stats add to core stats
- `05-gacha-and-currencies.md` — Hero Essence, Dupe Shop
- `06-idle-and-shrine.md` — Sacred Stones source
- `09-campaign-structure.md` — Element / Skill Stone drop tables

7. Tuning Knobs

- Level growth coefficients (3.5%, 4%)
- Skill rank cost curve (3.5)
- Awakening stat boost (5% per rank, cap 30%)
- Promotion fodder counts (currently 5 per tier)
- Promotion material counts (10/20/30/50)
- Hero Essence conversion rate (3.4)
- Element Crystal tier ratios (10:1)

8. Acceptance Criteria

- ✓ A hero promoted from 4★→5★ at level 50 retains their level (does not reset).
- ✓ Skill rank 10 cannot be reached without Awakening 6 on the affected skill.
- ✓ A duplicate received past Awakening 6 deterministically converts to Hero Essence at the documented rate.
- ✓ Promotion fails cleanly with a user-facing error if Apex Catalyst is missing.
- ✓ Total gold cost from L1 to L60 is approximately 200k (within ±5%).
- ✓ Skill Stones are obtainable through at least 4 distinct content paths (campaign, dailies, Tower, Dupe Shop).

GDD 04 — Equipment

Highlighted (<mark>) blocks are new vs. the source PDF.

1. Overview

Each hero equips 7 items: Weapon, Helmet, Chest, Gloves, Boots, Accessory 1 (Ring), Accessory 2 (Necklace). Each piece has a **main stat** (fixed by slot + rarity) and 2–3 random **sub-stats** rolled at drop. Pieces also belong to a **set**, with 2-piece and 4-piece bonuses if multiple set members are equipped together.

2. Player Fantasy

Finding a piece with the perfect sub-stats is its own reward — and committing to a Berserker set on my Assassin actually changes how she plays.

3. Detailed Rules

3.1 Equipment Slots

Slot	Main stat	Sub-stat eligibility
Weapon	ATK% or flat ATK	ATK%, CritRate, CritDmg, Accuracy
Helmet	HP% or flat HP	HP%, DEF%, Accuracy, Speed
Chest	DEF% or flat DEF	DEF%, HP%, Accuracy, Speed
Gloves	CritRate%, CritDmg%, or ATK%	ATK%, CritRate, CritDmg, Speed
Boots	Speed (flat) or Speed%	Speed, HP%, DEF%, Accuracy
Ring (Accessory 1)	Any flat stat	Any sub-stat
Necklace (Accessory 2)	Any flat stat	Any sub-stat

Rationale: restricting main-stats per slot prevents trivial double-dipping (e.g. you can't put ATK% on Boots — Boots are for Speed). Accessories are open-slot for flexibility.

3.2 Armor Types (by class)

Armor type	Classes	Look/feel
Heavy	Tank	Plate, full helms, large gauntlets
Light	Assassin, Archer	Leather, cloth-lined, agile silhouette
Magic	Mage, Support	Robes, ornamental helms, ceremonial garb

Armor type is **purely cosmetic** at the system level — sub-stats can roll on any armor type. Heavy/Light/Magic is a visual identity bucket, not a mechanical gate. (Source PDF implied this; clarifying.)

3.3 Rarity Tiers

Rarity	Main stat ceiling	Sub-stat count	Drop weight
Common	Low	0–1	Very common
Uncommon	Low-mid	1–2	Common
Rare	Mid	2	Common
Epic	High	2–3 (3 unlocked at +12 enhance)	Uncommon
Legendary	Very High	3 (all 3 at drop)	Rare
Mythic (endgame)	Highest	3 + 1 <i>bonus</i> (max 4)	Very rare; Tower/Boss Rush only

The “Cosmic” tier from TBH is **not** used. Project I tops out at Mythic, with most of the curve centered on Epic / Legendary.

3.4 Main Stat Ladder

Main stats grow with enhancement level (0 → +15). Sample table for a Legendary Weapon:

Enhance	ATK% Main Stat
+0	18%
+3	25%
+6	32%
+9	41%
+12	50%
+15	60%

Enhancement increments: Every +3 enhances (“safe milestones”) unlock a sub-stat. Main stat grows continuously per +1 enhance. Failure at any enhance level **does not destroy** the item — it consumes the upgrade material only. Items cannot be destroyed by enhancement.

3.5 Sub-Stat System (NEW)



Sub-stat pool (rolls one per random slot):

Sub-stat	Note
ATK +%	additive to base ATK
HP +%	additive to base HP
DEF +%	additive to base DEF
Speed	flat additive
CritRate%	flat additive
CritDmg%	flat additive
Accuracy%	flat additive
Status Resistance%	flat additive (reduces incoming status duration / chance)
Lifesteal%	% of damage dealt returned as HP (caps at 25% per piece)
Effect Hit%	bonus to status apply roll (separate from Accuracy — designed for Mage builds)
Counter-Attack%	flat additive — chance to counter when hit by single-target attack

Roll mechanics: - Initial drop: 2 sub-stats on Rare/Epic, 3 on Legendary/Mythic. - A 3rd sub-stat slot unlocks at +12 enhancement (so Epic effectively has 3 slots after enhancement). - Each sub-stat **value** rolls within a range determined by rarity. Higher rarity = higher minimum / maximum. - At each +3 enhance milestone, an existing sub-stat receives a random additive roll (within its tier range). The player can lock 1 sub-stat per enhance attempt (using a **Lock Stone** material) to prevent it from being chosen for the roll-up. - Mythic items uniquely have a 4th sub-stat that is **fixed** (e.g. always Lifesteal). The fixed sub-stat is set by the boss / source that drops it.

Sub-stat reroll: - The **Forge** screen offers a sub-stat reroll using **Reroll Stones** (rare currency from Tower / Boss Rush / dupe shop). - A reroll completely re-randomizes ONE sub-stat slot (player picks which). It's destructive — the old value is gone. - A “preview-then-confirm” UI shows the new roll before committing — but the new roll cannot be re-rolled cheaply. Each reroll attempt costs the same Stones. - This makes endgame gear hunting about (a) farming the right item, (b) committing carefully to reroll which sub-stat, (c) accepting the variance.

3.6 Equipment Sets (NEW)



6 sets ship at launch. Each set has a 2-piece and 4-piece bonus.

Set	2-piece bonus	4-piece bonus	Theme
Berserker	+15% ATK	+30% Counter-Attack chance, +20% CritDmg	Aggressive DPS
Bastion	+20% DEF	+25% Shield Strength, +15% Damage Reduction (clamped to DEF stat in formula)	Tank specialist
Swiftwind	+25% Speed	+1 Action Gauge per ally-Speed-Up trigger	Speed-stack teams
Mystic	+15% Accuracy	+25% Effect Hit, debuffs last +1 turn	Mage / debuff
Vampire	+12% Lifesteal	+30% Healing on Lifesteal procs, ignores 20% enemy DEF	Sustain DPS
Lifegiver	+20% Healing Done	+1 cleanse per turn for the wearer, allies under 50% HP heal +25% from this hero	Support specialist

Set requirements: - 2-piece = any 2 pieces of the same set - 4-piece = any 4 pieces of the same set - A hero can wear 2 different 2-piece sets (e.g. 2 Berserker + 2 Bastion = both 2-piece bonuses active) OR commit to a 4-piece + 2-piece split (e.g. 4 Berserker + 2 Bastion). - The 4-piece bonus **replaces** the 2-piece bonus (it does not stack with it from the same set).

Sets do NOT apply to Accessories. Ring and Necklace are slot-agnostic and never count toward set count.

3.7 Enhancement & Salvage (NEW)



Enhancement: - Each +1 enhance costs gold + an Enhance Stone (item) of matching rarity. - Failure rate at +12+ enhances: 30% (item stays, materials consumed). Use Lucky Charms (premium item / event reward) to guarantee. - No item is destroyed by enhancement.

Salvage: - Sell any unwanted item for gold. - “Salvage” instead refunds part of the upgrade materials + adds Enhance Stones. - Bulk-salvage by rarity (Common / Uncommon / Rare all selectable in inventory). - Locked or Favorited items cannot be salvaged.

Salvage refund table:

Item rarity + enhancement	Refund
Common +0	10 Gold
Rare +6	200 Gold, 1 Common Enhance Stone
Epic +12	1,500 Gold, 1 Rare Enhance Stone, 1 Reroll Stone
Legendary +15	8,000 Gold, 1 Epic Enhance Stone, 2 Reroll Stones
Mythic +15	50,000 Gold, 1 Legendary Enhance Stone, 5 Reroll Stones

3.8 Quick-Equip & Loadouts



Quick-Equip “Recommended” button — in the Equipment menu, one tap auto-equips the highest-priority items per the hero’s class weights (see GDD 02 § 3.4).

Quick-Swap presets — each hero stores up to 3 named equipment loadouts (e.g. “Boss Fight”, “Idle Farm”, “PvE Speed”). One-tap swap.

Equipment Lock / Favorite — both flags exist: - **Locked** = cannot be sold/salvaged/fed into upgrades. Toggle in detail view. - **Favorited** = sort-priority flag, does not block actions. Used for “show favorites first” filters.

Bulk-select — multi-select up to 100 items at once for bulk-sell or bulk-salvage. Confirm dialog shows total return.

3.9 Equipped Item Display

The Equipment screen shows, for each slot: - Item icon + rarity color border - Main stat (highlighted) - All sub-stats (current values, color-coded by tier) - Enhancement level (0/+15) - Set name + current set-count progress (e.g. “Berserker 2/4”) - Locked / favorited indicator

Tapping an equipped item opens a comparison view showing alternative items in inventory.

4. Formulas

Concept	Formula
Effective stat with gear	$\text{BaseStat} \times (1 + \sum \text{percentGearBonus}) + \sum \text{flatGearBonus}$
Shield value when set bonus applies	$0.25 \times \text{wearer.maxHP} \times (1 + \text{bastion4pcBonus})$
Counter-attack roll	$\text{roll} < (\text{BaseCounterChance} + \sum \text{gearCounter})$
Lifesteal heal	$\text{damageDealt} \times (\sum \text{lifestealSubStat}) \times (1 + \text{vampire4pcBonus})$

5. Edge Cases

- **Same sub-stat rolled twice:** Treat as additive (e.g., two +5% ATK rolls = +10% effective).
- **All 3 sub-stat slots rolled into the same stat:** Possible. RNG. Diminishing returns are not enforced — the stack is real.
- **Reroll picks the same value:** Allowed. The reroll is destructive of the old value; if RNG produces the same number, the player just loses Stones for no gain.
- **4-piece set with mismatched main stats:** Allowed. Set bonuses care about set membership, not main stats.
- **Locked item attempts to be sold via bulk-select:** Excluded from the bulk operation, surfaced in the confirm dialog (“8 items selected, 1 locked — not selling”).
- **Mythic item’s fixed 4th sub-stat re-rolls?** No — the fixed slot cannot be rerolled. Only the 3 randomized slots can.

6. Dependencies

- `01-combat.md` — uses gear stats in damage formula
- `03-stats-and-progression.md` — defines core stat list and class weights
- `02-elements-and-classes.md` — class → armor type mapping
- `08-inventory-and-storage.md` — inventory slots and bulk actions
- `09-campaign-structure.md` — drop tables for gear by stage / boss
- `12-daily-weekly-loops.md` — Reroll Stones from weekly content

7. Tuning Knobs

- Sub-stat value ranges per rarity (3.5)
- Set bonus magnitudes (3.6)
- Enhancement failure rates at +12+ (3.7)
- Sub-stat pool (3.5) — can add new sub-stats post-launch
- Number of equipment sets — start with 6, add more in seasonal patches
- Reroll Stone cost per reroll (currently 1 per attempt)

8. Acceptance Criteria

- ✓ Each equipped item shows its main stat + sub-stats with values.
- ✓ Set 2-piece and 4-piece bonuses apply correctly when piece count is met.
- ✓ Mythic items always have exactly 4 sub-stats; one is fixed per drop source.
- ✓ Sub-stat reroll consumes 1 Reroll Stone and randomizes exactly one player-selected sub-stat.
- ✓ Enhancement failure never destroys the item.
- ✓ Bulk-salvage of 50 Common items returns the documented salvage value within 1% accuracy.
- ✓ Quick-Equip respects class stat priorities from `02-elements-and-classes.md`.
- ✓ Locked items cannot be salvaged via bulk-select.

GDD 05 — Gacha & Currencies

Highlighted (`<mark>`) blocks are new vs. the source PDF.

1. Overview

Project I uses a multi-currency gacha. Two banner types, two pity systems, three currencies, plus a Dupe Shop powered by Hero Essence.

Adds: Pity Tracker UI, Wishlist system, Beginner Step-Up Banner (replaces flat discount), Free Daily Pull, Hero Shard exchange.

2. Player Fantasy

Pulls feel exciting because rare hits really happen — and even when they don't, the pity counter is visibly moving, the wishlist is closing in, and dupes are turning into Essence I can spend on a hero I actually want.

3. Detailed Rules

3.1 Currencies

Currency	Tier	Source	Use
Gold	Progression	Stages, Shrine, dailies, idle	Level / promote / enhance
Gems	Free / paid soft	Quests, achievements, events, BP, paid packs	Buy Summon Tickets
Crystals	Paid only	Real-money purchases ONLY	Convert 1:1 to Gems; premium shop
Summon Ticket	Item	160 Gems each	Single pull on a banner
Hero Essence	Dupe soft	Duplicate conversion past Awakening 6	Dupe Shop
Hero Shard	Targeted dupe	Specific hero-shard drops from dupe conversion + monthly events	50 shards = 1 copy of a chosen Standard hero

Crystal → Gem conversion is one-way. Once converted, the Gems cannot be re-converted to Crystals. This protects the Crystal-only shop economy.

3.2 Banner Types

Limited Character Banner

- 1 featured Legendary, 2-3 featured Epics
- Duration: 21 days
- After expiry: featured Legendary leaves rotation, eventually returns in reruns
- Rates: Legendary 0.6%, Epic 7.2%, Rare 92.2%
- **Hard pity only at 50 pulls.** No soft pity — the rate stays at 0.6% on every pull until the 50th, at which point a Legendary is guaranteed.
- Carries over to NEXT Limited banner (does not reset on banner end)
- **Launch Limited Banner: Auron** (Water Archer, Frostbound Bracer). Follow-up Limited Banners over the live-ops calendar: Ragnar → Akari → Seraphina → Mei Lin → Ayame.

Standard Banner

- Permanent
- All Standard-pool heroes
- Rates: same as Limited
- **Hard pity only at 50 pulls (no soft pity).**
- **Standard Selector** unlocks after 210 Standard pulls (one-time, free Standard 5★ pick)
- **Launch Standard pool: 9 heroes** — Valdrik, Emberia, Kaida, Glacier, Marina, Neria, Sylphie, Lyria, Kael. After a Limited Banner hero's second rerun, they may rotate into the Standard pool (post-launch decision).

Beginner Step-Up Banner (NEW)



Replaces the source-PDF “first 50 pulls 20% off” with a structured step-up:

Step	Pulls	Cost	Reward
1	10 pulls	1,200 Gems (was 1,600 — 25% off)	Guaranteed Epic+ in batch
2	10 pulls	1,400 Gems	Guaranteed Legendary in batch
3	10 pulls	1,600 Gems	Guaranteed Standard Selector unlock + 1 free Standard 5★ pick

Available only to accounts ≤ 14 days old AND that haven't completed step 3 yet. Once step 3 is done, the banner disappears forever for that account.

Pity isolation: Step-Up Banner pulls have their **own pity counter** — they do NOT count toward Limited or Standard hard pity. The Step-2 guaranteed Legendary is a *bonus* on top of normal pity, not a consumption of it. This is intentional: it gives new players a strong first-week without burning their long-term pity counters.

Legendaries pulled on the Step-Up Banner ARE drawn from the **Standard Banner pool** (so Step 2's guaranteed Legendary is one of the 9 Standard heroes — not a Limited Banner hero).

3.3 Pity Tracker UI (NEW)



The Gacha screen always shows:

- **Current pulls toward hard pity (e.g., “27/50”) — single counter, no soft pity stage**
- Standard Banner cumulative count (toward 210 Selector unlock)

- Time remaining on Limited banner (countdown)

A small “pity will carry over” banner shows on the last day of a Limited banner so players don’t accidentally feel cheated.

3.4 Wishlist System (NEW)



In the Standard Banner UI, the player picks **3 Wishlist heroes** from the Standard pool. The hard-pity Legendary (every 50 pulls) is guaranteed to be **one of the wishlisted heroes** — random choice among the 3.

Rules: - Wishlist edits are free, but locked for 24 hours after a save (prevents abuse). - A hero that has reached max Awakening (6) is greyed out from the Wishlist — pulling them is wasteful. - Standard Selector unlock (every 210 pulls) is also auto-filtered to suggest Wishlist heroes first.

This single feature is a huge F2P quality-of-life lever. Modern players expect it (Genshin / Star Rail set the standard).

3.5 Free Daily Pull (NEW)



Every 24 hours since last reset, the player gets **one free pull on the Standard Banner** (does not apply to Limited). Counts toward both Standard pity and Standard Selector (210). Cannot be saved up — must be claimed daily.

This is small but compounding: 365 free pulls / year. Big retention lever.

3.6 Pull Types

Type	Cost	Notes
Single Pull	1 ticket	Independent roll
10-Pull	10 tickets	Epic guarantee in batch (1 Epic minimum, can be higher)

10-Pull pity progression is identical to 10 single pulls.

3.7 New Player Bonus (folded into Step-Up Banner)

The source-PDF “first 50 pulls 20% discount” is replaced by the **Beginner Step-Up Banner** (3.2). This is more impactful for players AND simpler to communicate.

Track flags: - `beginnerStepUp.step1Complete: bool` - `beginnerStepUp.step2Complete: bool` - `beginnerStepUp.step3Complete: bool`

3.8 Duplicate Conversion + Hero Shards



When a duplicate of a hero already at max Awakening (6) is pulled:

- The player receives **Hero Essence** (universal currency — see 3.1).
- The player **also** receives **Hero Shards specific to that hero**. 1 dupe = 5 Hero Shards.
- 50 Hero Shards = 1 copy of that hero (used for fusion, Awakening, fodder, or just unlocking heroes you haven't pulled).
- **Cross-conversion:** 50 Shards of any Standard hero can be exchanged for 50 Shards of any other Standard hero at a 2:1 cost (so 100 source → 50 target). Limited heroes cannot be cross-converted.

Why this matters: F2P players who never pull a specific hero can still slowly unlock them through Shard accumulation. Whales who over-pull get something useful for the next dupe instead of pure Essence.

3.9 Dupe Shop (Essence Shop)



A persistent shop using **Hero Essence** as the currency.

Stock (refreshes weekly except where noted):

Item	Cost	Limit
10x Skill Stones	500 Essence	5/week
1x Legendary EXP Crystal	1,000 Essence	3/week
1x Sacred Stone	2,000 Essence	2/week
1x Pristine Element Crystal (random element)	5,000 Essence	1/week
1x Apex Catalyst	100,000 Essence	1/month
100x Hero Shards of a player- chosen Standard hero	10,000 Essence	2/month
1x Mythic Reroll Stone	8,000 Essence	3/month
Cosmetic emote pack	3,000 Essence	Permanent

Tuning knob: cost values are adjustable post-launch via hot-config (see [gdd/21-liveops-and-telemetry.md](#)).

3.10 Roll Algorithm



1. Add 1 to pullCount for the banner
2. If hard pity hit (pullCount == 50): force Legendary roll
Else: roll d(10000) against baseRate (0.6% Legendary / 7.2% Epic / 92.2% Rare)
3. If Legendary:
 - Limited banner: pick the featured Legendary
 - Standard banner:
 - If hard pity (pullCount == 50): pick from Wishlist (random among up to 3)
 - Else: weighted random from Standard pool
4. If Epic: weighted random from Epic pool
5. If Rare: weighted random from Rare pool
6. If Legendary acquired (any source): reset pullCount to 0 for this banner
7. Save state

Rate stays flat at 0.6% Legendary on every pull until 50. No soft-pity ramp.

3.11 Account-Level Banner Data (save schema)

```
{
  "gacha": {
    "limited": {
      "pullCount": 27,
      "guaranteed": true // true if next Legendary is the featured one (Limited only)
    },
    "standard": {
      "totalPullCount": 184,
      "standardSelectorClaimed": false,
      "wishlist": ["valdrik", "lyria", "sylphie"],
      "wishlistLockedUntil": 1717612800
    },
    "freeDailyPull": {
      "lastClaimedAt": 1717526400
    },
    "beginnerStepUp": {
      "step1Complete": true,
      "step2Complete": false,
      "step3Complete": false
    },
    "essence": 23400,
    "heroShards": {
      "valdrik": 18,
      "kael": 50,
      "lyria": 6
    }
  }
}
```

4. Formulas

Concept	Formula
Pull rate	flat 0.6% Legendary / 7.2% Epic / 92.2% Rare on every pull
Hard pity guarantee	<code>pullCount == 50</code> → forced Legendary; pullCount resets to 0 on any Legendary pull
Wishlist pull (Standard hard pity)	random uniform among up to 3 wishlist heroes
Cross-conversion cost	50 source shards → 25 target shards (2:1 ratio)
Essence from 7th+ duplicate	$100 \times 1.1^{(\text{copy} - 7)}$, cap 200 (per GDD 03 § 3.4)

5. Edge Cases

- **Pity carry-over on banner expiry** — Limited pity carries to NEXT Limited banner. Standard pity is permanent (never resets).
- **Wishlist edit during a banner run** — allowed; takes effect on the next hard-pity Legendary.
- **Wishlist hero pulled normally before hard pity** — counts as a normal Legendary pull. Pity resets. Wishlist will trigger on next hard pity unless that hero already maxed.
- **Wishlist hero hits Awakening 6** — they are auto-removed from Wishlist and a notice appears asking the player to pick another.
- **Free daily pull during step-up banner** — both can trigger same day. Free pull goes on Standard banner; step-up pull is independent.
- **Step-up pull while pity is high** — the step-up rewards (forced Epic/Legendary) override the pity-counter behavior for that pull only.

6. Dependencies

- `03-stats-and-progression.md` — Hero Essence, Awakening, Element Crystals
- `04-equipment.md` — Reroll Stones from Dupe Shop
- `14-monetization.md` — Crystal purchases, Beginner Pack
- `16-tutorial-onboarding.md` — Day-1 free pulls, Step-Up funnel
- `17-save-system.md` — banner state persisted across sessions
- `20-ui-menus-tabs.md` — Gacha screen layout, Pity Tracker

7. Tuning Knobs

- Hard pity threshold (currently 50; no soft pity)
- Standard Selector unlock count (currently 210)
- Wishlist size (currently 3)
- Free daily pull cadence (currently 24h)
- Hero Shard ratio (1 dupe = 5 shards; 50 = 1 copy)
- Cross-conversion ratio (currently 2:1)
- Beginner Step-Up step costs and rewards
- Dupe Shop costs and weekly limits

8. Acceptance Criteria

- ✓ Pulling 50 times on Limited without hitting Legendary guarantees a Legendary on pull 50.
- ✓ The Standard hard pity Legendary is one of the player's 3 wishlisted heroes if the wishlist is set.
- ✓ **No soft pity:** Legendary rate stays at flat 0.6% on every pull until 50. Pull 50 is the only guaranteed Legendary.
- ✓ Pity carries over to the next Limited banner.
- ✓ Free daily pull awards exactly 1 Standard pull every 24 hours.
- ✓ Beginner Step-Up Banner is hidden for accounts >14 days old.
- ✓ Duplicate at Awakening 6 grants both Hero Essence and 5 Hero Shards.
- ✓ 50 Hero Shards convert to 1 hero copy.
- ✓ Crystal → Gem conversion is irreversible.
- ✓ Wishlist cannot include max-Awakening heroes.

GDD 06 — Idle & Shrine

Highlighted (<mark>) blocks are new vs. the source PDF.

1. Overview

Project I is an **idle** game first. Heroes fight automatically while the player works. Two distinct passive systems generate value: **Idle Combat Rewards** (loot from auto-battles) and the **Shrine** (Gold-only passive generator).

Adds: Idle Stage Selection UI, Idle ROI display, Active vs. Offline simulation rules.

2. Player Fantasy

I started the game when I sat down at my desk. Eight hours later I tab back in and there's a satisfying pile of loot waiting — enough to actually progress my heroes tonight.

3. Detailed Rules

3.1 Idle Combat Rewards

While the game is **open** (active background), heroes run the selected idle stage on loop and collect:

- **Gold** — based on stage Gold/clear × clears/hour
- **Hero EXP** — Small/Medium EXP Crystals only, no Legendary from idle
- **Equipment** — drops per stage table; Epic+ has small chance, Legendary only on idle every ~3 hours on appropriate stages
- **Enhancement Materials** — Common/Rare Enhance Stones
- **Summon Currency (Rare)** — small Gem trickle, ~30/hour at Stage 50+
- **Skill Stones** — small drip, gated to stages chapter 5+ (Hard difficulty unlocks higher rates)

Maximum idle storage: 12 hours. After 12h, no further accumulation until claimed.

Quality-of-life: If idle storage caps, a desktop notification fires (toggleable in settings). Compact window also shows a “STORAGE FULL” badge.

3.2 Active vs. Offline Idle (NEW)



Two distinct simulation modes:

Mode	Rate	When
Active	100%	Game window open (compact OR expanded), focused or unfocused
Background	100%	Game running in system tray (TBH-style)
Sleep / Closed	75%	Computer sleeping OR game fully closed; capped at 12-hour window

The 75% offline rate is a small “thank you for keeping the game open” incentive without punishing players for closing their PC.

Idle accumulation always caps at 12 hours of equivalent rate, so a player who’s offline for 4 days does NOT get 4 days of rewards on login.

Simulation determinism: offline accumulation is computed at login time using a deterministic seed (saved with the player profile). This means: (a) the player can’t cheat by changing system clock, (b) re-simulating offline rewards gives the same result. See `gdd/17-save-system.md`.

3.3 Idle Stage Selection (NEW)



The player can idle on **any previously cleared stage**. Compact and Expanded windows both surface a “Currently Idling On: Chapter 4-7 (Normal)” indicator with a **Change Stage** button.

The Idle Stage Picker shows a sortable list of all unlocked stages with:

Column	What it shows
Stage name	Chapter 4-7 Normal
Last-clear time	Mean battle time × 60 / clears-per-hour
Gold / hour	Estimated gold drop rate
EXP / hour	Estimated EXP drop
Drop preview	Top 3 items by drop chance with element / rarity badge
Success rate	Auto-mode win rate from last 20 attempts (e.g., “95%”)
3-star tier reached	Best clear tier badge

Recommendation badge: the highest-value stage that the current team can clear at $\geq 90\%$ rate gets a  “Recommended” badge. Players can ignore the recommendation and idle on whatever they want.

Pinned stages: the player can pin up to 3 stages for quick-swap. Pinning is local-save, not server-side.

3.4 Shrine

A persistent Gold-generation building. Generates Gold/hr while online or offline. Has a cap; collected manually (or auto-collected per setting). Levels 1-30 ship at launch (max 30 launch cap; expansions to 50+ later).

Level	Gold/hr	Cap	Materials needed
1	50	500	Free (start)
5	150	1,500	Gold only
10	500	5,000	Gold + Sacred Pebble × 20
15	1,000	10,000	Gold + Sacred Stone × 5
20	2,000	20,000	Gold + Sacred Stone × 20
25	4,000	40,000	Gold + Sacred Crystal × 5
30	8,000	80,000	Gold + Sacred Crystal × 30

Auto-collect toggle (setting): when ON, the shrine empties automatically into the player’s gold balance at the moment it hits the cap. Default: OFF (so the player retains the act of “going to claim” — a small dopamine moment).

Shrine collect animation: shrine collection triggers a brief gold-shower animation + satisfying SFX. Critical for feel.

3.5 Shrine Materials

See `gdd/03-stats-and-progression.md` § 3.7 for the full Sacred Stone ladder.

3.6 Daily Gameplay Loop

```

Open game / launch tray
  ↓
Auto-collect or claim Shrine
  ↓
Claim idle rewards
  ↓
Equip / level new gear
  ↓
Run new content (campaign / Tower / Expedition / dailies)
  ↓
Set idle target stage
  ↓
Tab away, work

```

3.7 Idle ROI Display (NEW)



In the Idle Stage Picker, a small “vs. current” comparison strip shows:

```

Current: Chapter 4-7 Normal – 230 G/hr, 95% success
Recommended: Chapter 5-3 Normal – 280 G/hr (+22%), 92% success
  ↳ "+50 G/hr but 3% more wipes"

```

This is purely informational. Helps min-maxers without forcing them.

4. Formulas

Concept	Formula
Idle gold/hr at stage S	$\text{goldPerClear}[S] \times (3600 / \text{avgClearSeconds}[S])$
Offline accumulation	$\min(\text{hoursAway}, 12) \times \text{idleRate} \times 0.75$ (Sleep / Closed)
Shrine accumulation	$\text{Shrine.rate} \times \text{elapsedHours}$, capped at Shrine.cap
Win rate estimate	Last 20 attempts on this stage with current team

5. Edge Cases

- **Player logs in after 24h:** they get 12h of accumulation (cap), not 24h. UI explicitly shows “Storage was full — collect more frequently to maximize.”
- **Idle target stage is deleted by patch:** auto-revert to last successfully cleared stage; flash a notice.
- **Team is too weak for current idle stage:** auto-wipe-downgrade (3 consecutive wipes) triggers per `01-combat.md` § 3.9.
- **Player changes idle stage mid-accumulation:** previous accumulated rewards are NOT lost; they’re calculated against the OLD stage at the moment of change. The new stage starts fresh.
- **Daylight savings / system clock jumps:** server-validated time (or local time + clock-tamper detection) prevents exploits.

6. Dependencies

- `01-combat.md` — combat simulation runs idle battles
- `03-stats-and-progression.md` — Shrine materials
- `09-campaign-structure.md` — drop tables per stage
- `15-desktop-window-ux.md` — compact-mode idle visualization, notifications
- `17-save-system.md` — deterministic offline simulation, anti-tamper
- `21-liveops-and-telemetry.md` — Shrine drop-rate hot-config

7. Tuning Knobs

- Idle cap (12 hours)

- Offline rate (75%)
- Shrine level table (3.4)
- Auto-collect default OFF
- ROI threshold (recommend stage when $\geq 10\%$ G/hr improvement with $\geq 85\%$ success rate)

8. Acceptance Criteria

1.  Player who closes the game for 8 hours gets exactly $8h \times 0.75 = 6h$ of equivalent rewards.
2.  Player closed for 30 hours gets capped at 12h of rewards.
3.  Idle target stage is configurable from compact AND expanded mode.
4.  Recommendation badge appears on highest-ROI stage with $\geq 90\%$ success rate.
5.  Shrine auto-collect toggle works correctly (when ON, collects at cap; OFF requires manual).
6.  Desktop notification fires when idle storage caps (if notifications enabled).
7.  Offline simulation is deterministic for the same seed (testable).

GDD 07 — Eggs & Fusion

1. Overview

The Egg system supplies a steady stream of 1★–3★ units used as **promotion fodder**, **Shrine workers** (passive Gold buff), and **Expedition crew**. Eggs are obtained via Gold purchase, story rewards, dailies, and events. They hatch on timers via the Hatchery building. **Adds: full hatch reward tables, fusion ladder confirmation, instant-hatch pricing.**

2. Player Fantasy

Between gacha sessions, there's always another egg ticking down. The small wins make the big wins feel earned.

3. Detailed Rules

3.1 Egg Tiers

Egg	Hatch time	Cost	Drop pool
Common Egg	1 hour	500 Gold	100% 1★
Uncommon Egg	4 hours	2,500 Gold	85% 1★, 15% 2★
Rare Egg	8 hours	15,000 Gold	60% 1★, 35% 2★, 5% 3★
Epic Egg (event/shop only)	12 hours	—	30% 2★, 60% 3★, 10% 4★

Epic Eggs are a NEW tier added beyond the source PDF. They are not Gold-purchasable — only granted from events, dailies, Tower milestones, and the Dupe Shop. This adds an aspirational tier without inflating the basic egg loop.

3.2 Hatchery

- Starts at 1 slot.
- Max 5 slots (additional unlocked via account level milestones — see `12-daily-weekly-loops.md`).
- Each slot has its own independent timer.

- An empty slot can accept any egg.

3.3 Hatch Process

1. Drag egg into open slot
2. Timer starts (per egg tier)
3. Notification when ready (per setting)
4. Player taps Hatch
5. Unit awarded → enters character storage

3.4 Instant Hatch

Spend Crystals to skip the timer. Cost scales with remaining time:

Remaining time	Crystal cost
0:00:00	0 (free claim)
0:15:00	5 Crystals
1:00:00	15 Crystals
4:00:00	30 Crystals
8:00:00	60 Crystals
12:00:00	90 Crystals

3.5 Egg Drop Pools (sample)

Common Egg pool: - 70% Slime (1★) - 10% Forest Sprite (1★) - 10% Mushroomling (1★) - 5% Tiny Wolf (1★) - 5% Baby Drake (1★)

(Full tables for all 4 egg tiers live in the data table — `data/eggs.json` . The above is illustrative only.)

3.6 1★ / 2★ / 3★ Unit Roles

These low-rarity units are **never trash**. Uses:

- **Promotion fodder** — fuse 5 same-star → 1 next-star
- **Shrine workers** — equip 1-3 low-rarity units to the Shrine for +5-15% Gold/hr bonus (max 3 workers, stacks)
- **Expedition crew** — many Expedition missions accept low-rarity heroes
- **Achievement / Codex completion** — collecting all units rewards titles + gems

3.7 Unit Fusion Ladder

Fuse	Cost	Result
5 × 1★	1,000 Gold	1 × 2★
5 × 2★	5,000 Gold	1 × 3★
5 × 3★	25,000 Gold	1 × 4★
5 × 4★	100,000 Gold	1 × 5★
5 × 5★	400,000 Gold + 1 Apex Catalyst	1 × 6★

Note: 5★ → 6★ via fusion gives a generic “fodder 6★” (random class/element) — not a specific Legendary hero. Useful for the 5★→6★ promotion of named heroes (you need fodder 5★ to consume; this is how you make them).

3.8 Egg Acquisition Sources

Source	Eggs
Gold shop (always available)	Common, Uncommon, Rare (Gold cost)
Story stages	Common eggs, 1 per first-clear
Shrine milestones (every 5 levels)	Uncommon egg
Daily missions	1-2 eggs depending on completion
Weekly missions	Rare egg per week tier
World Exploration (Expedition)	Random eggs
Achievements	One-time Rare or Epic egg rewards
Battle Pass tracks	1 Epic egg in premium track per BP
Events (post-launch)	Limited eggs / event eggs

4. Formulas

Concept	Formula
Instant-hatch cost	$\text{ceil}((\text{remainingSeconds} / 60) \times 0.15)$ Crystals (capped at 90 for 12h)
Shrine worker bonus	$\Sigma \text{workerBonus} = 5\% \times \text{workerCount}$ (max 3 workers, max 15% bonus)
Fusion success rate	100% (no failure for 1★–5★); 100% for 5★→6★ but Apex Catalyst required

5. Edge Cases

- **All hatchery slots full + waiting eggs:** eggs queue in inventory; player must manually slot them.
- **Character storage full when hatching:** hatch blocks; UI prompts to expand or fuse first.
- **Fusion with mismatched rarities:** rejected.
- **Instant-hatch right after starting:** 0 seconds remaining = 0 Crystals (just claim).
- **Shrine workers locked from being consumed for fusion:** auto-flagged. Player must manually unlink to free them.

6. Dependencies

- 03-stats-and-progression.md — star promotion uses fusion outputs as fodder
- 06-idle-and-shrine.md — Shrine workers
- 08-inventory-and-storage.md — egg/character storage slots
- 11-world-map-expedition.md — Expedition reward eggs

7. Tuning Knobs

- Egg costs (3.1)
- Hatch times (3.1)
- Drop pool distributions (3.5)
- Fusion costs (3.7)
- Instant-hatch crystal formula (4)
- Shrine worker bonus magnitude (4)

8. Acceptance Criteria

1.  Common Egg always hatches to a 1★ unit.
2.  Rare Egg has a 5% chance to hatch a 3★ unit (testable over 1000 runs).
3.  Instant-hatch correctly charges Crystals per the formula.
4.  Fusion consumes exactly 5 same-rarity units + Gold cost.
5.  Fusion of 5 × 5★ requires 1 Apex Catalyst.
6.  Locked units (e.g., Shrine workers) cannot be auto-consumed in fusion.

GDD 08 — Inventory & Storage

1. Overview

Two storage systems: Character Storage (heroes/units) and Item Inventory (gear/materials/consumables). Both have starting caps, expandable via Crystals + milestone rewards. Adds: Lock & Favorite flags, bulk-select, filters, search, in-place comparison.

2. Player Fantasy

I'm collecting hundreds of units and dozens of weapons — and I can find anything in two clicks.

3. Detailed Rules

3.1 Character Storage

- **Default:** 100 slots
- **Max:** 1000 slots
- **Expand:** +10 per purchase
- **Each character instance = 1 slot** (duplicates count separately until consumed/fused)

Purchase #	+Slots	Crystal cost
1–5	+10	50
6–10	+10	75
11–20	+10	100
21–40	+10	150
41+	+10	200

Free storage milestones (account level ups): - Account L10: +20 character slots - Account L20: +30 - Account L30: +30 - Total free: 80 slots (so a free-track player without spending Crystals can grow to ~180 slots before paying).

3.2 Item Inventory

- **Default:** 200 slots
- **Max:** 2000 slots
- **Expand:** +20 per purchase
- **Stackable items:** 1 slot per unique type (e.g., 5,000 EXP Crystals = 1 slot)
- **Non-stackable:** weapons, accessories, eggs — 1 slot each

Purchase #	+Slots	Crystal cost
1–5	+20	50
6–10	+20	75
11–20	+20	100
21–40	+20	150
41+	+20	200

Free storage milestones (battle pass + account level): - Account L5: +20 inventory slots - Account L15: +40
 - Battle Pass tier 25 (every season): +40 - Total free per season: 100 slots.

3.3 Storage Behaviors

When character storage is full: -  Cannot hatch, claim from mail, gacha, or receive character rewards - 
 Can fuse, awaken, sell low-rarity units, expand, or send rewards to **mailbox** (held for 30 days)

When item inventory is full: -  Cannot loot dropped gear (drops are sent to mailbox) -  Can salvage, sell, equip, expand

Mailbox overflow: mail holds claimables for 30 days. After that, unclaimed items are deleted. The UI surfaces this clearly with a warning at 7 days remaining.

3.4 Lock / Favorite



Lock (per item or character): - Prevents accidental sell, salvage, or fusion-consume. - Toggleable in detail view. - Locked items show a  icon in lists.

Favorite (per item or character): - Sort-priority flag, does NOT block actions. - Shown with  icon. - “Show Favorites First” filter available.

A common pattern: Favorite all your “good but uncertain” items, then Lock only your “definitely keeping” items.

3.5 Filters / Search / Sort



Available across both storage screens:

Filters (multi-select): - Rarity (Common → Mythic) - Element (Fire / Water / Wind / Neutral) - Class (Tank / Assassin / Archer / Mage / Support) - Equipment type (Weapon / Helmet / Chest / Gloves / Boots / Ring / Necklace) - Equipment set - Equipped status (equipped / unequipped / on which hero) - Locked / Favorited

Sort (single-select): - Level (high → low) - Rarity (high → low) - Name (A-Z) - Acquisition date (recent → old) - Power (computed from stats × class weights)

Search (text): - Character / item name match - Set name match (e.g., “berserker”)

3.6 Bulk Actions



In Item Inventory, the **Bulk Select** mode allows: - Multi-select up to 100 items at once - Bulk sell → total Gold preview before confirm - Bulk salvage → total Gold + materials preview before confirm - Bulk lock / unlock / favorite

Locked items are auto-excluded from destructive bulk operations with a clear “X items locked, will not be affected” tooltip.

In Character Storage, Bulk Select supports: - Bulk-feed-for-promotion (auto-fills with the optimal fodder when you click Promote) - Bulk-lock / unlock / favorite

3.7 Comparison View



When tapping an item: - Side-by-side comparison with whatever the currently-selected hero has equipped in that slot. - Stat deltas shown in green (improvement) or red (downgrade). - Quick “Equip on Hero” button if the item is an improvement.

4. Formulas

Concept	Formula
Item power (sort)	$\sum \text{classWeight}[\text{stat}] \times \text{statValue} \times \text{rarityMod}$ (rarityMod: Common=1, Mythic=5)
Storage expansion cost	per 3.1 / 3.2 tables

5. Edge Cases

- **Storage full + mail full:** rewards are queued and surface as a critical notification. Items don't disappear — the queue blocks any further reward generation until cleared.
- **Sort by acquisition date** when items have identical timestamps: tie-break by item UUID (deterministic).
- **Equipped item attempted to bulk-sell:** auto-excluded from selection.
- **Filter combinations with zero matches:** show “No items match these filters” + Clear Filters button.

6. Dependencies

- `03-stats-and-progression.md` — character data structure
- `04-equipment.md` — sub-stats and sets shown in filters
- `05-gacha-and-currencies.md` — mail delivery for pulled heroes when storage full
- `13-battle-pass.md` — free storage from BP tier 25
- `14-monetization.md` — Crystal cost for expansion

7. Tuning Knobs

- Storage starting caps (100 / 200)
- Storage max caps (1000 / 2000)
- Expansion costs (3.1 / 3.2)
- Mail expiry (currently 30 days)
- Bulk-select max items (currently 100)
- Free storage milestones

8. Acceptance Criteria

- ✓ Character storage shows used / total slot count.
- ✓ Expanding storage with Crystals immediately reflects in the UI.
- ✓ Lock prevents bulk-sell from removing the item.
- ✓ Bulk-sell of 50 items preview total Gold within 1 Gold accuracy.
- ✓ Storage-full state correctly blocks gacha pulls and routes rewards to mailbox.
- ✓ Mailbox auto-deletes items 30 days after delivery (with 7-day-warning UI surface).
- ✓ Filter and search work on both storage screens.

GDD 09 — Campaign Structure

Highlighted (<mark>) blocks are new vs. the source PDF.

1. Overview

The campaign is the spine of all PvE progression. It defines the world (“acts” of the journey), the stages where idle farming happens, the bosses that gate progress, and the difficulty tiers that extend the campaign’s lifespan.

2. Player Fantasy

Each chapter introduces a new theme, a new enemy archetype, a stronger boss — and four difficulty tiers means once I’ve cleared it, there’s a Hard/Nightmare/Hell version waiting.

3. Detailed Rules

3.1 Structure



Layer	Count	Purpose
Acts	3 (launch)	Major narrative beats; thematic biome change
Chapters	4 per Act = 12 launch	Sub-region; each ends in a chapter boss
Stages	10 per Chapter = 120 stages	Wave-based combat encounters
Difficulty tiers	4 (Normal / Hard / Nightmare / Hell)	Apply across the whole campaign

Total stage permutations: 120 stages × 4 difficulties = **480 distinct stage instances**.

3.2 Acts (high-level themes)



Act	Theme	Element Bias	Boss archetype
Act 1 — Awakening	Forests, hillside ruins, early adventure	Wind dominant; mixed	Forest guardian, beast lord, fallen knight, wind drake
Act 2 — Ashes & Embers	Volcanic ranges, burning ruins	Fire dominant	Magma elemental, ember tyrant, phoenix scion, crimson dragon
Act 3 — Tides & Frost	Sea coast, ice mountains	Water dominant	Tidal beast, frost giant, abyssal kraken, leviathan

This element bias guarantees players need at least 2 elements to progress comfortably — preventing mono-element teams from steamrolling the whole game.

3.3 Stage Format

Each stage is a sequence of waves:

Stage type	Waves	Boss?
Standard stage (1-9)	3 waves of 1-5 enemies	No (chapter-mid boss optional on stages 5, 7)
Chapter boss stage (10)	2 trash waves + 1 boss wave	Yes

3.4 Difficulty Tiers



Difficulty	Enemy HP/ ATK	Element Penalty	Gold Reward	First-clear Bonus	Drop Quality
Normal	1.0×	-25%	1.0×	Standard	Common→Rare
Hard	1.5×	-25%	1.3×	+Skill Stones x20	Common→Epic
Nightmare	2.5×	-40%	1.7×	+Sacred Crystal x5 + Pristine Element Crystal	Common→Legendary
Hell	4.0×	-50%	2.5×	+Apex Catalyst (1, ONCE per chapter) + Cosmetic	Rare→Mythic (Mythic only from chapter bosses)

Difficulty unlock gates: - Hard: clear chapter 6 on Normal - Nightmare: clear chapter 9 on Hard - Hell: clear chapter 12 on Nightmare

3.5 First-clear vs. Repeat-clear Rewards



First-clear rewards (per stage, per difficulty): - Standard: Gems (10-100 depending on stage depth), 1 Common/Uncommon egg, 1 piece of gear at stage tier - Bonus 3-star rewards (see 3.6) - First time clearing a chapter boss also grants 50-500 Gems + difficulty-tier specific bonuses (see 3.4)

Repeat-clear rewards: - Standard gold table for the stage - Standard drop table (gear) - Sometimes EXP Crystals, materials - NO Gems, NO eggs, NO milestone rewards

This sharp distinction prevents players from feeling like they need to grind early stages forever.

3.6 3-Star Rating System (NEW)



Each stage clear is rated 0-3 stars based on three independent conditions:

1. **Clear the stage** (★1)
2. **All heroes survived** (no KO during stage) (★2)
3. **Cleared in under N turns** (varies by stage; typically 8-15 turns) (★3)

A 3-star clear awards bonus rewards. 3-starring all stages in a chapter awards a chapter completion bonus (10 Gems, 1 piece of Sacred Stone, profile badge).

Recorded as best-ever: the player can't "lose" stars by re-running. Best is always shown. UI marker on the stage select map.

3-star clears do not need to happen on the same difficulty — 3-starring a Nightmare run satisfies the chapter completion bonus for ALL difficulties of that stage (gracefully).

3.7 Boss Design



Each chapter boss has: - ~5-10× HP of a standard enemy - 2-4 unique skills (vs. a normal enemy's 1-2) - A signature mechanic (e.g. Chapter 3 boss “Wind Drake” applies AoE Speed Down every 4 turns; Chapter 7 boss “Phoenix Scion” revives once at 50% HP) - An **enrage timer** (see 01-combat.md § 3.11) - An **element match** (sometimes resistant to certain elements) - **Scout-eligible** (see 01-combat.md § 3.16)

Drops: - Difficulty-tier specific gear (Mythic drops only from Hell-mode chapter bosses) - Promotion materials (Sacred Stones, Element Crystals at appropriate difficulty) - First-clear Gems + eggs

3.8 Mid-week Event Stage (NEW)



Every Wednesday, one of the campaign chapters becomes the “Featured Chapter of the Week” with: - +50% Gold drop - +25% gear drop chance - 2× Skill Stone drops in that chapter - A rotating mini-quest (“Clear 5 stages in featured chapter”) with Gem reward

The featured chapter cycles deterministically (week 1 = chapter 1, week 2 = chapter 2, ...). This re-engages players with old content and keeps farming targets fresh.

3.9 Drop Table Discipline

Each stage has an explicit drop table (data-driven). Tables follow consistent principles: - Higher stage = higher chance of higher-rarity drops - Boss stages drop the “set-specific” items (e.g., chapter 5 boss only drops Berserker set on Hard+) - No stage drops every set — players must visit specific stages for specific sets

4. Formulas

Concept	Formula
Enemy HP at stage S, difficulty D	$\text{BaseHP}[S] \times \text{DifficultyMultiplier}[D]$
Enemy ATK at stage S, difficulty D	$\text{BaseATK}[S] \times \text{DifficultyMultiplier}[D]$
Repeat-clear gold	$\text{BaseGold}[S] \times \text{DifficultyMultiplier}[D].\text{goldBonus} \times \text{idleClearRate}[S]$
3-star turn budget	$8 + \text{floor}(S / 20) + 1 \times \text{difficultyTierIndex}$
Featured chapter cycle index	$((\text{weekNumberSinceLaunch}) \% 12) + 1$

5. Edge Cases

- **First-clear bonus already claimed:** subsequent runs reward only the repeat-clear table.
- **3-star turn budget edge:** turn N exactly = qualifies (use `<=`).
- **Hero death late in the stage:** ★2 is lost forever for that run. To re-earn, must clear without any KO.
- **Featured chapter overlap with player's current chapter:** no special rule — bonus stacks.
- **Skipping difficulty tiers:** not allowed. Hard unlocks only after Chapter 6 Normal clear.

6. Dependencies

- `01-combat.md` — uses difficulty modifiers in damage/HP/Speed
- `02-elements-and-classes.md` — element bias per Act
- `03-stats-and-progression.md` — Element Crystals, Sacred Stones drop sources
- `04-equipment.md` — gear drop tables, set distribution
- `06-idle-and-shrine.md` — idle stage selection includes campaign stages
- `13-battle-pass.md` — chapter-completion missions feed BP

7. Tuning Knobs

- Difficulty multipliers (3.4)
- First-clear vs repeat rewards (3.5)
- 3-star turn budgets (4)

- Featured chapter rotation period
- Boss enrage thresholds (inherited from `01-combat.md`)
- Drop table per stage (data-driven, hot-config)

8. Acceptance Criteria

1. ✓ 120 stages × 4 difficulties = 480 distinct stage instances are defined.
2. ✓ Difficulty unlock gates work (Hard requires Ch.6 Normal clear, etc.).
3. ✓ 3-star rating per stage is recorded and never decreases.
4. ✓ First-clear rewards are granted exactly once; repeat-clear table works on subsequent runs.
5. ✓ Chapter boss enrage applies per `01-combat.md` § 3.11.
6. ✓ Featured chapter rotates weekly, deterministic from launch week.
7. ✓ Mythic drops are restricted to Hell-mode chapter bosses.
8. ✓ Element bias per act matches the documented theme.

GDD 10 — Tower of Trials

Entire GDD is NEW. All content highlighted.

1. Overview

The **Tower of Trials** is a 100-floor endgame mode that resets monthly. Each floor is a fixed encounter that escalates in difficulty. Players use any 3 heroes from their roster (heroes lock between floors — can't reuse). Floors 1-20 are easy entry; 21-50 require build-craft; 51-80 require optimization; 81-100 are flex / wall-pushing endgame.

2. Player Fantasy

I built a deep roster for a reason. The Tower keeps asking me to bring out heroes I never use, climb a few more floors than last month, and prove the depth I've earned.

3. Detailed Rules

3.1 Structure



Floor band	Difficulty	Reward focus
1-10	Tutorial — auto-clearable	Skill Stones, Gold, Tower Coin
11-25	Mid-build challenge	Sacred Stones, Tower Coin, gear
26-50	Optimization wall	Epic gear, Sacred Crystal, Pristine Element Crystal, Tower Coin
51-80	Advanced strategy	Legendary gear (forced sub-stat at floor 60), Reroll Stones, Apex Catalyst (1 at floor 70)
81-100	Endgame wall	Mythic gear, profile badges, monthly leaderboard ranking

3.2 Hero Lock Rule



Each hero used to clear a floor cannot be used on the next floor.

Exception: every 10th floor (10, 20, 30, ...) is a “Rest Floor” — all heroes unlock and become usable again.

This forces depth-of-roster play. A player with only 3 strong heroes will struggle past floor 10. A player with 15 well-built heroes can sprint to floor 50+.

3.3 Monthly Reset



- Tower resets on the **1st of each month at 00:00 UTC**.
- Player’s highest cleared floor is preserved as **personal best** (leaderboard cache).
- Per-floor rewards reset (so a player can re-climb the rewards).
- Boss floor (every 10th) cosmetic / one-time rewards are also re-earnable each month.

3.4 Floor Modifiers



Some floors apply **active modifiers** that change combat rules:

Modifier	Effect	Example floor
Burning Ground	All combatants take 5% maxHP burn per turn	Floor 15
Reverse Element	Element advantage becomes disadvantage (and vice versa)	Floor 28
One-Hero Force	Only one hero in the party can act per turn (cycle)	Floor 35
Ultimate Lockout	Ultimates disabled	Floor 47
Crit-Only	Only critical hits deal damage; non-crits deal 1	Floor 55
Double Speed	All combatants act every turn (no gauge)	Floor 64
Counter Field	All single-target attacks have 50% counter chance	Floor 73
Boss Phase Shift	Boss has 3 HP phases with different element each	Floor 80, 90, 100

Mods are revealed in the floor scout panel before committing.

3.5 Tower Coin (currency)



Awarded for clearing floors. Sources: - Floor first-clear: 10-150 Tower Coin (scales) - Monthly milestone (every 10 floors): bonus Tower Coin - Top-10% leaderboard (post-monthly close): bonus Tower Coin

Tower Shop (Tower Coin-only):

Item	Cost	Limit (per month)
Sacred Crystal	50	5
Apex Catalyst	2,000	1 (1 per Tower season = 1 per monthly reset)
Reroll Stone	100	10
Pristine Element Crystal (chosen element)	200	5
Mythic gear ticket	500	1
Tower exclusive cosmetic	1,000	1
Skill Stones x100	50	5
10 × Hero Shards (chosen Standard hero)	600	3

3.6 Leaderboard



- Visible to player: their personal rank within their server / region.
- No PvP combat — leaderboard is purely “highest floor cleared.”
- Top 100 ranks earn end-of-month bonus rewards.
- Friends ranking visible if friends list exists (deferred to post-launch).
- Anti-cheat: max-floor claim is validated against server-side combat log (see `21-liveops-and-telemetry.md`).

3.7 Tower UI



- Vertical-scrolling 100-floor map (visually evocative — a literal tower silhouette)
- Current floor highlighted, prior cleared floors shown with checkmarks
- Heroes “used this run” panel — shows which heroes are locked for the next floor
- Rest Floor markers (10, 20, ...) clearly visible
- Modifier preview on each floor
- “Reset run” button (resets all hero locks, doesn’t re-clear floors)

4. Formulas

Concept	Formula
Floor HP/ATK	$100 \times (1 + 0.15 \times \text{floorNumber})$ baseline; modifiers apply on top
Tower Coin per floor	$10 + 2 \times \text{floorNumber}$
Monthly milestone (10-floor) bonus	$\text{floor} / 2 + 50$ Tower Coin
Top-10% bonus	$100 \times (\text{rank percentile factor})$

5. Edge Cases

- **All heroes used and no Rest Floor reached:** the player cannot proceed; must claim “Reset run” which un-locks heroes but doesn’t unwind cleared floors.
- **Mid-floor crash / disconnect:** floor state is saved at start; current floor restarts on relaunch.
- **Monthly reset during active run:** player’s progress carries forward; rewards already claimed don’t re-grant.
- **Hero deleted mid-tower (via sale or fusion):** they are removed from the run team if used. If used on a current locked floor, the lock entry persists (treated as locked but unusable).
- **Modifier conflicts with hero passive:** modifier always wins (e.g., Ultimate Lockout disables even Akari’s Phoenix Festival).

6. Dependencies

- `01-combat.md` — combat engine
- `02-elements-and-classes.md` — element matchups
- `03-stats-and-progression.md` — hero stat baselines
- `09-campaign-structure.md` — drops align with campaign tier; Apex Catalyst overlap
- `17-save-system.md` — monthly reset, leaderboard persistence
- `21-liveops-and-telemetry.md` — anti-cheat validation

7. Tuning Knobs

- Floor HP/ATK scaling coefficient (currently 0.15)
- Tower Coin reward formula
- Modifier assignment per floor

- Tower Shop costs and weekly limits
- Hero lock rule (Rest Floor cadence)

8. Acceptance Criteria

1.  100 floors are populated with mod and difficulty at launch.
2.  Hero lock rule enforces no-reuse between floors except at Rest Floors.
3.  Monthly reset clears floor rewards but preserves leaderboard.
4.  Tower Coin balance and shop limits work per the table.
5.  Modifier effects apply correctly (testable on Floor 15 Burning Ground, etc.).
6.  Leaderboard top-10% rewards distribute on monthly close.

GDD 11 — World Map & Expedition

Entire GDD is NEW. All content highlighted.

1. Overview

World Map / Expedition is a passive-dispatch system. The player sends parties of low-rarity or off-team heroes on **timed missions** (1h / 4h / 8h / 12h) to gather materials, gold, EXP, gear, and rare drops. Up to 4 simultaneous expeditions. No combat — pure passive timer with reward roll on completion.

2. Player Fantasy

The heroes I'm not actively using still earn me real rewards. Even my 1★ Slime is a working member of the team.

3. Detailed Rules

3.1 Structure



- **Expedition Slots:** start with 1, expand to 4 via account level / BP rewards.
- **Mission Types:** 1h / 4h / 8h / 12h — each tier with several mission variations.
- **Crew Requirements:** each mission specifies hero count (1-4) and any class/element requirements.

3.2 Mission Examples



Mission	Duration	Crew	Rewards
Forest Scouting	1h	1 hero	200 Gold, 50 Small EXP Crystals
Sea Patrol	1h	1 Water hero	200 Gold, 50 Small EXP Crystals, 1 chance Water Crystal
Ember Run	4h	2 heroes incl. 1 Fire	1,500 Gold, 1 Common Egg, Lesser Fire Crystal
Volcano Tomb	8h	3 heroes	4,000 Gold, 1 Epic gear roll, 2 Sacred Stone
Boss Site Recon	8h	1 hero L40+	4,000 Gold, 50 Gems, possible Reroll Stone
Lost Sanctum	12h	4 heroes	8,000 Gold, 2 Epic gear roll, Pristine Element Crystal (random), 1 chance for Hero Shard pack

3.3 Mission Rotation



A pool of ~30 missions exists. **3 are active at any given time**, refreshing every 6 hours. Players who time their logins around refreshes can chain missions efficiently.

Each refresh cycle picks 3 missions weighted by: - Player chapter progress (later missions unlock at chapter milestones) - Time-tier variety (always at least one short, one medium, one long mission available) - Element variety (rotates which elements are required across the cycle)

3.4 Crew Selection



The player selects heroes to dispatch. Constraints:

- Heroes used on an expedition **cannot be used in active combat** until the mission completes.
- Heroes on the **idle target stage team** ARE allowed to be expedition crew (idle team is separate).
- Heroes used as **Shrine workers** are excluded from expedition (must be unassigned first).
- Crew must meet **mission requirements** (count, class, element).

Crew bonuses: - Higher-rarity crew = +X% reward chance (e.g., 5★ crew member = +25% Gold) - Element match (crew element matches mission “element required”) = +20% main-loot chance - All crew members have the same element = +30% Element Crystal chance

3.5 Quick-Send Logic



Players who don't want to manage crew can tap **Quick-Send** to auto-assign: - Selects appropriate heroes by class + element + level fitness - Avoids using heroes currently in the main battle team - Confirm preview shown before commit

3.6 Completion

- Mission timer counts down even while game is closed.
- Notification on completion (per setting).
- Player must manually **claim** rewards (no auto-claim). Crew is freed at claim time.
- Reward roll uses a deterministic seed (testable, anti-cheat) per `gdd/17-save-system.md`.

3.7 Expedition vs. Idle Combat

Aspect	Idle Combat	Expedition
Active heroes	3 hero team	1-4 hero crew
Outputs	Stage drops, EXP, gold, gear	Specific mission rewards
Active simulation?	Yes (combat ticks)	No (just timer)
Hero overlap allowed?	No (combat team and expedition crew can't overlap)	—
Best for	Gold, gear, EXP grind	Specific materials, eggs, Gems

The two systems complement each other — players running an active combat team can also send 4 expeditions on idle heroes simultaneously.

4. Formulas

Concept	Formula
Crew bonus (rarity)	+5% per crew star tier above 1★, capped at +25%
Element-match bonus	flat +20% main reward chance
All-element bonus	flat +30% Element Crystal chance
Mission refresh time	every 6h since launch UTC

5. Edge Cases

- **Crew hero deleted mid-mission:** mission completes normally, but rewards are reduced by 25% per missing crew member.
- **All slots full + mission expires:** rewards are mailed to the player (claim within 30 days).
- **Cross-mission element bonus stacking:** an Element Match bonus applies per-mission only; the All-Element bonus stacks with element-match.
- **Quick-Send picks a hero already on idle combat team:** Quick-Send respects this — skips that hero, picks next-fit.
- **Mission requirements not met:** dispatch button disabled with tooltip explaining which requirement is unmet.

6. Dependencies

- `03-stats-and-progression.md` — hero rarity, EXP rewards
- `06-idle-and-shrine.md` — Shrine workers exclusion, idle team
- `07-eggs-and-fusion.md` — eggs as reward
- `13-battle-pass.md` — expedition slot expansion from BP
- `17-save-system.md` — deterministic mission completion roll

7. Tuning Knobs

- Mission pool size (currently ~30)
- Refresh interval (currently 6h)
- Mission duration tiers (1/4/8/12h)
- Reward magnitudes per mission tier
- Crew bonus magnitudes (4)
- Number of expedition slots (1 → 4)

8. Acceptance Criteria

- ✓ 4 expedition slots are unlockable via account progression.
- ✓ Missions refresh every 6 hours.
- ✓ Crew bonuses apply correctly to reward rolls.
- ✓ Quick-Send finds valid crew without overlapping active battle team.
- ✓ Reward roll uses deterministic seed.

6.  Hero locked on expedition cannot be used in active combat until claim.

GDD 12 — Daily, Weekly, Hero Trials, Boss Rush, Achievements

Entire GDD is NEW. All content highlighted.

1. Overview

The daily, weekly, and special endgame content loops keep the game **alive after the initial campaign clear**.
Five sub-systems:

- **Daily Missions** (reset 04:00 local)
- **Weekly Missions** (reset Monday 04:00)
- **Login Bonus** (7-day rotating)
- **Hero Trials** (weekly single-hero showcase stages)
- **Boss Rush** (weekly chapter-bosses gauntlet)
- **Achievements** (permanent, one-time)

2. Player Fantasy

Every day has a checklist. Every week has a goal. Every level-up unlocks something to grind. The game never feels empty.

3. Detailed Rules

3.1 Daily Missions (NEW)



8 daily missions, with point totals. Hitting threshold milestones grants escalating rewards.

Mission	Points
Clear 5 stages	100
Spend 1,000 Gold	50
Hatch 1 egg	50
Collect Shrine 1×	50
Dispatch 1 expedition	50
Perform 1 gacha pull (any banner)	100
Win a battle in <8 turns	100
Upgrade 1 skill	75

Total per day: 575 points possible.

Thresholds:

Points	Reward
200	50 Gems
400	100 Gems + 1 Common Egg
575 (all complete)	200 Gems + 1 Rare Egg + 1 Reroll Stone

3.2 Weekly Missions (NEW)



6 weekly missions:

Mission	Reward
Clear 50 stages (any difficulty)	200 Gems
Clear 20 Hard+ stages	400 Gems + 1 Sacred Stone
Defeat 5 chapter bosses	1 Pristine Element Crystal
Complete 5 expeditions	2 Sacred Stones + 100 Skill Stones
Reach Tower floor 25+	1 Apex Catalyst (only once per month; subsequent weekly clears give 50 Tower Coin)
Hatch 5 eggs	1 Epic Egg

3.3 7-Day Login Bonus (NEW)



Day	Reward
1	100 Gems
2	1 Common Egg + 200 Gold
3	5 Skill Stones
4	1 Standard Summon Ticket
5	1 Rare Egg
6	10 Reroll Stones
7	1 free Standard 5★ Hero pick from the Curated 6 (see below)

The Day 7 reward exists ONCE per account; subsequent login streaks award a 7-day “Veteran” cycle that’s smaller (50 Gems × 7) and resets every 7 days.



Curated 6 — Free-pick hero pool

The Day 7 login bonus AND the Battle Pass Tier 50 free track reward share **the same Curated 6 list**:

#	Hero	Element	Class	Why
1	Valdrik	 Fire	Tank	Frontline anchor — pairs with Lyria's fragile back-line
2	Glacier	 Water	Tank	Alternative Tank — Frost ecosystem starter
3	Sylphie	 Wind	Support	Premier healer — same-element synergy with Lyria
4	Marina	 Water	Support	Alternative Support — sustain + cleanse
5	Emberia	 Fire	Mage	AoE wave-clear partner for single-target Lyria
6	Kael	 Wind	Assassin	Wind-team boss-killer pairing with Lyria

Lyria is excluded — every player already owns her from the tutorial.

Kaida and Neria are excluded — they overlap roles with Kael (assassin) and Emberia (mage). Players who want them can still pull on Standard banner.

This list is curated for **starter team variety** — picking any of the 6 gives the player a complete 3-hero core alongside Lyria.

3.4 Hero Trials (NEW endgame mode)



A **weekly rotating stage** built around a single hero. Each week, one of the available launch heroes is “featured” — the player can only bring that hero (plus optionally 1-2 partner heroes per trial design).

At launch the rotation cycles through the 10 available heroes (9 Standard + Auron). As future Limited Banner heroes release, they automatically enter the rotation.

Each trial has 3 difficulty tiers (Bronze / Silver / Gold). Completing all three is the “Mastery” reward.

Tier	Difficulty	Reward
Bronze	Easy — feature hero L40+ recommended	50 Skill Stones, 1 Reroll Stone, 100 Gems
Silver	Medium — feature hero must be 4★+	100 Skill Stones, 2 Reroll Stones, 200 Gems
Gold	Hard — feature hero must be 5★+	1 Pristine Element Crystal (matching hero’s element), 5 Hero Shards of the feature hero, 500 Gems

Rotation: week N’s featured hero = `availableHeroIndex[(N - 1) % availableHeroCount]`. Predictable rotation so players know when to “play their favorites.” `availableHeroCount` = 10 at launch, grows as Limited Banner heroes release.

Encourages: building a deep roster (eventually every hero gets attention), and earning Hero Shards of specific heroes you may not own.

3.5 Boss Rush (NEW endgame mode)



A weekly mode where the player fights all 12 chapter bosses **in sequence** with a single party of 3 heroes. No team swaps. Cooldowns persist across boss fights (mostly — they reset between bosses).

Bosses Cleared	Reward
4 (first quarter)	1 Sacred Stone, 200 Gems
8 (mid)	1 Sacred Crystal, 1 Pristine Element Crystal (random), 1 Reroll Stone
12 (full clear — Normal/Hard)	500 Gems, 1 Mythic gear ticket, profile badge
12 (full clear — Nightmare , monthly first-time only)	1 Apex Catalyst + 500 Gems + 1 Mythic gear ticket + profile badge

Difficulty tiers: Normal / Hard / Nightmare (Hell unlocks post-launch). Each tier is its own clear; rewards stack.

Boss Rush resets every Monday with the weekly missions.

3.6 Achievements (NEW)



Permanent one-time goals. Categories:

Category	Examples	Rewards
Collection	“Own 5 unique heroes”, “Own all currently-released 5★ heroes”, “Awaken 1 hero to max”, “Own all 15 4★ heroes”	Gems, Hero Shards
Combat	“Deal 1M damage in a single hit”, “Survive 50 turns in one battle”, “Crit 100x”	Gems, eggs
Progression	“Reach account L20”, “Clear chapter 12 Hell”, “3-star all stages chapter 1”	Storage expansion, Gems
Tower	“Reach floor 50”, “Reach floor 100”, “Top 100 in monthly leaderboard”	Cosmetics, Tower Coin
Boss Rush	“Full clear on Normal”, “Full clear on Nightmare”	Gems, badge
Gacha	“1,000 lifetime pulls”, “Pull a Legendary on a 10-pull batch”	Gems, dupe shop voucher
Exploration / Codex	“Defeat each enemy type once”, “Discover all chapters”	Gems, profile titles

Estimated ~150 achievements at launch. ~25 are “secret” (hidden until triggered).

3.7 Codex / Hero Gallery / Bestiary (NEW)



Three browse-only collections:

- **Hero Gallery:** all 15 5★ heroes (incl. ones you don’t own — locked with silhouette). Future-banner heroes show as “Coming Soon” with a date estimate. 4★ heroes also displayed (when released). Tapping shows kit, splash, element, class. Used for wishlist research.
- **Bestiary:** every enemy encountered in campaign + Tower. Records kill count and drop rates observed.
- **Item Codex:** all gear types, sets, and their bonuses.

Codex completion is achievement-tied (see 3.6).

4. Formulas

Concept	Formula
Daily mission progress	additive points
Weekly mission reset	Monday 04:00 local timezone
Daily mission reset	04:00 local timezone
Hero Trial week index	$\text{floor}((\text{daysSinceLaunch} + 0) / 7) \% \text{availableHeroCount}$ (10 at launch, grows over time)
Boss Rush reset	Monday 04:00 local
Login bonus pre-claim grace	last claim $\geq$ 4h ago in the same day

5. Edge Cases

- **Daylight savings:** mission resets use UTC offset captured at first launch + DST adjustments.
- **Server time vs. local time:** local time used for resets (intentional, allows players to claim “their morning” instead of arbitrary UTC).
- **Login streak missed:** if a player misses a day, streak resets to Day 1 (no penalty, just no Day 7 5★ pick that week — they re-earn it after another 7).
- **Hero Trial featured hero is locked:** if the player doesn’t own them, Hero Shards can be earned via Bronze/Silver tiers using the 1-2 partner allowed slots. Gold tier locked.
- **Boss Rush mid-clear: party wipe:** the run ends; rewards earned so far are kept. Re-attempt allowed same week with a fresh team.

6. Dependencies

- `01-combat.md` — battle execution
- `09-campaign-structure.md` — Boss Rush uses chapter bosses
- `10-tower-of-trials.md` — Achievements integrate with Tower milestones
- `13-battle-pass.md` — BP progress earned from daily/weekly missions
- `17-save-system.md` — reset scheduling, achievement state
- `20-ui-menus-tabs.md` — Achievement screen, Codex screen

7. Tuning Knobs

- Daily / weekly point thresholds (3.1, 3.2)
- Login bonus content (3.3)
- Hero Trial rotation order (3.4)
- Boss Rush milestone rewards (3.5)
- Achievement count and reward sizes (3.6)

8. Acceptance Criteria

- ✓ 8 daily missions tracked, points awarded correctly.
- ✓ Daily mission rewards distribute at 200/400/575 points.
- ✓ Weekly missions reset Monday 04:00 local.
- ✓ Day-7 login bonus delivers a free Standard 5★ hero pick once per account.
- ✓ Hero Trial rotates deterministically each week through all currently-released 5★ heroes (10 at launch, growing as banner heroes release).
- ✓ Boss Rush 12-clear awards Apex Catalyst + Mythic ticket.
- ✓ ~150 achievements at launch, including hidden ones.
- ✓ Codex / Bestiary tracks completion and unlocks rewards.

GDD 13 — Battle Pass

Entire GDD is NEW. All content highlighted.

1. Overview

The Battle Pass is a 50-tier seasonal progression track. **Free Track** for everyone, **Premium Track** unlocked by paying (\$9.99), **Premium-Plus** for whales (\$19.99 — includes +10 tier skips + exclusive cosmetics). 60-day season. Tiers earned via campaign play, dailies, weeklies, gacha pulls, achievements.

2. Player Fantasy

Every battle, every pull, every dailies completion makes my Battle Pass meter creep up. By season's end, I've earned a hero I'd otherwise have to pull for — and I never felt forced.

3. Detailed Rules

3.1 Season Cycle



Element	Value
Season length	60 days
Total tiers	50
Free Track	Always available
Premium Track	\$9.99 unlock
Premium-Plus	\$19.99 (+10 tier skips + exclusive title/avatar)
Mid-season tier purchase	\$0.99 per tier (caps at 5 purchases per season)
End-of-season grace	7 days to claim missed rewards

3.2 BP XP Sources

Source	XP
Daily mission completion (per category)	50 each
Daily 575-point milestone	+200
Weekly mission complete	500 each
Chapter boss first-clear	1,000
Tower floor 25 clear	500
Tower floor 50 clear	1,000
Hero Trial Gold clear	750
Boss Rush full clear	1,500
Achievement unlock	250 each
Gacha pull	5 XP per pull

Tier XP cost: flat **1,500 XP per tier** (so 50 tiers × 1,500 = **75,000 XP per season**). Tuned so committed daily players hit tier 50 around day 45-55 of a 60-day season — leaves comfortable margin without trivializing the climb. Casual players reach ~tier 35-40, still getting most of the Free Track value.

3.3 Free Track Rewards (sample)

Tier	Reward
1	100 Gems
5	1 Common Egg
10	1 Standard Summon Ticket
15	50 Skill Stones
20	1 Sacred Stone
25	40 Free Inventory Slots
30	1 Standard Summon Ticket
35	100 Gems
40	1 Rare Egg
45	200 Gems
50	1 free Standard 5★ Hero pick from the Curated 6 (see gdd/12-daily-weekly-loops.md § 3.3 — same list as Day 7 login bonus)

3.4 Premium Track Rewards (sample, in addition to free track)

Tier	Reward
1	200 Crystals + Premium Badge
5	1 Limited Summon Ticket
10	1 Pristine Element Crystal
15	1 Reroll Stone × 5
20	200 Crystals
25	1 Apex Catalyst
30	1 Limited Summon Ticket
35	1 Mythic gear ticket
40	BP-exclusive Cosmetic Skin (TBD) — applies to one Standard hero per season. Season 1 hero / design is TBD. Returns as a paid-shop cosmetic 1 season later for players who missed it.
45	5 Limited Summon Tickets
50	1 BP-exclusive cosmetic (Avatar + Animated Frame) + 500 Crystals

Premium Track recoup: ~\$9.99 of value delivered in Crystals + Limited Tickets, plus the BP-exclusive hero and cosmetic. Strong value proposition.

3.5 Premium-Plus Bonuses



- Unlocks Premium Track + grants +10 instant tier skips
- Exclusive title displayed in profile + chat (post-launch)
- Exclusive animated avatar frame
- Exclusive emote pack
- One-time bonus: 1 free Pristine Element Crystal at season start + 500 Crystals

3.6 BP-Exclusive Skin (replaces “BP Hero” design)



Each season’s BP includes a unique **BP-exclusive Cosmetic Skin** at Premium Tier 40. This skin: - Applies to one specific Standard hero (which hero rotates per season — Season 1 TBD) - Is purely cosmetic — no stat

impact - Available ONLY through that season's Premium BP at launch - Returns as a paid-shop cosmetic ONE season later for players who missed it (so the BP holder retains exclusivity for ~60 days)

This pattern (cosmetic-only BP exclusive) is safer than a BP-exclusive hero — F2P players are not locked out of meta-power, and the cosmetic feels valuable. Worse player perception risk of “pay to win” → zero.

Season 1 skin: TBD — to be designed before launch.

3.7 BP UI



- Horizontal scrolling row showing all 50 tiers
- Current tier highlighted; XP progress bar to next tier
- Toggle: Free Track / Premium Track / Premium-Plus
- Lock icons on Premium rewards if not purchased
- “Buy Premium” CTA always visible
- Tier-skip buy button per tier
- “Claim all” button if multiple rewards uncollected

4. Formulas

Concept	Formula
BP XP needed per tier	flat 1,500 (75,000 total per 50-tier season)
Tier skip cost	\$0.99 per tier, max 5 purchases
End-of-season grace	7 days
BP-exclusive hero re-release	next season's Standard pool addition

5. Edge Cases

- **Mid-season Premium upgrade:** player who buys Premium at tier 30 retroactively unlocks all tier 1-30 Premium rewards.
- **Premium purchased on last day:** all unlockable rewards delivered to mailbox immediately for claim within grace period.
- **Tier skip purchase when at tier 50:** rejected with refund (or simply disabled in UI).
- **BP XP earned past tier 50:** capped (no carryover into next season).
- **BP hero already owned via re-release:** no duplicate; conversion to Hero Essence + 5 Hero Shards (per `gdd/05-gacha-and-currencies.md`).

6. Dependencies

- `05-gacha-and-currencies.md` — Crystals, Limited Tickets, BP-exclusive hero
- `12-daily-weekly-loops.md` — XP sources from missions
- `14-monetization.md` — pricing
- `17-save-system.md` — BP state persistence across seasons
- `20-ui-menus-tabs.md` — BP screen

7. Tuning Knobs

- Season length (60 days)
- Tier count (50)
- Premium price (\$9.99)
- Premium-Plus price (\$19.99)
- Tier-skip price (\$0.99, max 5)
- XP per source / XP per tier
- Reward content per tier

8. Acceptance Criteria

1.  50-tier BP completes in ~45-55 days of daily play (75% engagement) within the 60-day season.
2.  Premium track rewards backfill correctly when purchased mid-season.
3.  Free-track Tier 50 grants 1 Standard 5★ choice.
4.  BP-exclusive skin is available ONLY via Premium Tier 40.
5.  Following season makes the prior BP skin available in the paid cosmetics shop.
6.  Grace period of 7 days post-season allows reward claim.

GDD 14 — Monetization

Entire GDD is NEW. All content highlighted.

1. Overview

Standard gacha monetization. F2P-friendly progression with paid skip / convenience / chase mechanics. **All monetization is optional** — no progression is gated permanently behind paywall.

2. Player Fantasy

I can earn everything for free if I play enough. But \$5 a month makes the grind kinder, and one-time \$20-\$30 packs let me chase a hero I really want without going broke.

3. Detailed Rules

3.1 Pricing Tiers



Item	Price	Description
Pebble Crystal Pack	\$0.99	100 Crystals
Small Crystal Pack	\$4.99	550 Crystals (10% bonus)
Medium Crystal Pack	\$9.99	1,150 Crystals (15% bonus)
Large Crystal Pack	\$19.99	2,400 Crystals (20% bonus)
Mega Crystal Pack	\$49.99	6,300 Crystals (25% bonus)
Ultra Crystal Pack	\$99.99	13,500 Crystals (35% bonus)

3.2 Monthly Card



Price: \$4.99 / month

Day-of purchase: 600 Crystals immediately

Daily for 30 days: 100 Crystals daily login bonus (3,000 total over 30 days = 3,600 Crystals total value).

Plus extras: - +1 expedition slot active for 30 days - +2 daily summon ticket purchases at half-price - Monthly badge

Player perception: \$5/month for ~\$25 of value over the month. Very high LTV stickiness.

3.3 Beginner Pack



Available: account $\leq$ 14 days old. One-time purchase.

Price: \$4.99

Contents: - 600 Crystals - 5 Limited Banner Tickets - 1 Standard 5★ select pick (from curated 6-hero list) - 1 BP Premium Track activation (current season only) - 5 Reroll Stones

Value: ~\$40 of in-game content for \$4.99. Strong conversion lever for new players.

3.4 Limited Banner Bundles



For each Limited Banner, a featured bundle is available:

Featured Hero Starter Bundle (\$14.99): - 800 Crystals - 10 Limited Banner Tickets - 50 Skill Stones - 1 Reroll Stone

Featured Hero Power Bundle (\$29.99): - 1,800 Crystals - 30 Limited Banner Tickets - 1 Apex Catalyst - 10 Reroll Stones - 1 Pristine Element Crystal

3.5 Dupe Shop

Already documented in `gdd/05-gacha-and-currencies.md` § 3.9.

3.6 Storage Expansion (recap)

Crystals-based per `gdd/08-inventory-and-storage.md` . NOT a primary monetization driver.

3.7 Cosmetics (no-pay-to-win)



Cosmetics are PURE vanity — no stat impact:

Type	Source	Description
Avatar frames	BP, Tower, Achievements, paid bundles	Around player avatar
Profile titles	Achievements, BP, events	Display in profile
Hero costumes (post-launch)	Premium shop, events	Alternate hero skins; cosmetic only
Compact window themes	Premium shop (\$1.99 each)	Different visual frames for the desktop window
Animated emotes	Events, BP, Dupe Shop	UI flair

3.8 ARPU Targets (industry-standard reference)



Metric	Target
D1 conversion rate (any spend)	2-3%
D7 conversion rate	4-5%
D30 conversion rate	6-8%
Average ARPU (lifetime, paying)	\$30-60
Average ARPU (whale, top 5%)	\$300+
Median ARPU (all paying)	\$15-25
% of revenue from BP + Monthly	~40% (steady, predictable revenue)
% from Limited Banner bundles	~35% (spiky, event-driven)
% from Crystal packs	~20%
% from cosmetics + misc	~5%

Targets based on Epic Seven / AFK Arena benchmarks. Project I's PC-only audience may skew older / higher ARPU.

3.9 Anti-Predatory Principles



Project I will NOT use: - **Energy / stamina gates** — no waiting walls - **Power creep treadmills** — old heroes remain viable - **Misleading bundle math** — actual rewards always exceed sticker price - **Sub-100% guaranteed pity** — pity is real - **FOMO-driven daily timers on monetization** — only on content (events, BP) - **Loot boxes with cash purchase** — gacha is gem-currency, not direct cash - **Mystery boxes** — no “mystery” purchases - **Pay-to-skip gear grinds** — sub-stat rerolls cost RNG-currency, not cash

These constraints define the “fair gacha” brand for Project I.

3.10 Refund Policy



- **Steam:** standard 2h / 14-day Steam refund policy applies.
- **In-game currency / gacha pulls:** non-refundable once consumed.
- **Cosmetics:** non-refundable.
- **Battle Pass:** refundable within 24h of purchase IF no premium rewards have been claimed.

Customer support response time target: 48h for billing issues, 7 days for non-billing.

4. Formulas

Concept	Formula
Crystal-to-Gem conversion	1 Crystal = 1 Gem (one-way)
Daily Monthly Card payout	100 Crystals × 30 days
Bundle “value” calculation	Sum of in-game-equivalent prices ÷ bundle price
BP daily milestone XP	per <code>gdd/13-battle-pass.md</code>

5. Edge Cases

- **Crystals purchased + then game shuts down:** Steam handles refund; in-game Crystals are restored if refund issued.
- **Monthly Card 30-day countdown:** counts real-world days, not active play days. If player doesn’t login for 5 days, they lose 5 of 30 daily Crystals.
- **Limited Banner bundles after banner ends:** the Limited Tickets persist but cannot be used (no active Limited banner). They remain in inventory for the next Limited banner.
- **Beginner Pack purchased on account day 14:** allowed (up to and including day 14).
- **Refund of BP after rewards claimed:** denied. UI surfaces this clearly before purchase.

6. Dependencies

- `05-gacha-and-currencies.md` — Crystal / Gem economy
- `13-battle-pass.md` — BP pricing
- `21-liveops-and-telemetry.md` — IAP receipt validation
- `17-save-system.md` — purchase history persistence
- `20-ui-menus-tabs.md` — Shop screen

7. Tuning Knobs

- All prices (3.1–3.4)
- Daily monthly card payout (3.2)
- Bundle contents (3.4)
- Cosmetic prices (3.7)
- ARPU targets (3.8) — analytics calibration

8. Acceptance Criteria

- ✓ 1. Crystal packs deliver the documented quantities.
- ✓ 2. Monthly Card delivers 100 Crystals/day for 30 days post-purchase.
- ✓ 3. Beginner Pack is hidden after 14 days OR after purchase.
- ✓ 4. Limited Banner Bundles are available only while a Limited banner is active.
- ✓ 5. All purchases route through Steam IAP and validate receipts (see `21-liveops-and-telemetry.md`).
- ✓ 6. Refund flows match documented policy.
- ✓ 7. No content is permanently gated behind paywall (every Premium item is eventually available via free progression or future Standard banner).

GDD 15 — Desktop Window UX

Highlighted (<mark>) blocks are new vs. the source PDF.

1. Overview

Project I is a Unity desktop game with three window modes: **Compact**, **Expanded**, and **Tray**. The compact window is the game's identity — small, peripheral, always playable. Adds: full settings menu (sub-tabbed), accessibility suite, hotkey rebinding, audio sliders, notifications, multi-monitor, performance modes.

2. Player Fantasy

The game lives on my desktop like a clock or a music player — useful, unobtrusive, but I can grab it when I want to manage things.

3. Detailed Rules

3.1 Window Modes

Mode	Default size	Behavior
Compact	480 × 160 px	Default. Borderless. Always visible. Toggleable always-on-top.
Expanded	1280 × 720 px	Resizable. Full UI. Standard window chrome.
Tray	—	No window. Game runs in system tray with right-click menu.

3.2 Required Window Features (from PDF)

- Always-on-top toggle (default OFF)
- Click-through mode (default OFF)
- Window position lock (default OFF)

- Position memory across sessions (compact + expanded saved separately)
- Reset shortcuts (Shift+F11 = reset scale, Shift+F12 = reset position, Ctrl+Shift+T = click-through, Ctrl+Shift+M = mode switch)

3.3 Compact Mode Display

Visible elements: - Active party chibis - Current enemy chibis - Damage numbers - Loot drops (tickerized — small floating text) - Gold counter, EXP counter, current stage name - Shrine ready indicator (small icon, badge) - Egg ready indicator - Boss warning icon - Expand button, settings button, close button (top-right cluster)

Loot Ticker (NEW): scrolling text at the bottom of the compact window showing the last 3 loot drops. Hovering pauses the scroll. Click opens the inventory.

3.4 Expanded Mode UI

See `gdd/20-ui-menus-tabs.md` for the full menu structure.

3.5 Tray Mode



System tray icon menu: - “Show Compact Mode” - “Show Expanded Mode” - “Pause Idle Combat” - “Mute” (toggle) - Hero / Inventory / Gacha quick-launch buttons (open in expanded mode) - “Settings” - “Exit”

Game continues idle simulation while in tray. CPU/GPU footprint is minimal — same as background combat (no animations rendered).

3.6 Hotkeys (all rebindable)



Default hotkeys:

Action	Default	Rebindable
Toggle compact  expanded	Ctrl+Shift+M	Yes
Toggle always-on-top	Ctrl+Shift+A	Yes
Toggle click-through	Ctrl+Shift+T	Yes
Lock window position	Ctrl+Shift+L	Yes
Reset window position	Shift+F12	Yes
Reset window size	Shift+F11	Yes
Open settings	Ctrl+,	Yes
Quick claim Shrine	Ctrl+Shift+S	Yes
Quick claim idle rewards	Ctrl+Shift+R	Yes
Open Gacha	Ctrl+G	Yes
Pause combat (idle)	Ctrl+Shift+P	Yes

Conflict detection: if two actions map to the same combo, UI warns and disables both until resolved.

3.7 Settings Menu (NEW — fully restructured)



Settings is organized into **9 sub-tabs**, each with focused options:

Audio

- Master Volume (slider, 0-100)
- Music Volume (slider, 0-100)
- SFX Volume (slider, 0-100)
- Voice Volume (slider, 0-100, even though no voice ships v1 — UI ready)
- UI Sound Volume (slider, 0-100)
- Mute all (override toggle)

Display

- Compact Mode FPS (15 / 30 / 60)
- Expanded Mode FPS (30 / 60 / 90 / 120 / Uncapped)
- Tray / Background FPS (5 / 10 / Simulation-only)
- Display Scale (UI 100% / 125% / 150%)

- Damage Numbers (Detailed / Simplified / Off)
- VFX Intensity (Off / Low / Medium / High)
- Animation Speed in Expanded Mode (1× / 1.5× / 2×) — separate from combat speed
- Reduce screen shake (toggle)

Window

- Start in Compact Mode (toggle)
- Start in Expanded Mode (toggle)
- Always-on-top (toggle)
- Click-through (toggle)
- Lock compact window position (toggle)
- Reset window position (button)
- Reset window size (button)
- Multi-monitor: remember per-monitor (toggle, default ON)
- Snap-to-edge (toggle)

Gameplay

- Combat Speed Default (1× / 2× / 3× / 4×)
- Auto-collect Shrine when full (toggle, default OFF)
- Auto-collect idle rewards on login (toggle, default OFF)
- Auto-equip best gear when leveling (toggle, default OFF)
- Auto-sell items below rarity (Off / Common / Uncommon / Rare)
- Confirm-on-spend (premium currency) (toggle, default ON)
- Auto-AI Bias (Aggressive / Balanced / Conservative)
- Pause-when-unfocused (toggle, default OFF)
- Pause-when-PC-sleeps (toggle, default OFF — offline rewards always work)

Controls

- Full hotkey list with rebind UI
- Reset all hotkeys to default (button)
- Mouse hover preview in compact mode (toggle)

Accessibility

- Color blind mode (Off / Protan / Deutan / Tritan)
- High contrast UI (toggle)
- Text size (Small / Normal / Large / XL)

- Reduce motion (toggle — disables non-essential animations)
- Reduce flashing (toggle — disables critical hit flash, etc.)
- Subtitles (toggle — for future VO)
- Screen reader hooks (toggle, opt-in; partial support v1; full v2)

Notifications

- Desktop notifications (toggle, master)
- Shrine ready alert (toggle, default ON)
- Idle storage cap alert (toggle, default ON)
- Egg ready alert (toggle, default OFF — avoids spam)
- Boss-stage available (toggle, default ON)
- Sound on notification (toggle, default ON)

Account

- Cloud save sync (toggle, default ON)
- Last sync timestamp (display)
- Manual sync now (button)
- Export save data (button — produces a `.savedata` file)
- Reset tutorial / replay onboarding (button — confirmation required)
- Sign out (button — for cloud save providers)
- Delete account (button — destructive; opens confirmation modal with 3-step verify)
- Discord Rich Presence (toggle, default ON)
- Telemetry / analytics (toggle, default ON; opt-out)

About

- Game version
- Build number
- Patch notes link
- Credits
- Open source licenses
- Privacy policy link
- Terms of service link
- Customer support link
- Help / FAQ link

3.8 Accessibility (deeper detail)



Color blind modes: UI palette swaps for Protanopia, Deuteranopia, Tritanopia. Combat HUD (element icons, damage numbers, status effects) get distinct shapes + symbols beyond color (e.g., Fire = flame icon + red, but also has a flame *outline*; Water has wave outline; Wind has triangle outline). This makes element identification possible without color.

High contrast UI: thicker borders, monochrome backgrounds, larger text. Standalone toggle from color blind mode.

Text size: scales all text. Layouts adapt — long text wraps, doesn't overflow.

Reduce motion: disables idle animations on the compact window. Combat animations still play (essential to feedback).

Reduce flashing: disables critical hit flash, ultimate impact flash, level-up screen flash. Required compliance for photosensitive accessibility.

Screen reader: v1 supports tab navigation + aria-label-equivalent metadata on critical buttons. Full screen reader support is v2.

3.9 Performance Modes



Mode	Compact FPS	Expanded FPS	Tray FPS	When
Performance	30	60	Simulation-only	Default for desktops
Battery Saver	15	30	Simulation-only	Auto on battery laptops
Ultra	60	120	10	Manual high-end

Battery Saver auto-engages when system reports < 20% battery AND on battery power. Player can override.

3.10 Compact Mode Mouse Interactions



Mouse action	Result
Left-click anywhere (non-button)	(with click-through OFF) Toggles between drag-mode and click-mode
Right-click	Opens compact mode context menu (pause, settings, expand, exit, lock window)
Hover over hero chibi	Tooltip: hero name, HP%, current buffs
Hover over loot ticker	Pauses scroll, expands to show detail
Hover over Shrine icon	Tooltip: current Gold accumulated / cap
Double-click	Triggers Expand (if click-through OFF)

3.11 Multi-Monitor



- Window remembers which monitor it was last on.
- If saved monitor is disconnected at launch, window snaps to primary monitor.
- “Move to next monitor” hotkey: Ctrl+Shift+M+→ (rebindable).
- Per-monitor display scaling respected.

4. Formulas

Concept	Formula
Compact mode CPU cap	Target < 2% CPU on a modern desktop
Tray mode CPU cap	Target < 0.5% CPU
Loot ticker scroll speed	30 px/s, pause on hover

5. Edge Cases

- **Click-through ON + window dragged to off-screen:** Shift+F12 reset position rescues it.
- **Hotkey rebind conflict:** UI prevents save until conflict resolved.
- **All monitors disconnected with window on a removed monitor:** snap to primary.
- **DPI change mid-session:** UI re-renders; saved window position preserved by % not pixels.
- **Battery saver toggles on, then user plugs in:** restore to Performance after 30s of stable AC.

6. Dependencies

- `01-combat.md` — combat speed, auto-AI bias
- `06-idle-and-shrine.md` — auto-collect toggle, notifications
- `17-save-system.md` — settings persistence
- `19-localization-plan.md` — text size + localization
- `20-ui-menus-tabs.md` — Settings opens from any screen
- `21-liveops-and-telemetry.md` — Telemetry opt-in/out

7. Tuning Knobs

- Default FPS per mode (3.9)
- Default volume settings (50/100 across all sliders)
- Default hotkey bindings (3.6)
- Default toggles (auto-collect OFF, confirm-on-spend ON, etc.)
- Battery saver auto-engage threshold (20%)

8. Acceptance Criteria

1.  Settings menu has 9 sub-tabs as documented.
2.  All audio sliders affect their respective channels independently.
3.  Compact mode position is saved across sessions.
4.  All 11 hotkeys are rebindable.
5.  Color blind mode swaps palettes + adds shape distinctions.
6.  Battery saver auto-engages below 20% on AC battery.
7.  Window mode switch (Ctrl+Shift+M) works from any mode.
8.  Multi-monitor saved correctly across reboots.
9.  Click-through OFF allows drag; ON prevents all interaction.

GDD 16 — Tutorial & Onboarding

Entire GDD is NEW. All content highlighted.

1. Overview

9-stage scripted onboarding flow. Designed for **reroll-friendly speed** — a player can reach the guaranteed Legendary in under 25 minutes. No real-money pressure for the first 7 days. Each stage introduces exactly one new system.

2. Player Fantasy

In 20 minutes I went from “what’s this?” to “I love this team, I want to keep building.” Nothing felt forced; everything I unlocked felt earned.

3. Detailed Rules

3.1 Tutorial Sequence



Stage	Subject	Time	Mechanics introduced
1	First Battle	2 min	Combat basics, Speed, basic attack, Skill 1
2	Element Triangle	2 min	Element advantage, Skill 2 cast
3	Ultimate & Defend	3 min	Ultimate use, Defend action
4	Idle Rewards Intro	3 min	Idle accumulation explained, Shrine intro
5	First Gacha — Guaranteed Lyria	3 min	Pull animation, gacha currency, pity tease. First 10-pull is scripted: Lyria is guaranteed to appear. Lyria is the main character — every player gets her as their starter.
6	Level & Promote	5 min	EXP crystals, leveling, promotion (1★→2★ tutorial fodder provided)
7	Equipment	4 min	Equip first piece of gear, sub-stats explained, set bonus highlight
8	Compact Mode Intro	1 min	Switch to compact, idle reward demo
9	Daily / Weekly / BP / Tower / Expedition unlock teaser	3 min	Brief tour of the meta game, all systems unlocked

Total tutorial time: ~26 minutes for a thorough first run.

3.2 Reroll-Friendly Pacing



- Tutorial 5★ is **guaranteed**.
- Day-1 free gems = ~3,000 (enough for ~18 pulls). Combined with tutorial = ~30 pull equivalents.
- Account reset is fast: settings → “Reset tutorial / start fresh” → re-tutorial in 20 min.
- This makes reroll the genre-standard 20-30 min loop. Don’t fight it; embrace it.

3.3 Onboarding Sequence



Day 0 (first session): - Tutorial → Beginner Step-Up Banner Step 1 unlock → Day 1 login bonus - Account L5 unlock: Inventory expansion +20

Day 1-3: - Daily missions introduced (Day 1 reward) - 7-day login streak active - Hatchery unlocked (Day 1) - Expedition unlocked (Day 2) - Tower of Trials introduced (Day 3, account L10 gate)

Day 4-7: - Battle Pass introduced (Day 4) - Hero Trials introduced (Day 5) - Boss Rush introduced (Day 7)

Day 7 login bonus: the free Standard 5★ pick.

Day 14: Beginner Step-Up Banner expires.

3.4 First-Time-User Hints



Inline hint tooltips appear once per system on first use:

- “Tap a hero to see their detail screen!”
- “Sort options are in the filter dropdown!”
- “Skill cooldowns reset between stages.”
- “Set bonuses activate at 2 or 4 pieces.”
- “Hero Trials reward Hero Shards — collect 50 to unlock a hero!”
- “Pity carries over. Don’t reset, it’s tracked across banners!”

All hints can be **dismissed** and never re-shown. Bypass available via “Reset Tutorials” in settings.

3.5 Tutorial UI



- Modal overlay with pulsing “next” button.
- “Skip Tutorial” button surfaces only after stage 3 (no skip in the first 3 stages — critical for ensuring players understand combat).
- “Replay Tutorial” available in settings → Account.
- Localized text for all hint strings (per `gdd/19-localization-plan.md`).

3.6 First Hero — Lyria (Main Character)



Lyria is the main character and starter hero — every player receives her at Stage 5 of the tutorial. No choice / no reroll-pick. Lyria’s tutorial moment is scripted to feel celebratory: full splash art, name reveal, kit-tour modal.

After Lyria is added, the tutorial auto-builds a starter team: - **Lyria** (Wind Archer — primary DPS) - **Sylphie** (Wind Support — partner; granted as a 3★ promotable to 4★ via tutorial fodder) - **Valdrik** (Fire Tank — partner; granted as a 3★ promotable to 4★)

The two 3★ partner units exist so the player has a functioning party immediately. The player can replace them with summoned heroes as soon as they pull better options.

Why no choice? Project I uses Lyria as its mascot / identity. Letting players reroll past her dilutes that identity. Players who want a different starter can still pull on the Beginner Step-Up Banner (3.3) which gives ~2 additional Legendaries quickly.

4. Formulas

Concept	Formula
Total day-1 reachable gem count	~3,000 (gives ~18 pulls)
Guaranteed Legendary timing	Stage 5 of 9
Beginner Step-Up Banner expiry	account day 14

5. Edge Cases

- **Tutorial interrupted:** state saves at each step; resume on next launch.
- **Tutorial skipped after stage 3:** the player can still earn Day 1 rewards normally; only the in-game guided experience is shorter.
- **Tutorial reset after major progress:** confirmation modal explicitly warns about gem / Gold loss. Reset does NOT refund Crystals.
- **Reroll account creation:** new account = clean state. Old account on same Steam ID may not be reusable (post-launch policy).

6. Dependencies

- `01-combat.md` — tutorial uses simplified combat
- `05-gacha-and-currencies.md` — Step-Up Banner, free pulls
- `06-idle-and-shrine.md` — Shrine introduction in stage 4
- `12-daily-weekly-loops.md` — login bonus, daily missions
- `13-battle-pass.md` — BP introduction in day 4
- `19-localization-plan.md` — tutorial strings

7. Tuning Knobs

- Tutorial stage count (currently 9)
- Tutorial total time (target 26 min)
- Day-1 gem allowance (~3,000)

- Hero choice for starter pick (3.6 list)
- Hint trigger system

8. Acceptance Criteria

1.  Tutorial completes in under 30 minutes (95% of players).
2.  Tutorial stage 5 awards Lyria as the guaranteed starter Legendary (no choice).
3.  Sylphie and Valdrik are granted as 3★ partner units so player has a functioning starter team.
4.  Day-1 free gems total at least 3,000.
5.  Tutorial can be reset from settings; resetting does NOT refund Crystals.
6.  All hints can be dismissed permanently.

GDD 17 — Save System

Entire GDD is NEW. All content highlighted.

1. Overview

Local JSON save + optional cloud sync hook. Deterministic seeds for idle simulation. Tamper-resistant via checksum + obfuscation. Designed for fast cloud-restore on a new PC.

2. Player Fantasy

If my PC dies, my progress isn't gone. If I upgrade to a new machine, my game just works.

3. Detailed Rules

3.1 Save Locations



Location	Use	Format
%APPDATA%/ProjectI/save/profile.json	Primary	Plain JSON (obfuscated)
%APPDATA%/ProjectI/save/profile.bak	Backup	Last successful save (rotated)
Steam Cloud	Cloud sync	Same JSON, uploaded on save
Manual export .savedata	User backup	JSON file (player can email / re-import)

On Mac (future): ~/Library/Application Support/ProjectI/save/

3.2 Save Data Schema (top-level)




```

{
  "version": "1.0.0",
  "playerId": "uuid-v4",
  "lastSaveAt": "ISO-8601",
  "lastSyncAt": "ISO-8601",
  "checksum": "sha256",
  "playerProfile": { ... account level, gems, gold, crystals ... },
  "heroes": [ ... per-hero state ... ],
  "inventory": [ ... ],
  "shrineState": { ... },
  "hatcheryState": { ... },
  "expeditionState": { ... },
  "gacha": { ... per `gdd/05-gacha-and-currencies.md` $ 3.11 ... },
  "campaignState": { ... stage-clear records ... },
  "towerState": { ... },
  "bossRushState": { ... },
  "heroTrialState": { ... },
  "missions": { "daily": [...], "weekly": [...] },
  "battlePass": { ... },
  "achievements": [ ... ],
  "settings": { ... 9 sub-tab settings ... },
  "idleState": {
    "lastUpdateAt": "ISO-8601",
    "deterministicSeed": "uint64",
    "currentStage": "chapter-stage-difficulty",
    "rngState": "uint64"
  },
  "telemetryOptIn": true,
  "purchaseHistory": [ ... receipt IDs ... ]
}

```

3.3 Save Cadence

Trigger	Action
Significant change (purchase, gacha pull, hero upgrade)	Immediate save
Periodic	Every 60 seconds during active play
On window close	Final save
On crash	Recovery save (via OS signal handler if reachable)

Saves are atomic: write to `profile.json.tmp` then rename. The previous `profile.json` becomes `profile.bak`.

3.4 Cloud Sync Strategy



Steam Cloud is the v1 cloud target.

- Save uploaded on every “Save” event
- Last-modified timestamp + checksum used for conflict detection
- Conflict resolution: prefer the save with the higher `lastSaveAt` . On true ambiguity (same timestamp, different checksum), prompt the player with a “Choose local or cloud” modal showing high-level stats from each save.

Cross-platform sync (future): - Mac/Windows share Steam Cloud - Other platforms (EGS) would require platform-specific cloud sync

Offline tolerance: - If Steam Cloud unavailable, game continues with local save - Sync attempted on every subsequent save; notification when sync resumes

3.5 Idle Simulation Determinism



When the game launches after time has elapsed, idle accumulation is simulated using a **deterministic RNG seed** stored in `idleState.deterministicSeed` . The simulation:

1. Computes `elapsedHours = now - lastUpdatedAt` (capped at 12h)
2. Adjusts for online/offline rate (`0.75` if Sleep/Closed)
3. Runs the loot drops for the configured idle stage using the stored seed
4. Updates `rngState` (advance the seed)
5. Saves the new state

Because the seed and rate are deterministic, **re-running the same simulation produces identical results**. This is critical for: - Anti-cheat (server can validate) - Bug reproduction (testers can replay scenarios) - Player trust (no hidden “RNG manipulation”)

Anti-tamper: - The save file is obfuscated (XOR with a hashed key per playerId) - A SHA-256 checksum of the save is included; mismatched checksum → save rejected; backup loaded - Modified saves are detected; in normal F2P this would warrant a soft ban, but since Project I is single-player, **modified saves block gacha pulls + cloud sync only**. The player can keep playing but pull data refuses to upload.

3.6 Save Conflict UI



If a true conflict exists:

```

Save conflict detected

Local save:
  Last played: 2 hours ago
  Account L42, 18 heroes, 23k Gems

Cloud save:
  Last played: 5 minutes ago
  Account L41, 12 heroes, 8k Gems

[ Use Local ] [ Use Cloud ]

```

The chosen save replaces the other. The discarded save is preserved as `profile-conflict-{timestamp}.json` for 30 days in case of mistakes.

3.7 IAP Receipt Storage



Every IAP receipt is stored in `purchaseHistory[]` :

```

{
  "txId": "steam-receipt-id",
  "productId": "small_crystal_pack",
  "amountUSD": 4.99,
  "purchasedAt": "ISO-8601",
  "validated": true,
  "validatedAt": "ISO-8601",
  "fingerprint": "checksum"
}

```

Receipts are validated against Steam's web API on first launch after purchase. Failed validation → entitlements held until next online sync.

3.8 Account Delete



“Delete Account” in settings → Account flows:

1. Confirmation modal #1: “Are you sure?”
2. Type-the-word verification (player types “DELETE”)
3. Confirmation modal #2 with full impact summary
4. 7-day grace period — account marked for delete, can be cancelled
5. After 7 days: local save deleted, cloud save deleted, telemetry tokens purged

This implements GDPR / similar right-to-erasure compliance.

3.9 Backup & Export



Player can manually export their save:

- Settings → Account → Export Save
- Produces `ProjectI-{playerId}-{timestamp}.savedata`
- File is encrypted with the same obfuscation
- Re-importing on a new account replaces all state (with confirmation)

4. Formulas

Concept	Formula
Checksum	<code>SHA-256(canonicalJSON)</code>
Save obfuscation key	<code>SHA-256(playerId + buildSecret)</code> then XOR cipher
Idle elapsed hours	<code>min((now - lastUpdatedAt), 12)</code>
Conflict tiebreak	higher <code>lastSaveAt</code> wins

5. Edge Cases

- **System clock manipulation:** If `now - lastUpdatedAt < 0` (clock moved backward), treat as `0` (no rewards). Surface a warning toast.
- **Save file corrupted:** load backup. If both fail, prompt user with “restore from Steam Cloud” or “start fresh” options.
- **Steam Cloud quota exceeded:** save locally; surface warning. Cloud sync resumes when player clears Cloud space.
- **First launch ever:** generate `playerId` (UUID v4), initial save written.
- **Save data > version mismatch:** load with migration script (each version increment has a migrator); reject if migration fails.

6. Dependencies

- `01-combat.md` — combat state preserved across battles

- `06-idle-and-shrine.md` — idle accumulation determinism
- `10-tower-of-trials.md` — monthly reset state
- `12-daily-weekly-loops.md` — mission state, reset timers
- `13-battle-pass.md` — BP state, season tracking
- `14-monetization.md` — purchase history
- `15-desktop-window-ux.md` — settings persistence
- `21-liveops-and-telemetry.md` — server-side validation of saves

7. Tuning Knobs

- Save cadence (60s default)
- Cloud sync attempt frequency
- Idle simulation cap (12h)
- Conflict grace preservation (30 days)
- Account delete grace (7 days)
- Obfuscation key derivation (cannot be changed without breaking existing saves)

8. Acceptance Criteria

1.  Save file is atomic — no half-written corruption.
2.  Backup save (`profile.bak`) is loaded if primary is corrupted.
3.  Idle simulation is deterministic — same seed + elapsed = same loot.
4.  Cloud sync conflict UI presents both saves; player choice replaces the other.
5.  Account delete enters a 7-day grace period; can be cancelled.
6.  Save export produces a valid, re-importable `.savedata` file.
7.  Modified save blocks gacha pulls (no soft progression for editors).
8.  IAP receipts are validated against Steam API on first sync.

GDD 18 — Audio Direction

Entire GDD is NEW. All content highlighted. Placeholder content — user is providing assets later; this is the specification.

1. Overview

Project I's audio is intentionally minimal and unobtrusive — players are running the game in the background while working. Combat is musically simple, satisfying SFX-driven; music swells only on bosses, ultimates, and key moments. **No voice acting in v1.**

2. Player Fantasy

I can have the game open all day without my ears getting tired. When something exciting happens — a critical, an Ultimate, a 5★ pull — I notice it.

3. Detailed Rules

3.1 SFX Taxonomy



UI sounds: - Button hover, button press, tab open, modal open/close, menu navigation - Notification chime (subtle), error tone (subtle)

Combat sounds: - Basic attack (per weapon type — sword, bow, staff, etc.) - Skill cast (per element — Fire whoosh, Water splash, Wind sweep) - Hit impact (light, heavy, critical, blocked) - Damage number (small jingle on crit only) - Status effect application (each effect has signature sound) - Hero KO (low thud) - Wave clear (mild flourish) - Boss appearance (distinct cue) - Boss enrage (warning sting)

Reward sounds: - Loot pickup (rarity-tiered — Common = soft, Mythic = celebratory) - Egg hatch (distinctive crack + fanfare for Epic+) - Shrine collect (gold-shower sound, satisfying) - Level up (bright chord) - Skill upgrade (twinkle) - Achievement unlock (medium fanfare)

Gacha sounds: - Pull start (drum roll-equivalent) - Pull result reveal — rarity-tiered: - Rare: subtle bell - Epic: warm chord - Legendary: massive celebratory cue (extends into the splash art reveal)

Window / system sounds: - Window snap to edge (subtle click) - Compact  Expanded transition (brief whoosh) - Tray show / hide (subtle)

3.2 Music



Music cues: - **Main menu / loading** — atmospheric, soft, looping - **Compact mode background** — extremely soft ambient layer (toggleable per player; default OFF for performance) - **Expanded mode menus** — slightly more present ambient - **Campaign combat** — per-act theme (Act 1 = forest/whimsical, Act 2 = volcanic/intense, Act 3 = oceanic/somber) - **Boss fights** — unique theme per chapter boss - **Tower of Trials** — escalating theme that intensifies every 10 floors - **Boss Rush** — gauntlet theme - **Hero Trials** — per-element theme

No voice acting in v1. Hero “voice triggers” are reserved in UI layout but no audio shipped. Sound design substitutes for voice (e.g., on Ultimate cast, a distinctive elemental flourish plays instead of a hero shout).

3.3 Mixing Bus



Bus	Default Volume	Player-adjustable
Master	100	Yes
Music	60	Yes
SFX	80	Yes
Voice (placeholder)	80	Yes
UI	70	Yes

Mixing rules: - SFX has slight ducking when Ultimate animation plays (so the Ultimate cue is the focal point) - Music dips during boss enrage (so the warning is heard) - Critical hit SFX has a distinct attack envelope (sharp transient) - Compact mode audio at 70% of expanded volume by default (background-friendly)

3.4 Asset Specs (for production)



Asset type	Format	Duration	Sample rate
UI sounds	.wav / .ogg	< 1s	44.1 kHz
Combat SFX	.wav / .ogg	< 2s	44.1 kHz
Music loops	.ogg	1-3 min loops	44.1 kHz, stereo
Boss themes	.ogg	2-4 min	44.1 kHz, stereo

Compression target: ~64-128 kbps OGG Vorbis for music, lossless or 192k for SFX.

3.5 Localization-Aware

For v1, no voice acting means localization audio scope is minimal. If voice acting is added post-launch (it likely will be for at least the popular Asian-market locales), the system needs:

- VO triggers wired in code at all hero-ability points
- VO key naming convention: `hero.{heroId}.{abilityKey}.{lineIndex}`
- Fallback to silent if locale doesn't have VO

(See `gdd/19-localization-plan.md`.)

4. Formulas

Concept	Formula
Effective output volume	$\text{MasterVol} \times \text{BusVol} \times \text{IndividualSourceVol}$ (all 0-100 scaled)
SFX ducking under Ultimate	-30% for the duration of the Ultimate animation
Music dip on boss enrage	-20% for 5 seconds

5. Edge Cases

- **Multiple SFX overlap:** mixer caps simultaneous voices at 8 SFX channels; oldest is dropped.
- **System audio device change:** Unity auto-detects and continues. No save state needed.
- **All audio muted:** game functions normally; UI silent toggle indicator visible.
- **Music file fails to load:** silent fallback; log warning.

6. Dependencies

- 15-desktop-window-ux.md — volume sliders, mute toggle
- 19-localization-plan.md — future VO localization
- 01-combat.md — combat SFX triggers
- 05-gacha-and-currencies.md — gacha pull audio cues
- 06-idle-and-shrine.md — Shrine collect SFX

7. Tuning Knobs

- Default volume per bus (3.3)
- Ducking magnitudes (3.3)
- Music loop lengths (3.2)
- Compact-mode music default (currently OFF)

8. Acceptance Criteria

1.  All UI sound sliders adjust the corresponding bus.
2.  Master mute silences all audio.
3.  Different rarity gacha pulls produce distinct cues.
4.  Per-act music themes play in the correct chapters.
5.  Boss enrage music dip works.
6.  Compact mode music respects its independent default.

GDD 19 — Localization Plan

Entire GDD is NEW. All content highlighted.

1. Overview

V1 ships English only, but the entire codebase is built for localization from day one. **No hardcoded strings.** All text is resolved through a localization key system. Key naming convention is strict so future translators can work without seeing English.

2. Player Fantasy

When the Japanese / Korean / Chinese launch happens, the game feels native — not bolted-on.

3. Detailed Rules

3.1 Architecture



```
Source code:   LocalizationManager.Get("hero.lyria.skill1.name")
               ↓
String table:  { "hero.lyria.skill1.name": "Gale Piercer", ... }
               ↓
UI displays:   "Gale Piercer"
```

Per-locale string table files: - `data/strings/en.json` (English) - `data/strings/ko.json` (Korean — post-launch) - `data/strings/ja.json` (Japanese — post-launch) - `data/strings/zh-cn.json` (Simplified Chinese — post-launch) - `data/strings/zh-tw.json` (Traditional Chinese — post-launch) - `data/strings/es.json` , `data/strings/fr.json` , `data/strings/de.json` (EU rollout)

3.2 Key Naming Convention



```
{category}.{subcategory}.{specificKey}[,subKey]
```

Examples: - `hero.lyria.name` - `hero.lyria.title` - `hero.lyria.skill1.name` -
`hero.lyria.skill1.description` - `hero.lyria.passive.description` - `combat.damageType.crit`
 - `ui.menu.gacha.title` - `ui.button.confirm` - `ui.tooltip.element.fire` -
`gacha.banner.limited.title` - `event.boss_rush.welcome` - `settings.audio.master` -
`error.network.connection_lost`

Categories: `hero`, `enemy`, `stage`, `item`, `equipment`, `set`, `ui`, `combat`, `gacha`, `tower`,
`expedition`, `bp`, `mission`, `achievement`, `tutorial`, `settings`, `notification`, `error`.

3.3 Pluralization



Some locales (Russian, Polish, Arabic, etc.) have complex plural forms. The system uses ICU MessageFormat:

```
{
  "ui.inventory.item_count": "{count, plural, one {# item} other {# items}}"
```

Even if v1 ships English only, this format is used from day one to avoid retrofitting.

3.4 Right-to-Left (RTL) Support



Arabic and Hebrew are RTL languages. UI layout system must support: - Automatic mirror of horizontal layouts (gacha banner → flips orientation) - Text alignment flip (left-aligned [?] right-aligned) - Combat UI: party still on left; enemies on right (combat orientation is gameplay-canonical, not text-flow)

V1 does NOT ship Arabic. The framework supports it (not a retrofit).

3.5 Text Variable Substitution



Damage formula tooltips and other dynamic text use named substitutions:

```
{
  "skill.damage_tooltip": "Deals {damageMultiplier}× ATK ({damageRange}) damage."
```

Variables are passed at call time:

```
LocalizationManager.Get("skill.damage_tooltip", {  
    damageMultiplier: "2.20",  
    damageRange: "1200-1800"  
})
```

3.6 Translator Briefing



When a locale is added, translators receive:

- The full English string table (source of truth)
- A glossary of game-specific terms (elemental names, class names, currency names)
- Style guide for tone (Project I is “earnest fantasy”, not ironic; serious but warm)
- Character voice notes (e.g., Sylphie speaks gently; Kael is taciturn)
- Length constraints per string (UI fits, etc.)
- Cultural sensitivity flags (any region-specific concerns)

3.7 Locale-Specific Considerations



Locale	Notes
Japanese (ja)	Hero names may use both katakana and kanji; choose per-hero
Korean (ko)	Honorifics matter for hero dialogue
Chinese (zh-cn)	Simplified characters; some elemental terms have established translations
Chinese (zh-tw)	Traditional characters; Taiwanese gaming market is large
Spanish (es)	“Latin Spanish” (LATAM) is default; “Spain Spanish” (ES) optional later
Portuguese (pt-BR)	Brazilian market; separate from Continental Portuguese
French (fr)	Single locale at launch (no Quebec split)
German (de)	Direct translations work; some terms borrowed (e.g., “Pity”)
Russian (ru)	Complex pluralization; post-launch
Arabic (ar)	RTL; post-launch

3.8 Asset Localization



In-image text (e.g., logo, UI ornaments) should be **avoided where possible**. If an image contains text: - Maintain per-locale versions - Or use overlaid runtime text (preferred)

Audio (future VO) follows separate localization spec — see `gdd/18-audio-direction.md`.

3.9 Localization Testing



For each locale before release: - All UI screens visited in test mode with that locale - Long-text scenarios tested (German is typically +30% length) - Text-truncation testing - RTL flipping verified (Arabic) - Cultural review by native speaker

Project I uses a “locale tester team” of 1-2 per major locale for QA.

4. Formulas

Concept	Formula
Fallback rule	If key missing in current locale → fallback to English → fallback to key string itself
Plural form selection	ICU CLDR rules

5. Edge Cases

- **Missing translation:** falls back to English. UI shows the key string itself if English also missing (clearly broken — easy to catch in dev).
- **Locale switch mid-session:** UI re-localizes immediately. Game state is locale-independent.
- **Locale-specific date / number formatting:** uses platform locale APIs.
- **Translation longer than UI accommodates:** auto-resize containers; otherwise show “...” truncation with full-text tooltip.

6. Dependencies

- `15-desktop-window-ux.md` — Language selector setting
- `18-audio-direction.md` — Future VO localization
- All other GDDs — all text references LocalizationManager
- `17-save-system.md` — locale preference persists

7. Tuning Knobs

- Locale list (currently EN only at launch)
- Fallback chain rules
- Tester team size per locale
- Translation budget per quarter

8. Acceptance Criteria

1.  No hardcoded strings in the codebase — every player-facing text uses LocalizationManager.
2.  Key naming follows the documented convention.
3.  Pluralization works correctly using ICU MessageFormat.

4.  Locale fallback chain works (current → English → key).
5.  RTL layout flips correctly when Arabic locale is loaded (testable in dev mode).
6.  Variable substitution works for dynamic strings.
7.  Locale switch updates UI without restart.

GDD 20 — UI / Menus / Tabs

Entire GDD is NEW. All content highlighted.

1. Overview

This GDD defines the **full navigation tree** for both Compact and Expanded modes. Adds 18 new tabs/screens to the source PDF's 9. Includes Mail/Inbox, Achievements, Codex, Hero Trials, Boss Rush, Stage Select map, settings sub-tabs, and Help/FAQ.

2. Player Fantasy

I can find anything in two clicks. Nothing is buried. New content surfaces with a clear badge.

3. Detailed Rules

3.1 Compact Mode UI

Already documented in `gdd/15-desktop-window-ux.md`. Minimal — combat view + a single expand button.

3.2 Expanded Mode — Top Navigation



Persistent top tab bar (left to right):

Tab	Icon	Subtabs / Notes
Home / Dashboard		Daily check-in, missions summary, news
Stage Select		Chapter map, difficulty toggle, scout
Heroes		Roster, hero detail, fusion, awakening, skill upgrade
Inventory		Gear, materials, eggs, codex
Gacha		Limited, Standard, Step-Up; pity tracker; wishlist
Tower		100-floor map, scout, shop
Expedition		Active expeditions, mission board
Hero Trials		Weekly featured hero, 3-tier trials
Boss Rush		Weekly gauntlet, difficulty toggle
Battle Pass		Free / Premium tracks, claim
Shop		Crystal packs, Monthly Card, Beginner Pack, Dupe Shop
Achievements		List, completion %, claim
Mail		Pending claims, expiry warnings
Settings		9 sub-tabs (per <code>gdd/15-desktop-window-ux.md</code>)

3.3 Home / Dashboard (NEW)



Landing screen after tutorial. Shows: - Daily mission progress bar (with one-tap “open mission list”) - Login bonus indicator (claim button if available) - Featured chapter of the week banner - News carousel (patch notes, events — post-launch) - Quick-actions: Resume Idle, Pull Banner, Open Tower - Active expedition timer summary - Mailbox unread count badge

3.4 Stage Select Map (NEW)



- Visual chapter map (Acts 1 / 2 / 3 with chapter nodes)
- Tap chapter → expanded view of 10 stages
- Difficulty toggle (Normal / Hard / Nightmare / Hell)
- 3-star indicator per stage
- “Idle here” button per cleared stage
- “Scout” button on boss stages

- Featured Chapter highlighted

3.5 Heroes Hub (NEW structure)



Three sub-tabs:

Roster (browse) - Grid of owned heroes - Filter / sort (per `gdd/08-inventory-and-storage.md`) - Lock / favorite icons - “Recently obtained” badge

Hero Detail (per hero) - Splash art, card, all stats - Equipment slots (tap to manage) - Skill levels (with Skill Stone investments) - Auto-AI overrides - Awakening status, level, star tier - Promotion progress (next milestone) - Comparison vs another owned hero

Hero Gallery (codex view, all 16) - All 15 5★ heroes — owned and unowned (silhouetted); future-banner heroes show “Coming Soon” date estimate; 4★ heroes (when released) displayed alongside - Filter by element / class / banner pool - Used for wishlist research

3.6 Inventory Hub (NEW structure)



Three sub-tabs:

Equipment - Filter by slot / set / rarity / equipped status - Comparison view - Bulk select for salvage / sell - Forge (enhance, reroll sub-stat) - Set Bonus inspector (which heroes have active set bonuses)

Materials - All consumable items - Grouped by category (EXP, Promotion, Skill, Sacred, Currency, Tickets) - Each shows current count + sources for re-acquisition

Codex (NEW) - Hero Gallery link - Bestiary (enemies encountered) - Item Codex (all gear types + sets) - Set Bonus catalog

3.7 Gacha Hub



Three banner tabs + pity tracker + wishlist:

Limited Banner - Featured hero splash - Banner duration countdown - Pity progress (X/30 soft, X/50 hard, X/30 carryover indicator) - Rates display, single / 10-pull buttons

Standard Banner - Carousel of all Standard heroes - Pity progress (X/30 soft, X/50 hard, X/210 Selector) - Wishlist editor (3 picks, 24h lock) - Single / 10-pull / Free Daily Pull buttons

Step-Up Banner (only visible to accounts ≤ 14 days) - 3-step progression - Step rewards shown - Cost per step

3.8 Mail / Inbox (NEW)



- All pending claimables (gacha overflow, expedition rewards, BP rewards, achievement rewards, compensation gifts)
- Each entry shows: source, expiry countdown, items contained, claim button
- “Claim All” button at top (with confirm if any items would overflow inventory)
- 30-day expiration with 7-day-warning UI
- “View claimed history” tab — last 90 days

3.9 Achievements (NEW)



- Categorized tabs (Collection / Combat / Progression / Tower / Boss Rush / Gacha / Codex / Secret)
- Progress bars per achievement
- Hidden achievements show as “???” until triggered
- Claim button when completed
- Profile-title-of-the-month showcased on player profile

3.10 Settings (per `gdd/15-desktop-window-ux.md`)

Already documented. 9 sub-tabs.

3.11 Help / FAQ (NEW)



- Categorized articles (How to Play / Account / Purchases / Technical / Combat / Gacha)
- Search box
- “Contact Support” — opens email template or in-game ticket form (post-launch)
- Patch notes browser
- Game version / build info quick-display

3.12 New Content Badges



Red dot badges on tabs when: - Mail has unclaimed rewards - Daily/weekly missions completed but not claimed - BP tiers unlocked but not claimed - New banners go live (24h fade) - Achievement completed - Tower / Boss Rush has un-claimed milestones

Players love the “all dots cleared” satisfaction loop. Don’t overdo it — show only meaningful badges.

3.13 Team Builder



In Hero menu, “**Team Presets**” sub-tab: - 5 named team slots - Drag-drop heroes into slot configurations - Save preset - One-tap activate - “Recommended team for next stage” button (auto-builds based on next stage’s element + class needs)

4. Formulas

Concept	Formula
Badge unread count	sum of unclaimed items / unread missions
New content badge expiry	24h after first sighting OR until interacted
Recommended team	compute team power × element-fit × class-coverage

5. Edge Cases

- **Badge persistence after claim:** badge clears immediately upon claim. UI is reactive.
- **Mail expiring during play session:** notification on first session after expiry; expired items deleted at session start.
- **Settings open mid-combat:** combat pauses (in expanded mode; compact mode continues running).
- **Tab not yet unlocked:** grayed out with “Unlocks at chapter X” tooltip.

6. Dependencies

- 08-inventory-and-storage.md — filter/sort/search across screens
- 12-daily-weekly-loops.md — daily/weekly UI surface, achievements
- 13-battle-pass.md — BP screen
- 15-desktop-window-ux.md — Settings sub-tabs
- 17-save-system.md — UI state persistence (last visited tab)

7. Tuning Knobs

- New content badge expiry (24h)

- Mail expiry warning threshold (7 days)
- Settings tab order
- Badge thresholds (e.g., show badge on Mail when ≥ 1 unclaimed item)

8. Acceptance Criteria

1.  All 14 top-level tabs are reachable from Home.
2.  Mail/Inbox shows pending claims with countdown.
3.  Badges appear on tabs with new content, clear after interaction.
4.  Stage Select Map shows 3-star indicators.
5.  Gacha screen shows pity tracker + wishlist.
6.  Hero Detail shows skill levels, equipment, awakening progression.
7.  5 Team Presets save and switch in one tap.
8.  Help / FAQ is reachable from Settings.

GDD 21 — Live-Ops, Telemetry, Security

Entire GDD is NEW. All content highlighted.

1. Overview

Architecture for **running the game as a service**, even at launch with no live events. Covers: hot-config (server-driven balance), event calendar, maintenance broadcasts, telemetry events, anti-cheat basics, IAP receipt validation.

2. Player Fantasy

The game changes and grows over time without me having to download a patch every time the meta shifts.

3. Detailed Rules

3.1 Hot-Config (server-driven balance)



A set of game-balance values lives in a JSON config file fetched on launch (and periodically refreshed):

```

{
  "version": "2026-06-15-001",
  "gacha": {
    "rates": { "legendary": 0.006, "epic": 0.072, "rare": 0.922 },
    "hardPity": 50,
    "freeDailyPullEnabled": true
  },
  "shrine": {
    "auto_collect_default": false,
    "levels": [ { "level": 1, "rate": 50, "cap": 500 }, ... ]
  },
  "battlePass": {
    "currentSeasonId": "season-1",
    "tierXpCost": 1000,
    "endsAt": "2026-08-15T04:00:00Z"
  },
  "currentLimitedBanner": {
    "heroId": "kaida",
    "endsAt": "2026-07-06T04:00:00Z"
  },
  "featuredChapterOfTheWeek": 5,
  "missions": { ... },
  "shopPrices": { ... },
  "dropTables": { ... }
}

```

Fetches on: game start, and every 4 hours during play. **Fallback:** last cached config if server unreachable.

Validation: SHA-256 signature verified against embedded public key — prevents config injection.

This means: balance tuning, banner rotation, BP season transitions, event calendar all happen **without an app update**. Critical for live-ops at scale.

3.2 Event Calendar (post-launch)



Even though no events ship at launch, the calendar architecture is in place:

```

{
  "events": [
    {
      "id": "event-summer-2026",
      "startsAt": "2026-07-01T04:00:00Z",
      "endsAt": "2026-07-15T04:00:00Z",
      "type": "limited-stage",
      "config": { ... }
    }
  ]
}

```

Events can be: - **Limited stages** (with own drop tables) - **Bonus rate** (e.g., +50% Gold for 48h) - **Free pull** (1 free Limited pull during event) - **Cosmetic-only** (limited cosmetics) - **Maintenance** (server-side notice) - **Compensation** (mass mail to all players)

Each event runs through the same config system. UI surfaces upcoming events in the Home dashboard.

3.3 Maintenance Broadcast



When server pushes a maintenance notice, all clients receive on next launch:

```
{
  "maintenance": {
    "active": true,
    "message": "Balance patch in progress; expect downtime until 04:00 UTC",
    "expectedEndsAt": "2026-06-15T08:00:00Z"
  }
}
```

Banner UI overlays the game with a dismissible notification. If `maintenance.active = true` AND the game requires server validation (gacha pulls, IAP), block those actions with a clear message.

3.4 Telemetry Events



Game emits telemetry events to a backend (e.g., GameAnalytics, PostHog, or custom):

Funnel events: - `app.launch` - `app.tutorial.started` - `app.tutorial.step.completed` (n) - `app.tutorial.completed` - `app.first_pull` (banner, result rarity) - `app.first_purchase` (productId, amountUSD) - `app.account.day_n_active` (n: 1, 3, 7, 14, 30)

Combat events: - `combat.battle.started` (stage, difficulty) - `combat.battle.outcome` (win/loss/wipe, turns, deaths) - `combat.idle.simulation` (elapsedHours, lootValue)

Progression events: - `progression.level_up` (herold, fromLevel, toLevel) - `progression.promote` (herold, fromStar, toStar) - `progression.awaken` (herold, awakeningLevel) - `progression.skill_upgrade` (herold, skillId, fromRank, toRank)

Economy events: - `economy.gem_spend` (amount, sink) - `economy.crystal_spend` (amount, sink) - `economy.crystal_purchase` (productId, amountUSD) - `economy.shrine.collect` (gold) - `economy.idle.claim` (gold, exp, items)

Engagement events: - `engagement.daily.completed` (points) - `engagement.weekly.completed` - `engagement.bp.tier_up` (tier, premium) - `engagement.tower.floor_cleared` (floor) - `engagement.expedition.dispatched` / `completed`

Privacy: - Player can opt-out via Settings → Account → Telemetry - Even with opt-in, no PII is collected (no email, no IP-mapped location, no install fingerprint beyond Steam ID) - All events are batched and sent on session end / 5-min cadence

3.5 Anti-Cheat Basics



Project I is **single-player**, but anti-cheat still matters because: 1. IAP is real money. Cheaters who edit saves to grant themselves Crystals avoid payment. 2. Tower of Trials has leaderboards. Cheaters can rank artificially. 3. Achievement Steam-integration — fake completes pollute the Steam ecosystem.

Mitigations: - Save file checksum + obfuscation (see `gdd/17-save-system.md`) - Steam Cloud sync — only synced saves can earn ladder/Steam achievements - Server-side validation for: gacha pulls, IAP grants, Tower floor claims, Boss Rush clears - Tower leaderboard: combat replays validated server-side (deterministic re-simulation must match claimed clear) - Cheaters: not banned (single-player), but their ladder entries are excluded and Steam achievements rejected

Cheat Engine detection (passive): - Memory tampering watcher on critical values (Gem balance, Crystal balance, current gacha pity) - If tampering detected → flag account for offline review (no immediate consequence; manual review for ladder/achievement exclusion)

3.6 IAP Receipt Validation



All IAP purchases follow:

1. Player clicks "Buy" → Steam dialog opens
2. Player confirms → Steam processes payment
3. Steam returns transaction ID + receipt
4. Client validates receipt against Steam Web API
5. If valid: grant in-game items, store receipt in purchaseHistory
6. If invalid: reject grant, log incident, surface error to player

Refund handling: - Steam sends webhook → game tracks receipt status - If refund detected: revoke granted items if possible (Crystals reduced; Gems removed; if already spent, balance goes negative — block further purchases until balance balances) - Player notified via in-game mail

3.7 Server / Backend Stack (minimum viable)



Required backend services for v1:

Service	Provider	Purpose
Config Server	Custom (AWS/GCP) or Firebase Remote Config	Hot-config delivery
Telemetry	GameAnalytics / PostHog	Event ingestion
Steam API	Steam Web API	Receipt validation, achievement grants
Crash Reporting	Sentry / Unity Cloud Diagnostics	Stability monitoring
Customer Support	Helpscout / Zendesk	Ticket handling (post-launch)
Leaderboard	Steam Leaderboard API	Tower monthly leaderboard

Latency target: Config fetch < 500ms; telemetry batching < 5 min cadence; receipt validation < 2s.

3.8 Compensation System



For when things go wrong (server outage, balance bug, lost rewards):

1. Operations identifies affected players (could be all, or a segment)
2. Bulk mail with explanation + compensation items
3. Items expire 30 days
4. Player sees clear “Apology Gift” mail entry

Standard compensation packages: - **Minor:** 100 Gems + 1 Standard Ticket - **Medium:** 500 Gems + 1 Limited Ticket + 50 Skill Stones - **Major:** 1,500 Gems + 5 Limited Tickets + 1 Pristine Element Crystal

3.9 News / Patch Notes



Patch notes published via: - In-game news carousel (Home dashboard) - Steam announcement - Website (post-launch)

Each patch tagged with: - Version number - Changelog (player-friendly summary) - Compensation for bugs being fixed (if any) - New content list - Balance change list

4. Formulas

Concept	Formula
Config refresh cadence	every 4 hours
Telemetry batch cadence	every 5 minutes
Compensation tier formula	based on bug severity (Minor/Medium/Major)
Leaderboard server validation	re-simulate combat with stored seed; compare outcome

5. Edge Cases

- **Offline first launch:** use embedded default config until first sync.
- **Telemetry endpoint unreachable:** queue events locally; flush on next sync.
- **Steam Web API down during purchase:** purchase completes (Steam handles it), grant deferred until next sync. Player notified.
- **Two simultaneous IAP attempts from same account:** deduplicated by transaction ID.
- **Compensation mail expires:** items lost. UI showed 7-day warning. Player can re-request via support (manual review).

6. Dependencies

- 05-gacha-and-currencies.md — gacha pull server validation
- 10-tower-of-trials.md — leaderboard validation
- 13-battle-pass.md — BP season transitions via config
- 14-monetization.md — receipt validation
- 15-desktop-window-ux.md — Telemetry opt-out setting
- 17-save-system.md — purchase history, save validation

7. Tuning Knobs

- Config refresh interval (4h)
- Telemetry batch size / cadence
- Anti-cheat sensitivity (passive monitoring thresholds)
- Compensation tier amounts
- Leaderboard re-simulation cadence

8. Acceptance Criteria

1.  Config fetches on launch and re-fetches every 4 hours.
2.  Banner / BP / event schedule changes apply via config without app update.
3.  Telemetry events fire correctly (testable via debug log).
4.  IAP receipts validate against Steam Web API.
5.  Failed IAP validation surfaces clear error.
6.  Save tampering blocks gacha + ladder entries (but allows continued single-player play).
7.  Compensation mail can be sent to all players via ops dashboard (or scripted).
8.  Telemetry respects opt-out setting.

GDD 22 — Boss Design (PLACEHOLDER)

Status: PLACEHOLDER. User is finalizing boss themes / kits separately. This file holds the boss-slot structure and design constraints. Fill in each boss row when designs land.

1. Overview

12 chapter bosses + Tower of Trials phase bosses + Boss Rush gauntlet. Each chapter boss is the **defining encounter** of that chapter — the most memorable moment in the player’s run.

2. Required per boss

When designing each boss, fill in:

- **Name** (e.g. “Embertongue Wyrn”)
- **Title** (e.g. “Tyrant of the Crater”)
- **Chapter** (1-12)
- **Element** ( /  /  / Neutral)
- **Visual theme** (3-5 lines)
- **HP scale** (relative to standard enemies — typically 5-10x)
- **2-4 unique skills** (cooldown, mechanic)
- **Signature mechanic** (the “thing” — e.g. “revives once at 50% HP”, “every 4 turns spawns 2 adds”)
- **Drop table** — promotion material, equipment set hint, first-clear bonus
- **Element resistance** (optional — some bosses are neutral, some have a counter-element)
- **Hell variant** (does the boss gain new mechanics on Hell, or just bigger numbers?)

3. Per-Act element guidance

Per `gdd/09-campaign-structure.md` § 3.2:

Act	Theme	Suggested boss elements
Act 1 — Awakening	Forests, ruins	Mostly Wind + Neutral; final boss could be Wind-heavy
Act 2 — Ashes & Embers	Volcanic	Mostly Fire + Neutral; final boss is Fire
Act 3 — Tides & Frost	Sea + ice	Mostly Water + Neutral; final boss is Water

4. Boss slot stubs (TO FILL)

Act 1 — Awakening

#	Chapter	Boss Name	Element	Theme	Status
1	1	TBD	 Wind / Neutral	Forest guardian / first major foe	 Design pending
2	2	TBD	 Wind	Beast lord	 Design pending
3	3	TBD	Neutral	Fallen knight (story tone shift)	 Design pending
4	4	TBD	 Wind	Wind drake / Act 1 climax	 Design pending

Act 2 — Ashes & Embers

#	Chapter	Boss Name	Element	Theme	Status
5	5	<i>TBD</i>	 Fire	Magma elemental / introduction	 Design pending
6	6	<i>TBD</i>	 Fire	Ember tyrant	 Design pending
7	7	<i>TBD</i>	 Fire	Phoenix scion (signature: revives once)	 Design pending
8	8	<i>TBD</i>	 Fire	Crimson dragon / Act 2 climax	 Design pending

Act 3 — Tides & Frost

#	Chapter	Boss Name	Element	Theme	Status
9	9	<i>TBD</i>	 Water	Tidal beast	 Design pending
10	10	<i>TBD</i>	 Water (ice-themed)	Frost giant	 Design pending
11	11	<i>TBD</i>	 Water	Abyssal kraken	 Design pending
12	12	<i>TBD</i>	 Water	Leviathan / final boss of campaign	 Design pending

5. Common boss patterns to reuse (template library)

When designing bosses, mix-and-match from these proven patterns:

- **Phase shift** — boss has 3 HP phases (100→66%, 66→33%, 33→0%) with different attack patterns
- **Add summoning** — boss summons 2 minions every N turns
- **Damage gate** — boss is shielded until you break its shield within X turns
- **DPS check** — boss enrages at turn N (already in `gdd/01-combat.md` § 3.11)
- **Element shift** — boss changes element each phase (forces team flexibility)

- **Revive mechanic** — boss comes back once at 50% HP (e.g. Phoenix Scion)
- **AoE pulse** — boss does AoE every 3 turns; force player to bring sustain
- **Damage focus** — boss targets the highest-DPS hero in the party
- **Cleanse immunity** — boss debuffs can't be cleansed (forces resistance gear)
- **Heal-block** — heals on the team are reduced by 50% during boss fight

6. Tower of Trials phase bosses

Floors **10 / 20 / 30 / 40 / 50 / 60 / 70 / 80 / 90 / 100** are boss floors. These can reuse campaign bosses with intensified mechanics, OR be Tower-exclusive bosses.

- **Floors 10-50:** chapter boss variants (re-themed)
- **Floors 60-100:** Tower-exclusive bosses (final 5 should feel like raid-level encounters)

Floor 100 boss is THE flagship endgame encounter — should have a unique kit, custom art, custom music.

7. Boss-design constraints

- Must respect the **enrage timer** (`gdd/01-combat.md` § 3.11)
- Must be **scoutable** (`gdd/01-combat.md` § 3.16) — kit visible after first defeat
- Should be **counterable by 2+ team comps** (avoid hard-counter-only bosses)
- Should drop **predictable rewards** (players plan farm runs)
- Hell-mode variants must add **at least one new mechanic** — not just stats

8. Dependencies

- `gdd/01-combat.md` — combat engine, enrage system
- `gdd/09-campaign-structure.md` — chapter context
- `gdd/10-tower-of-trials.md` — Tower phase bosses
- `gdd/12-daily-weekly-loops.md` — Boss Rush uses campaign bosses
- Art direction (TBD) — boss aesthetic guard rails

9. Acceptance Criteria (once filled)

1.  All 12 chapter bosses have name + element + 2-4 skills + signature mechanic
2.  All 10 Tower phase bosses defined
3.  Hell-mode variants documented per boss

4.  Each boss has at least one “fair” counter team and one “ideal” counter team
5.  Drop tables specified

GDD 23 — Art Direction (PLACEHOLDER)

Status: PLACEHOLDER. User is producing art separately and will provide assets later. This file documents the **questions to answer** + the **constraints already implied** by existing design.

1. Overview

Project I’s visual identity is the foundation of the desktop window paradigm. The compact mode is what players see most — it must feel charming and unobtrusive. The expanded mode is where premium art lives.

2. Constraints already implied by existing design

These are NOT decisions yet — they’re inferred from existing hero/system docs. The art director should treat them as starting hypotheses.

Implication	From which doc
Stylized chibi sprites for combat	<code>gdd/15-desktop-window-ux.md</code> — chibi animations referenced
Full splash art for each 5★ hero	Source PDF + hero sheets
Compact window 480×160 px	<code>gdd/15-desktop-window-ux.md</code>
Background size 960×320 (compact-mode strips)	Source PDF asset list
Splash art uses anime-stylized illustration (implied by source PDF descriptions)	Hero sheets
3 elemental color identities (Fire = red/orange, Water = blue/teal, Wind = green/white)	All hero sheets
Element + class icons exist	Source PDF
UI must scale 100% / 125% / 150%	<code>gdd/15-desktop-window-ux.md</code> § 3.7

3. Areas needing definition

3.1 Overall aesthetic

- Anime-stylized? Semi-realistic? Cute-chibi? Painterly?
- Lighting tone — soft & warm, neon & saturated, dark & moody?
- Target reference games for the art lead to look at? (e.g., Honkai Star Rail, AFK Arena, Disgaea, etc.)

3.2 Compact mode visuals

- Pixel art vs. low-resolution rendered sprites vs. high-res sprites downsampled?
- How much animation budget per chibi (frame count, particle density)?
- Background art treatment — parallax layers? Static? Looping ambient?
- Damage number style — flat? Animated? Color-coded by crit?

3.3 Expanded mode visuals

- HD chibi sprites (different from compact) OR same chibis up-rezzed?
- Splash art always visible on character menu? In what frame treatment?
- Skill icons style — flat, embossed, painterly?
- Status effect icons style

3.4 UI visual language

- Button shape — rounded rectangle? Plate-style? Hexagonal?
- Typography — display font (logo / headers), UI font (buttons), body font (descriptions)
- Border treatments — ornamental, minimal, glassmorphic?
- Color palette — primary background, primary accent, gold/legendary glow, error red, success green
- Hero card layout template (must work for all 15+ 5★ + 15 4★ heroes)
- Gacha pull animation aesthetic — crystals, lights, banners?
- Menu transitions — cuts, fades, parallax?

3.5 VFX style

- Particle density — sparse (low-end PC friendly) or rich (showpiece)?
- Element VFX language:
 -  Fire — flames? embers? phoenix imagery? lava?
 -  Water — waves? bubbles? ice crystals? oceanic glow?
 -  Wind — leaves? wind lines? feathers? spirit butterflies?
- Critical hit VFX — screen-flash? screen-shake? slow-mo zoom?

- Ultimate VFX — full-screen takeover? Hero portrait reveal? Skill-banner card?
- Boss appearance VFX — distinct camera punch?

3.6 Iconography

- Skill icons (one per skill, ~24 unique per launch hero × 4 = 60+ icons)
- Element icons (3 — variants for small / large display)
- Class icons (5)
- Status effect icons (~17 from `gdd/01-combat.md` § 3.13)
- Equipment slot icons (7)
- Sub-stat icons (~11)
- Set bonus icons (6)
- Currency icons (Gold, Gems, Crystals, Hero Essence, Hero Shards, Tower Coin, plus material icons)

3.7 Animation principles

- Idle animation style (subtle breathing, looping action, dynamic pose?)
- Skill animation timing per skill type (basic / skill 1 / skill 2 / ultimate)
- Ultimate “camera moment” — Y/N?
- Death animation poetic vs. functional?

3.8 Branding / marketing assets

- Game logo
- Steam capsule art
- Splash screen art (game launches with...)
- Trailer style guide
- Social media tile templates

4. Asset budget guidelines

(Recommended targets — art director to confirm)

Asset type	Per hero	For 15 5★	For 15 4★
Splash art	1 (3000×4000 px reference)	15	15
Hero card	1 (600×900 px)	15	15
Chibi idle sheet	1 (4 frames)	15	15
Chibi attack sheet	1 (5-6 frames)	15	15
Chibi skill 1 sheet	1 (5-6 frames)	15	15
Chibi skill 2 sheet	1 (5-6 frames)	15	15
Chibi ultimate sheet	1 (6 frames)	15	15
Chibi death sheet	1 (5 frames)	15	15
Ultimate art / banner	1	15	optional for 4★
Skill icons	4 (basic + 3 skills)	60	60

Total per-hero art: ~7-8 sheets + 1 splash + 1 card + icons.

5. Open questions for the art director

Before art production begins:

1. Reference games? Three art-style references the team agrees represents the target.
2. Style consistency check — chibi style vs. splash style — should they look “by the same artist”?
3. Color palette per element — exact hex values?
4. Logo / brand identity — does Project I have a final logo by launch?
5. Compact-mode visual identity — is the compact window itself a styled “frame” with skin choices? (See `gdd/14-monetization.md` — compact window themes as a cosmetic item.)

6. Dependencies

- `gdd/15-desktop-window-ux.md` — compact + expanded sizes and constraints
- `gdd/18-audio-direction.md` — pairs with visual identity for tone
- Every hero sheet — visual identity per character
- `gdd/22-boss-design.md` — boss visual theme guard rails (also TBD)

7. Acceptance Criteria (once filled)

1.  Art Bible document exists with style references and core palettes
2.  Reference reels assembled (chibi, splash, UI, VFX)
3.  All asset templates locked (hero card, skill icon, status icon, etc.)
4.  Pipeline / file format standards documented
5.  Cosmetic skin pipeline plan defined (for post-launch BP rewards)

GDD 24 — Day-1 Launch Content (Lockdown)

Locks every piece of content shipping on Day 1 of Project I's launch. Pulls together decisions scattered across the other 23 GDDs and 15 hero sheets into a single content-manifest reference.

1. Overview

Day 1 = the moment the game goes live on Steam. Players who download the game from the Steam page get **everything in this document** and nothing more.

Post-launch live-ops adds banners, events, balance patches via hot-config + small client patches.

2. Hero Roster on Day 1

5★ heroes available (10 of 15)

Hero	Element	Class	Banner
Lyria	 Wind	Archer	Tutorial-guaranteed starter — every player owns her
Valdrik	 Fire	Tank	Standard
Emberia	 Fire	Mage	Standard
Kaida	 Fire	Assassin	Standard
Glacier	 Water	Tank	Standard
Marina	 Water	Support	Standard
Neria	 Water	Mage	Standard
Sylphie	 Wind	Support	Standard
Kael	 Wind	Assassin	Standard
Auron	 Water	Archer	Launch Limited Banner (21 days)

Not in Day-1 roster (5 future banner heroes, post-launch release)

- Ragnar — Fire Archer
- Akari — Fire Support
- Seraphina — Water Assassin
- Mei Lin — Wind Tank
- Ayame — Wind Mage

4★ heroes

15 designs **TBD** — user is producing separately. 4★ heroes drop from Epic gacha tier, Rare/Epic eggs, and Expedition rewards.

3. Banner Lineup — Day 1

Standard Banner (permanent, live at launch)

- 9 heroes (see above), permanent rotation
- Rates: Legendary 0.6% / Epic 7.2% / Rare 92.2%
- Hard pity at pull 50, no soft pity
- Standard Selector unlocks at 210 cumulative pulls
- Free Daily Pull active
- Wishlist (3 picks) active

Launch Limited Banner — Auron's Frostbound Dawn (live at launch)

- **Hero:** Auron (Water Archer)
- **Duration:** 21 days from launch
- **Featured Epics:** TBD — pick 2-3 from the Epic pool to rate-up (data table)
- **Bundles available:**
 - Auron Starter Bundle (\$14.99): 800 Crystals + 10 Limited Tickets + 50 Skill Stones + 1 Reroll Stone
 - Auron Power Bundle (\$29.99): 1,800 Crystals + 30 Limited Tickets + 1 Apex Catalyst + 10 Reroll Stones + 1 Pristine Element Crystal
- Pity carries over to next Limited Banner

Beginner Step-Up Banner (live at launch, account-gated)

- Active for accounts ≤ 14 days old
- 3 steps:
 - Step 1: 10 pulls @ 1,200 Gems — guaranteed Epic+ in batch

- Step 2: 10 pulls @ 1,400 Gems — guaranteed Legendary in batch (from Standard pool)
- Step 3: 10 pulls @ 1,600 Gems — guaranteed Standard Selector unlock + 1 free Standard 5★ pick
- Isolated pity counter

4. Battle Pass — Season 1

Configuration

- **Season length:** 60 days
- **Total tiers:** 50
- **XP per tier:** 1,500 (75,000 total)
- **Premium price:** \$9.99
- **Premium-Plus price:** \$19.99 (+10 instant tier skips + exclusive title/avatar/emote)
- **Mid-season tier purchase:** \$0.99/tier (max 5)
- **End-of-season grace:** 7 days to claim missed rewards

Free Track full table (locked for Season 1)

Tier	Reward
1	100 Gems
2	1 Common Egg
3	200 Gold × 50 (10k)
4	5 Skill Stones
5	1 Common Egg + 100 Gems
6	1 Sacred Pebble × 5
7	100 Small EXP Crystals
8	200 Gems
9	1 Common Egg
10	1 Standard Summon Ticket
11	5 Skill Stones
12	100 Gems
13	1 Sacred Stone
14	5 Medium EXP Crystals
15	50 Skill Stones
16	1 Uncommon Egg
17	100 Gems
18	1 Reroll Stone
19	1 Sacred Stone
20	1 Sacred Stone × 2
21	5 Medium EXP Crystals
22	100 Gems
23	1 Common Egg
24	50 Skill Stones
25	40 Free Inventory Slots
26	1 Reroll Stone

Tier	Reward
27	200 Gems
28	5 Sacred Pebbles
29	1 Uncommon Egg
30	1 Standard Summon Ticket
31	100 Gems
32	5 Sacred Stones
33	1 Common Egg
34	100 Skill Stones
35	100 Gems
36	1 Reroll Stone
37	3 Medium EXP Crystals
38	1 Sacred Stone
39	100 Gems
40	1 Rare Egg
41	50 Skill Stones
42	1 Sacred Stone
43	200 Gems
44	1 Uncommon Egg
45	200 Gems + 1 Reroll Stone
46	50 Skill Stones
47	1 Rare Egg
48	200 Gems
49	1 Sacred Stone
50	1 free Standard 5★ Hero pick from Curated 6 (see gdd/12-daily-weekly-loops.md)

Free track totals (approx): ~2,800 Gems + 2 Standard Tickets + 4 Reroll Stones + 1 free 5★ pick + lots of progression materials.

Premium Track full table (Tier 1-50, in addition to free)

Tier	Reward
1	200 Crystals + Premium Badge
2	1 Reroll Stone
3	5 Sacred Stones
4	100 Skill Stones
5	1 Limited Summon Ticket
6	200 Crystals
7	1 Pristine Element Crystal (random)
8	1 Reroll Stone
9	100 Gems
10	1 Pristine Element Crystal (random)
11	200 Crystals
12	50 Skill Stones
13	1 Reroll Stone
14	1 Sacred Crystal
15	5 Reroll Stones
16	200 Crystals
17	100 Skill Stones
18	1 Pristine Element Crystal (random)
19	1 Reroll Stone
20	200 Crystals
21	1 Sacred Crystal
22	50 Skill Stones
23	1 Reroll Stone
24	200 Crystals
25	1 Apex Catalyst
26	100 Skill Stones

Tier	Reward
27	1 Sacred Crystal
28	1 Reroll Stone
29	200 Crystals
30	1 Limited Summon Ticket
31	1 Pristine Element Crystal (chosen element)
32	50 Skill Stones
33	200 Crystals
34	1 Reroll Stone
35	1 Mythic gear ticket
36	100 Skill Stones
37	1 Pristine Element Crystal (chosen)
38	1 Reroll Stone
39	200 Crystals
40	BP-exclusive Cosmetic Skin (Season 1 design TBD — applies to one Standard hero)
41	1 Sacred Crystal × 2
42	100 Skill Stones
43	200 Crystals
44	1 Reroll Stone
45	5 Limited Summon Tickets
46	1 Pristine Element Crystal (chosen)
47	100 Skill Stones
48	200 Crystals
49	1 Reroll Stone
50	1 BP-exclusive Avatar Frame + Animated Title + 500 Crystals

Premium track value (additional to free): ~2,000+ Crystals, 2 Limited Tickets + 5 Limited Tickets at tier 45 = 7 Limited Tickets, 1 Apex Catalyst, 1 Mythic ticket, BP-exclusive skin, cosmetic frame + title. Easily exceeds the \$9.99 buy-in.

Premium-Plus exclusives

- +10 tier skips
- Season 1 exclusive title
- Animated avatar frame (premium-tier exclusive)
- Premium emote pack (3 emotes)
- 500 bonus Crystals at season start
- 1 free Pristine Element Crystal at season start

5. Tower of Trials — Season 1

Configuration

- 100 floors
- Monthly reset (resets first of each calendar month at 00:00 UTC)
- Hero lock between floors except every 10th floor (Rest Floor)
- Tower Coin earned per floor

Floor modifier schedule (key floors locked, filler floors generic)

Floor band	Modifier
1-9	None (tutorial difficulty)
10 (Boss)	Phase Shift — boss has 2 HP phases
11-14	None
15	Burning Ground — all combatants take 5% max HP burn per turn
16-19	None
20 (Boss)	Add Summoning — 2 minions every 4 turns
21-27	None
28	Reverse Element — element advantage becomes disadvantage
29	None
30 (Boss)	DPS Check — boss enrages at turn 20
31-34	None
35	One-Hero Force — only one hero acts per turn (cycles)
36-39	None
40 (Boss)	Phase Shift — 3 HP phases
41-46	None
47	Ultimate Lockout — Ultimates disabled
48-49	None
50 (Boss)	Phase Shift — 3 phases, each phase = different element
51-54	None
55	Crit-Only — only critical hits deal damage; non-crits deal 1
56-59	None
60 (Boss)	Add Summoning + Phase Shift
61-63	None
64	

Floor band	Modifier
	Double Speed — all combatants act every turn (no gauge)
65-69	None
70 (Boss)	Apex Catalyst milestone reward + Phase Shift + Add Summoning
71-72	None
73	Counter Field — all single-target attacks have 50% counter chance
74-79	None
80 (Boss)	Phase Shift — 3 element shifts
81-89	Random combinations of prior modifiers (data table)
90 (Boss)	All prior modifiers compressed into one fight
91-99	Elite filler (high HP/ATK, no modifiers)
100 (FINAL BOSS)	Tower Apex — unique kit + custom art + custom music (final flagship)

Tower Shop Season 1 stock (resets monthly)

(per `gdd/10-tower-of-trials.md` § 3.5, updated for Apex tightening)

Item	Tower Coin	Monthly Limit
Sacred Crystal	50	5
Apex Catalyst	2,000	1 per Tower season (monthly)
Reroll Stone	100	10
Pristine Element Crystal (chosen element)	200	5
Mythic gear ticket	500	1
Tower exclusive cosmetic (rotating)	1,000	1
Skill Stones × 100	50	5
10 × Hero Shards (chosen Standard hero)	600	3

6. Hero Trials — Week 1

- **Featured Hero: Lyria** (every player owns her — best onboarding for the system)
- **Trial Tiers:** Bronze / Silver / Gold
- **Rotation:** week N's featured hero cycles through the 10 available heroes ($(N-1) \% 10$)

7. Boss Rush — Week 1

- Available on **Normal difficulty only** at launch (Hard / Nightmare unlock as players progress)
- 12 chapter bosses (TBD per `gdd/22-boss-design.md`) fought in sequence
- Resets Monday 04:00 local
- Rewards per `gdd/12-daily-weekly-loops.md`

8. 14-Day Login Bonus (Launch Promo)

Stronger than the standard 7-day login — used as a launch promo:

Day	Reward
1	300 Gems (launch boost) + Welcome Mail
2	1 Common Egg + 500 Gold
3	10 Skill Stones
4	1 Standard Summon Ticket
5	1 Uncommon Egg
6	20 Reroll Stones
7	1 free Standard 5★ Hero pick from Curated 6
8	200 Gems
9	1 Common Egg
10	1 Standard Summon Ticket
11	50 Skill Stones
12	1 Rare Egg
13	200 Gems
14	1 Apex Catalyst (first-Apex onboarding gift)

After Day 14, the 7-day “Veteran” cycle takes over (per `gdd/12-daily-weekly-loops.md` § 3.3).

9. Daily Missions Config (Day 1)

All 8 daily missions per `gdd/12-daily-weekly-loops.md` § 3.1 active:

- Clear 5 stages — 100 pts
- Spend 1,000 Gold — 50 pts
- Hatch 1 egg — 50 pts
- Collect Shrine 1× — 50 pts
- Dispatch 1 expedition — 50 pts
- Perform 1 gacha pull — 100 pts
- Win a battle in <8 turns — 100 pts
- Upgrade 1 skill — 75 pts

Thresholds at 200 / 400 / 575 points unchanged.

10. Weekly Missions Config (Day 1)

All 6 weekly missions per `gdd/12-daily-weekly-loops.md` § 3.2 active. Tower / Boss Rush mission tiers gated by progression as documented.

11. Launch Achievements (50 starter)

Of the ~150 total achievement target, **50 launch achievements** ship Day 1:

Collection (10): Own 3 heroes / 5 / 7 / 9 / Own Auron / Awaken any hero to 1 / 3 / 6 / Own all 9 Standard / Own 4★ TBD

Combat (10): Win 10 / 100 / 1000 battles / Deal 1M damage in one hit / Crit 100× / Survive 50 turns / Kill 100 / 1000 / 10000 enemies / Win without losing any hero

Progression (10): Account L5 / L10 / L20 / L40 / Clear Chapter 1 / 6 / 12 / Clear Chapter 1 Hard / Promote first 5★ / Reach Skill Rank 10

Tower (5): Reach floor 10 / 25 / 50 / 75 / 100

Boss Rush (5): Complete Boss Rush Normal / Hard / Nightmare / Without deaths / Without using ultimates

Gacha (5): First pull / 100 pulls / 500 pulls / Pull Auron during launch banner / Complete Step-Up Banner

Codex (5): Discover 25% / 50% / 75% / 100% of enemies / Discover all unlocked heroes

Remaining 100 achievements add post-launch with content drops.

12. Idle Content

Players can idle on **any cleared stage**. Chapter 1 stages are available immediately post-tutorial. Drop tables per stage are configured (production-phase data work).

13. Expedition Mission Pool (Day 1)

All ~30 missions from `gdd/11-world-map-expedition.md` § 3.2 available. Refresh cycle every 6h begins at launch UTC.

14. Shop (Day 1 IAP catalog)

Active per `gdd/14-monetization.md` :

- **Beginner Pack** (\$4.99) — account $\leq$ 14 days only
- **Monthly Card** (\$4.99)
- **Crystal packs** — all 6 tiers (\$0.99 to \$99.99)
- **Battle Pass** (\$9.99 / \$19.99)
- **Auron Starter Bundle** (\$14.99)
- **Auron Power Bundle** (\$29.99)
- **Dupe Shop** (Hero Essence currency, full stock)
- **Tower Shop** (Tower Coin)

15. Launch Compensation Mail

Every player who logs in within the first 7 days receives:

Subject: Welcome to Project I!
 Gift: 1,000 Gems, 1 Common Egg, 50 Skill Stones, 5 Reroll Stones

16. Day-1 Telemetry Targets

Per `gdd/21-liveops-and-telemetry.md` :

-  Tutorial funnel events firing
-  First-pull / first-purchase tracked
-  D1 / D7 retention markers active
-  Crash reporting (Sentry) live
-  IAP receipt validation working

17. Day-1 Out-of-Scope (deferred to post-launch)

-  Events
-  Limited cosmetic shop rotation (beyond Tower exclusive)
-  4★ heroes (TBD designs)
-  Mac client (Windows only at launch)
-  Localization (EN only at launch)

18. Acceptance Criteria

1.  All 9 Standard heroes pullable on Standard Banner Day 1
2.  Auron available on Limited Banner Day 1
3.  Beginner Step-Up Banner active for new accounts
4.  BP Season 1 all 50 tiers configured (this doc)
5.  Tower Season 1 all 100 floors configured (this doc)
6.  14-Day Login Bonus configured (this doc)
7.  50 launch achievements seeded
8.  Daily / Weekly missions active
9.  Expedition pool rotating from Day 1
10.  All Day-1 IAP products in Steam catalog
11.  Telemetry pipeline live

GDD 25 — Enemy AI

1. Overview

Enemy AI scales with **enemy level** — higher-level enemies make smarter decisions. This creates a natural difficulty curve where Chapter 1 mobs feel mindless but Chapter 12 / Tower / Boss enemies play strategically. AI tiers are deterministic and testable.

2. Player Fantasy

Early enemies are fodder. By Chapter 5, they start using skills. By Chapter 10, they're targeting my squishies. Bosses play like a real opponent.

3. Detailed Rules

3.1 AI Tiers

Enemy AI is gated by **AI Tier**, which is set per-enemy via spawn data (not strictly by level — designers can tag a “smart” enemy in Chapter 2 if they want a difficulty spike).

By default, AI Tier is determined by enemy level:

Enemy Level	Default AI Tier	Behavior name
1-10	Tier 0	Mindless
11-25	Tier 1	Reactive
26-40	Tier 2	Tactical
41-55	Tier 3	Strategic
56+	Tier 4	Optimal

Bosses, Hell-mode enemies, and Tower floor 50+ enemies are always **Tier 4** regardless of level.

3.2 Tier 0 — Mindless

Used for very early-game enemies (Chapter 1 Normal stages).

Action selection: - 100% basic attack on cooldown - Skills never used (treat skills as unavailable) - No Ultimate (unless boss)

Targeting: - Random alive hero

Personality: “Just charging forward.” Predictable, easy to bait.

3.3 Tier 1 — Reactive

Mid-Chapter 1 and Chapter 2 enemies.

Action selection: - 70% basic attack, 30% Skill 1 (if off-cooldown) - Skill 2 never used - Ultimate never used unless HP < 30% (desperate move)

Targeting: - Random alive hero (basic attack) - Lowest current HP hero (skill 1 — finishing instinct)

Personality: “They’ve started swinging back.” Slight finishing-attack threat.

3.4 Tier 2 — Tactical

Chapters 3-7 normal enemies.

Action selection: - 50% basic attack - 30% Skill 1 (if off-cooldown) - 15% Skill 2 (if off-cooldown) - 5% Ultimate (if off-cooldown AND in last enemy alive — desperate)

Targeting: - Damage skills target **lowest HP%** alive hero - Buff skills (if enemy is support) target lowest-HP ally - AoE skills cast when 2+ heroes alive

Personality: “They’re picking off my back line.” Players need taunt or shield protection.

3.5 Tier 3 — Strategic

Chapters 8-12 normal, Hard mode enemies.

Action selection: - 35% basic attack - 25% Skill 1 - 25% Skill 2 - 15% Ultimate (if off-cooldown AND meaningful target available)

Targeting: - Damage targets **lowest effective DEF** hero first (squishies) - AoE skills cast when **2+ heroes have any active buff** (priority: strip buffs) - Debuff skills target the highest-ATK hero (cripple DPS) - Single-target Ultimates target the **highest-threat** hero (highest ATK × Speed) - Healing skills (if enemy is support-tagged) cast when ally HP < 50%

Personality: “They know what they’re doing.” Players need to think about formation + counter-comp.

3.6 Tier 4 — Optimal

Nightmare, Hell, Tower floor 50+, all bosses.

Action selection: - Action priority is **fully evaluated each turn** using a utility-based decision tree: 1. If Ultimate is off-cooldown AND (any hero is below 30% HP OR boss-tagged condition met): cast Ultimate 2. Else if Skill 2 is off-cooldown AND its expected damage/effect value is high (heuristic): cast Skill 2 3. Else if Skill 1 is off-cooldown: cast Skill 1 4. Else: basic attack

Targeting: - Damage targets **lowest effective HP** (current HP × DEF factor) — the “easiest kill” - AoE damage cast when **2+ heroes alive AND total expected damage > single-target option** - Element-mod aware: prefer striking a target with off-element penalty (where enemy has advantage) - Defensive plays: if a hero is dropping low and the enemy has a Heal-block / cleanse-resist style move, that move is prioritized - Cleave through Tank Taunts: if Tier-4 enemy has DEF-ignore skills, those are prioritized against the Taunting Tank

Personality: “They’re playing the game with me.” Top-tier content. Players need optimized teams + good gear.

3.7 Boss Special Rules

In addition to Tier 4 AI, **bosses** have:

- **Enrage timer** per `gdd/01-combat.md` § 3.11 — at turn N, boss buffs itself massively
- **Phase shift triggers** (per-boss design) — at HP thresholds, boss changes attack pattern
- **Signature mechanic** (per-boss design) — see `gdd/22-boss-design.md`

These are layered ON TOP OF Tier 4 AI. The base decision-making is still utility-driven.

3.8 Status Effect AI Behavior

Enemies aware of their own status effects:

If enemy has...	They will...
Burn / Regen (DoT on themselves)	Same actions; doesn't change AI
Stun / Freeze	Skip turn (forced)
Slow	Lower Speed → fewer turns in time
DEF Down (themselves)	Tier 3+: if a healer/buffer is available, target it first to cleanse
Shield breaking soon	Tier 4: prioritize protective skills
Crowd Control Resistance (themselves)	Tier 4: more aggressive plays (they know CC won't land)

3.9 AI in Idle / Auto-mode play

When the player is in **idle mode** (`gdd/06-idle-and-shrine.md`), the auto-AI runs on a deterministic seed. Enemy AI runs the same tier rules — no “easier” idle AI. This keeps idle results consistent with manual play of the same stage.

3.10 Tower & Boss Rush AI

- Tower floor 1-49: AI tier scales by floor (similar to level scaling)
- Tower floor 50+: always Tier 4
- All Tower bosses (every 10 floors): Tier 4 + boss rules
- Boss Rush: always Tier 4

3.11 Hell-Mode AI Tweaks

Hell mode applies **AI Tier +1** override: - Tier 0 → Tier 1 - Tier 1 → Tier 2 - ... - Tier 4 → Tier 4 + Hell-only modifier: enemy spends turn looking at *next* turn’s action and pre-positions accordingly (slight prediction)

3.12 AI Variability (anti-stale-meta protection)

To prevent players from solving “always cast skill 1 then skill 2 then ultimate” loops:

- Tier 2+ rolls a **5% chance** to deviate from optimal play and pick the 2nd-best move instead
- Tier 3+ rolls a **3% chance** to deviate
- Tier 4 rolls a **2% chance** to deviate

This variability is **seeded deterministically per-battle** (so a save replay produces identical results — see `gdd/17-save-system.md`).

4. Formulas

Concept	Formula
Default AI Tier from level	<code>clamp(floor((enemyLevel - 1) / 12), 0, 4)</code>
Deviation roll	<code>roll < deviationChance[tier]</code>
Targeting utility (Tier 4)	<code>damageValue × (1 + elementBonus) × (1 / max(1, defMitigationOf(target)))</code>
AoE cast threshold	<code>count(aliveTargets) ≥ 2 AND aoeMultiplier × count ≥ singleTargetMultiplier × 1.2</code>

5. Edge Cases

- **All heroes have identical stats** (training-room scenario) — targeting falls through to first-in-position
- **Tier 4 enemy with no off-cooldown skill** — basic attack the lowest HP target
- **All heroes Taunted by one Tank** — Tier 4 with DEF-ignore: still hits Tank but uses DEF-ignore skill; Tier 3 without DEF-ignore: still must hit the Tank (Taunt forces it)
- **Enemy is Stunned/Frozen on its turn** — skip turn entirely; cooldowns DO tick down
- **Enemy is Taunted (rare — debuff on enemy)** — must target the Taunter
- **Heal skill from enemy with no allies** — skip (skill not selectable)

6. Dependencies

- `gdd/01-combat.md` — combat engine; turn order; status effects
- `gdd/09-campaign-structure.md` — enemy level scaling per stage/difficulty
- `gdd/10-tower-of-trials.md` — Tower difficulty progression
- `gdd/17-save-system.md` — deterministic seed for AI variability
- `gdd/22-boss-design.md` — boss-specific overrides

7. Tuning Knobs

- AI Tier breakpoints by level (currently `floor((level-1)/12)`)
- Deviation chances per tier (5% / 3% / 2%)
- Utility weights in Tier 4 targeting
- Hell-mode tier override (currently +1)
- Action probability tables per tier

8. Acceptance Criteria

- ✓ Tier 0 enemies always basic-attack a random hero (testable)
- ✓ Tier 2+ enemies target the lowest-HP hero with damage skills (testable)
- ✓ Tier 4 enemies cast Ultimates only when meaningful (not on lone trash mob)
- ✓ Bosses always run Tier 4 + boss rules regardless of level
- ✓ Deviation roll is deterministic (same seed produces same deviation)
- ✓ Hell mode applies Tier +1 to all enemies in the stage
- ✓ Stunned enemies skip turn but tick cooldowns (testable)
- ✓ Tier 3+ Mages prioritize highest-ATK target with debuffs

9.  Healing enemies cleanse ally DEF Down at Tier 3+ (when available)

Hero Sheets

Hero Roster Index

Total launch roster: 15 5★ heroes + 15 4★ heroes (designs TBD). All 5★ heroes have full design sheets; 4★ heroes are documented at a placeholder level until user provides designs.

✓ Available at Launch (10 heroes)

Standard Banner Pool (9)

These are available in the Standard banner from Day 1 and form the Standard Selector pool (210-pull guaranteed pick) and Wishlist pool.

Hero	Element	Class	Weapon
Valdrik	 Fire	Tank	Warhammer & Tower Shield
Emberia	 Fire	Mage	Crimson Ember Staff
Kaida	 Fire	Assassin	Chained Twin Sickles
Glacier	 Water	Tank	Winterfang Gunlance
Marina	 Water	Support	Oceanheart Harp
Neria	 Water	Mage	Astral Grimoire
Lyria	 Wind	Archer	Zephyrvine Longbow
Sylphie	 Wind	Support	Verdant Spirit Staff
Kael	 Wind	Assassin	Tempest Fangs

Launch Limited Banner (1)

Available exclusively on the first paid Limited Banner — live at launch, runs 21 days.

Hero	Element	Class	Weapon
Auron	 Water	Archer	Frostbound Bracer

NEW Future Limited Banners (5 heroes, released post-launch)

Released one at a time over the live-ops calendar. Design drafts complete; polish pass pending.

Hero	Element	Class	Weapon	Suggested Release
Ragnar	 Fire	Archer	Inferno Recurve Bow	Banner #2
Akari	 Fire	Support	Phoenix Lantern	Banner #3
Seraphina	 Water	Assassin	Tidal Scythe	Banner #4
Mei Lin	 Wind	Tank	Jade Tower Shield	Banner #5
Ayame	 Wind	Mage	Tempest Ritual Fan	Banner #6

Release order is a recommendation — final live-ops schedule is set by `gdd/21-liveops-and-telemetry.md`.

4★ Roster (15 heroes — designs TBD)

See `4-stars-TBD.md` for the placeholder spec. User is producing 4★ designs separately.

Master Class × Element Matrix

Element	Tank	Assassin	Archer	Mage	Support
 Fire	Valdrik	Kaida	Ragnar	Emberia	Akari
 Water	Glacier	Seraphina	Auron	Neria	Marina
 Wind	Mei Lin	Kael	Lyria	Ayame	Sylphie

Bold = Standard pool (available at launch). Total = 15 unique element×class combinations covered (1 hero per cell).

Hero Design Conventions

All hero sheets follow this template (approved by user):

1. **Header table** — element, class, role, weapon, rarity, banner pool, status
2. **Appearance** — visual identity, color palette
3. **Stat Grades** — HP/ATK/DEF/SPD/CR/CD/ACC (S/A/B/C/D scale)
4. **Passive** — name, mechanics, max stacks
5. **Basic Attack** — name, effects
6. **Skill 1** — name, cooldown, multiplier, effects
7. **Skill 2** — name, cooldown, multiplier, effects
8. **Ultimate** — name, cooldown, multiplier, effects
9. **Character Identity** — strengths / weaknesses
10. **Chibi Animations** — idle / attack / skill 1 / skill 2 / ultimate / death
11. **Team Role** — description
12. **Recommended Partners** — list
13. **Auto-AI Defaults** — skill 1 / skill 2 / ultimate priority rules
14. **Quote**
15. **Status notes** (WIP heroes only)

See `../gdd/01-combat.md` for combat mechanics referenced in skill descriptions, and `../gdd/02-elements-and-classes.md` for element + class definitions.

Valdrik — The Flame Bulwark

Field	Value
Element	 Fire
Class	 Tank
Role	Main Tank / Team Protector
Weapon	Warhammer & Tower Shield
Rarity	★★★★★
Banner Pool	Standard (available at launch)

Appearance

- Veteran knight aesthetic — battle-worn ash-gray plate with smoldering fire runes
- Heavy gauntlets, full helmet with a tonsure visor
- Carries a massive warhammer + tower shield
- Tower shield engraved with phoenix sigil
- Smoldering ember particles drift from joints and weapon
- Color palette: ash gray, ember orange, gold accents, deep crimson

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
S	C	S	D	D	D	B

Passive — Veteran’s Resolve

Whenever Valdrik takes damage, gain 1 **Resolve** stack (max 5).

Each stack grants: - +4% DEF - +4% Shield Strength

At maximum stacks: - Gains **Crowd Control Resistance** (immune to Freeze, Stun, Taunt) - **Taunts applied by Valdrik cannot be removed** by enemy effects

Basic Attack — Ashen Hammer

- **Cooldown:** None
- Valdrik swings his massive warhammer in a heavy arc infused with smoldering fire.
- **Effects:**
 - Single Target Damage
 - 20% chance to inflict **Burn** (2 turns)
 - **Generates 1 Resolve stack**

Skill 1 — Blazing Bastion

- **Cooldown:** 2 turns | **Multiplier:** 1.5×
- Valdrik raises his tower shield and reinforces his allies with protective flames.
- **Effects:**
 - All allies gain **DEF Up** (+30%, 2 turns)
 - Grants a **Small Shield** to all allies based on Valdrik's HP (~10% of his max HP)

Skill 2 — Scorchguard Slam

- **Cooldown:** 3 turns | **Multiplier:** 1.8× (AoE)
- Valdrik charges forward and slams his massive tower shield into the enemy formation.
- **Effects:**
 - AoE Damage
 - **50% chance to Taunt** all enemies (2 turns)
 - Applies **Burn** to all hit enemies
 - **Generates 1 Resolve stack**

Ultimate — Inferno Fortress

- **Cooldown:** 5 turns | **Multiplier:** 2.6× (AoE damage component)
- Valdrik plants his tower shield into the battlefield and unleashes a blazing defensive fortress.
- **Effects:**
 - Massive **DEF Up** to all allies (+50%, 2 turns)
 - **Taunts all enemies** (2 turns)
 - Grants **Shield** to all allies based on Valdrik's HP (~25% of his max HP)
 - **Burns** all enemies (2 turns)

Character Identity

Strengths: - Highest DEF in the launch roster - Highest HP in the launch roster - Exceptional team protection
 - Reliable Burn application - Excellent boss tank - Powerful shield generation - Strong crowd control

Weaknesses: - Very slow - Low damage output - Limited offensive pressure - Vulnerable to DEF-ignoring effects - Relies on allies for finishing damage

Chibi Animations

- **Idle:** warhammer resting on shoulder, tower shield planted beside him, small embers drifting upward
- **Attack:** shield shove → heavy warhammer swing → returns to defensive stance
- **Skill 1 (Blazing Bastion):** raises shield overhead → protective fire barrier surrounds allies → ember particles burst outward
- **Skill 2 (Scorchguard Slam):** charges forward → shield slam impact → shockwave erupts → burning ground effect appears
- **Ultimate (Inferno Fortress):** plants tower shield into the ground → massive fire fortress appears behind him → protective barrier expands outward → burning runes circle the battlefield → fire erupts from fortress walls
- **Death:** warhammer drops → falls to one knee → shield crashes beside him → slowly collapses

Team Role

- Main Tank
- Primary Defender
- Frontline Protector
- Shield Specialist
- Defensive Anchor

Valdrik stands at the front of every battle, enduring every blow so his allies do not have to. His warhammer crushes those who threaten his companions, while his tower shield becomes an unbreakable wall between danger and those he swore to protect.

Recommended Partners

- **Emberia** — Valdrik tanks while Emberia melts the field with Burn stacks
- **Kaida** — Kaida ignores DEF; pairs with Valdrik's Taunt setup
- **Sylphie / Marina** — keeps Valdrik topped up while Resolve stacks build

- **Lyria** — back-row safety while Valdrik soaks front-row pressure
- **Akari** — buffs + Revival Flame insurance

Auto-AI Defaults

- **Skill 1 (Blazing Bastion):** Cast on cooldown when ally average HP < 80% OR team has no active shield
- **Skill 2 (Scorchguard Slam):** Cast on cooldown when 2+ enemies alive
- **Ultimate (Inferno Fortress):** Cast immediately at start of boss waves OR when any ally is below 40% HP

Quote

“A shield is not carried for yourself. It is carried for everyone standing behind you.”

Emberia — The Crimson Sorceress

Field	Value
Element	 Fire
Class	 Mage
Role	AoE DPS / Burn Specialist
Weapon	Crimson Ember Staff (legendary fire-mana focus with floating ember crystal)
Rarity	★★★★★
Banner Pool	Standard (available at launch)

Appearance

- Elegant red-haired sorceress with crimson eyes
- Sleeveless deep red robes with gold filigree, slit-leg battle dress
- Bare midriff with golden ember band
- Crimson ember staff floats beside her, surrounded by glowing fire rings
- Small ember particles continuously orbit her
- Color palette: deep red, crimson, gold, ember orange, black highlights

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
D	S	D	B	A	S	A

Passive — Crimson Dominion

Whenever Emberia applies **Burn** to an enemy, gain 1 **Crimson Flame** stack (max 5).

Each stack grants: - +5% ATK - +3% Burn damage dealt

At max stacks: Burn effects ignore 20% enemy DEF.

Basic Attack — Ember Bolt

- **Cooldown:** None
- Emberia launches a concentrated sphere of magical fire.
- **Effects:** Single Target Damage; 30% chance to inflict **Burn** (2 turns).

Skill 1 — Scarlet Nova

- **Cooldown:** 2 turns | **Multiplier:** 1.8× (AoE)
- Emberia conjures a blazing crimson explosion beneath all enemies.
- **Effects:** AoE Damage; applies **Burn** (2 turns); 25% chance per enemy for **ATK Down**.

Skill 2 — Infernal Bloom

- **Cooldown:** 3 turns | **Multiplier:** 1.8× (AoE)
- Crimson flames erupt across the battlefield in the shape of blooming fire flowers.
- **Effects:** AoE Damage; extends existing **Burn** duration by 1 turn; bonus damage to Burned enemies (+30%).

Ultimate — Crimson Cataclysm

- **Cooldown:** 5 turns | **Multiplier:** 2.6× (AoE)
- A colossal storm of magical fire engulfs the battlefield.
- **Effects:** Massive AoE Damage; applies **Burn** to all enemies; bonus damage for each active Burn effect (+15% per Burn stack across team).

Character Identity

Strengths: Highest AoE damage in Fire faction; exceptional Burn scaling; excellent wave clear; powerful against large groups.

Weaknesses: Low HP/DEF; vulnerable to assassins; relies on Burn setup; limited defensive utility.

Chibi Animations

- **Idle:** holds Crimson Ember Staff, ember particles, small magic circle below feet, fire rings orbit

- **Attack:** staff points forward → fire orb launches → ember trail follows
- **Skill 1:** raises staff overhead → crimson magic circle → fire explosion outward → burning petals
- **Skill 2:** multiple fire flowers bloom → spiraling flames → expanding fire rings
- **Ultimate:** massive crimson magic circle → giant firestorm → burning petals rain → battlefield consumed
- **Death:** staff slips from grasp → fire rings shatter → embers fade → collapses gracefully

Team Role

Primary AoE damage dealer. Burn spreader. Stacks Crimson Flame fast — by turn 5 she's deleting enemy teams.

Recommended Partners

- **Valdrik** — applies Burn from Taunts to start her stacks
- **Akari** — buffs + sustain
- **Kaida** — Kaida amplifies Burn damage via Phantom Pursuit
- **Sylphie** — sustains her back-line position

Auto-AI Defaults

- **Skill 1:** Cast on cooldown when 2+ enemies alive
- **Skill 2:** Cast when at least 1 enemy is Burned
- **Ultimate:** Cast on cooldown when 2+ enemies are Burned OR boss-tagged enemy alive

Quote

"Where I dance, ash follows."

Kaida — The Infernal Reaper

Field	Value
Element	 Fire
Class	 Assassin
Role	Single-Target DPS / Executioner
Weapon	Chained Twin Sickles
Rarity	★★★★★
Banner Pool	Standard (available at launch)

Appearance

- Lean, agile silhouette with crimson hunter aesthetic
- Twin sickles connected by enchanted chains drape across her body
- Dark crimson leathers with phoenix-feather trim
- Long black hair tied with crimson ribbon, golden eyes
- Constant subtle ember haze around her
- Color palette: black, crimson, ember orange, gold trim

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
D	S	D	S	A	S	B

Passive — Reaper’s Dance

Whenever Kaida lands a Critical Hit, gain 1 **Infernal Momentum** stack (max 5).

Each stack grants: - +4% Speed - +4% Critical Damage

At max stacks: ignore 20% enemy DEF on all attacks.

Basic Attack — Crimson Sever

- **Cooldown:** None
- Kaida darts forward with a rapid twin-sickle strike.
- **Effects:** Single Target Damage; 20% chance to inflict **Burn** (2 turns).

Skill 1 — Chain Rend

- **Cooldown:** 2 turns | **Multiplier:** 1.5× single-target
- Kaida swings her chained sickles outward before pulling herself into a devastating follow-up.
- **Effects:** Single Target Damage; applies **Burn**; +25% Critical Hit Chance for this attack.

Skill 2 — Phantom Pursuit

- **Cooldown:** 3 turns | **Multiplier:** 2.2× single-target
- Kaida vanishes into crimson smoke and reappears behind her target.
- **Effects:** Heavy Single Target Damage; +50% damage against Burned enemies; high Critical Damage scaling.

Ultimate — Infernal Reaping

- **Cooldown:** 5 turns | **Multiplier:** 3.5× single-target
- Kaida unleashes a relentless storm of chained sickle attacks ending in an execution strike.
- **Effects:** Massive Single Target Damage; applies **Burn**; +20% damage per active Burn on target (max +60%); ignores 30% enemy DEF.

Character Identity

Strengths: Fastest Fire hero; excellent boss killer; strong crit scaling; high burst damage; exceptional mobility.

Weaknesses: Low HP/DEF; limited AoE; vulnerable to crowd control; requires setup for maximum damage.

Chibi Animations

- **Idle:** spins one sickle around finger, chains sway, small embers drift

- **Attack:** quick cross-slash → chain whip follow-up → assassin stance
- **Skill 1:** sickles swing outward → chains wrap target area → rapid spinning slash
- **Skill 2:** dissolves into crimson smoke → reappears behind target → fast execution strike
- **Ultimate:** chains erupt outward → multiple infernal slashes → crimson reaper silhouette → massive finishing cross-cut
- **Death:** sickles fall from hands → chains collapse → embers fade → falls to one knee

Team Role

Primary boss-killer. Pair with Burn appliers (Emberia, Valdrik) so her Skill 2 bonus damage is always live.

Recommended Partners

- **Emberia** — provides Burn stacks for Kaida's bonus damage
- **Valdrik** — tanks for Kaida's fragile DPS
- **Akari** — Speed Up + Revive Flame for safety net
- **Lyria** — back-line counterpart for sustained crits

Auto-AI Defaults

- **Skill 1:** Cast on cooldown
- **Skill 2:** Cast on cooldown when target is Burned (prioritize over basic)
- **Ultimate:** Cast on the lowest-HP boss-tagged enemy OR on the highest-ATK enemy if no boss

Quote

"The faster the battle, the deadlier I become."

Glacier — The Frostwarden

Field	Value
Element	 Water
Class	 Tank
Role	Tank / Bruiser / Crowd Control
Weapon	Winterfang Gunlance (frost-infused gunlance / cryogenic artillery)
Rarity	★★★★★
Banner Pool	Standard (available at launch)
Visual identity note	Wherever “ice” or “frost” appears in Glacier’s design language (chibi, splash, particles, status effects like Frostbite / Freeze), it is mechanically Water . The element system has only Fire / Water / Wind. Glacier is the “icy face” of Water.

Appearance

- Imposing knight in glacial silver-blue plate
- Massive Winterfang Gunlance — half lance, half cryogenic cannon
- Frost mist seeps from his armor’s vents
- Snowflakes drift around him constantly
- Long platinum hair with ice-blue glow at the tips
- Color palette: glacial silver, ice blue, white frost, steel gray

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
S	A	S	D	C	B	B

Passive — Frozen Bulwark

Whenever Glacier takes damage, gain 1 **Frostguard** stack (max 5).

Each stack grants: - +4% DEF - (Implicit) +3% damage absorption built into the DEF formula

At max stacks: enemies that attack Glacier have a **25% chance to become Frozen** (1 turn).

Basic Attack — Winterfang Thrust

- **Cooldown:** None
- Glacier drives his gunlance forward, releasing freezing energy on impact.
- **Effects:** Single Target Damage; 15% chance to inflict **Frostbite** (-20% Speed, -10% DEF, 2 turns).

Skill 1 — Cryo Bastion

- **Cooldown:** 2 turns | **Multiplier:** 1.0× (defensive)
- Plants Winterfang into the ground and creates a frost barrier.
- **Effects:** Grants **Shield** to self equal to 30% of max HP; **DEF Up** (+30%) self; applies 1 Frostguard stack.

Skill 2 — Avalanche Shell

- **Cooldown:** 3 turns | **Multiplier:** 1.8× AoE
- Fires a cryogenic shell that explodes on impact.
- **Effects:** AoE Damage; applies **Frostbite**; 30% chance per enemy to **Slow** (1 turn).

Ultimate — Eternal Winter

- **Cooldown:** 5 turns | **Multiplier:** 2.6× AoE
- Glacier fires a glacier cannon blast that erupts into a catastrophic blizzard.
- **Effects:** Massive AoE Damage; applies **Frostbite** to all enemies; 40% chance per enemy to **Freeze** (1 turn); grants maximum Frostguard stacks to self.

Character Identity

Strengths: Extremely durable; excellent frontline presence; powerful crowd control; strong team utility; reliable Frost cycling.

Weaknesses: Slow turn cadence; lower critical potential; needs frontline positioning; limited burst damage.

Chibi Animations

- **Idle:** Winterfang planted beside him, frost mist drifting, snowflakes around feet
- **Attack:** heavy lance thrust → small cryogenic shell fires
- **Skill 1:** plants gunlance → ice barrier erupts upward → frost aura forms
- **Skill 2:** gunlance charges blue → fires glacier shell → ice explosion
- **Ultimate:** raises Winterfang skyward → massive frost cannon → glacier beam fires → blizzard engulfs → ice crystals erupt from ground
- **Death:** drops to one knee → Winterfang falls → frost mist fades → collapses on frozen ground

Team Role

Alternative Tank to Valdrik. Where Valdrik is pure defense + Taunt, Glacier brings frontline AoE + crowd control. Pairs well with crit DPS who exploit Frozen targets.

Recommended Partners

- **Auron** — Frozen enemies take massive damage from Auron's Frostbound abilities
- **Neria** — extends Glacier's control with Slow/Freeze
- **Marina** — keeps him sustained as Frostguard builds
- **Kaida** — exploits debuffed enemies

Auto-AI Defaults

- **Skill 1:** Cast on cooldown when self HP < 70% OR no Frostguard stacks active
- **Skill 2:** Cast on cooldown when 2+ enemies alive
- **Ultimate:** Cast on cooldown when boss-tagged enemy alive OR 3+ enemies alive

Quote

“The cold is patient. It always wins.”

Marina — The Tideborn Saintess

Field	Value
Element	 Water
Class	 Support
Role	Healer / Buffer / Cleanser
Weapon	Oceanheart Harp (sacred floating harp blessed by sea spirits)
Rarity	★★★★★
Banner Pool	Standard (available at launch)

Appearance

- Graceful saintess with long azure-blue hair and ocean-blue eyes
- White and aqua ceremonial robes with seafoam embroidery
- Pearl + sea-coral ornaments
- Sacred Oceanheart Harp floats beside her
- Small bubbles + water notes drift around her
- Color palette: ocean blue, pearl white, seafoam, gold accents

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
B	D	B	A	D	D	A

Passive — Song of Serenity

Whenever Marina uses a healing skill, all healed allies gain **Serenity** (max 3 stacks).

Serenity grants: - +5% Healing Received - +5% DEF - Duration: 2 turns

At max stacks: removes one debuff from the ally.

Basic Attack — Aqua Melody

- **Cooldown:** None
- Plucks a magical harp string, sending a condensed water wave at the target.
- **Effects:** Single Target Damage; small chance to reduce enemy **ATK** (-15%, 1 turn).

Skill 1 — Healing Symphony

- **Cooldown:** 2 turns | **Heal multiplier:** 0.6× ATK (large team heal)
- Marina performs a soothing melody that restores all allies.
- **Effects:** **AoE Heal**; removes 1 debuff from all allies; applies Serenity.

Skill 2 — Tidal Harmony

- **Cooldown:** 3 turns | **Multiplier:** 0.4× ATK (heal + shield)
- Harp resonates with ocean blessing, surrounding allies with protective waters.
- **Effects:** Grants **Shield** to all allies (15% of Marina's max HP); **DEF Up** (+30%, 2 turns); applies Serenity.

Ultimate — Ocean's Requiem

- **Cooldown:** 5 turns | **Heal multiplier:** 1.2× ATK
- Marina performs a divine symphony calling forth the ancient sea spirits.
- **Effects:**
 - Massive **Team Heal**
 - **Cleanses** all debuffs from all allies
 - Increases **Healing Received** (+30%, 2 turns)
- **Source PDF cleanup:** The original sheet listed "Restores a small amount of Energy" as a 4th effect. Since the Energy system was deleted (audit Q4), that effect is simply removed — no replacement. Marina's Ultimate is a pure heal / cleanse / healing-amplification tool.

Character Identity

Strengths: Exceptional healing; strong team sustain; reliable cleansing; defensive utility; excellent support scaling.

Weaknesses: Low damage; limited offensive presence; vulnerable if focused; team-dependent.

Chibi Animations

- **Idle:** Oceanheart Harp floating beside her, water notes drift upward, small bubbles around her
- **Attack:** plucks harp string → water projectile launches → harp remains visible
- **Skill 1:** gentle melody plays → healing waves spread → water lilies briefly appear
- **Skill 2:** protective water rings form → ocean sigils rotate around allies → harp glows
- **Ultimate:** Oceanheart Harp floats before her → grand symphony → water lilies bloom across battlefield → ocean spirits swirl around allies → healing notes fill the air
- **Death:** harp falls beside her → water notes fade → kneels → collapses

Team Role

Primary sustain Support. Best paired with fragile DPS lineups (assassins, mages) who need her cleanse to survive heavy debuff content.

Recommended Partners

- **Glacier** — Marina sustains while Frostguard builds
- **Neria** — Neria's debuffs are cleansed off the team by Marina
- **Auron** — fragile back-line that Marina keeps alive
- **Kaida / Kael** — assassins thrive with her sustain

Auto-AI Defaults

- **Skill 1:** Cast when 2+ allies are below 70% HP
- **Skill 2:** Cast when boss-tagged enemy alive OR ally has 1+ debuff
- **Ultimate:** Cast when team has 2+ active debuffs OR any ally < 30% HP

Quote

"My song mends what the world has broken."

Neria — The Tidal Prodigy

Field	Value
Element	 Water
Class	 Mage
Role	Magic DPS / Control
Weapon	Astral Grimoire (floating book of forbidden ocean magic)
Rarity	★★★★★
Banner Pool	Standard (available at launch)

Appearance

- Silver-blue hair with deeper ocean-blue highlights
- Sapphire-blue eyes
- Floating arcane grimoire beside her at all times
- Sea-themed robes with deep blue + silver embroidery
- Water rings, ocean glyphs, and floating bubbles orbit around her
- Color palette: deep ocean blue, silver, sapphire, pearl white

Visual rules: The Astral Grimoire remains visible in EVERY frame. Never replace with staff/wand/orb/tome — only the grimoire.

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
C	S	D	A	B	A	A

Passive — Tidal Wisdom

Whenever Neria applies a debuff, she gains 1 **Tide** stack (max 5).

Each stack grants: - +5% Skill Damage - +3% Accuracy

At max stacks: Neria's debuffs last +1 turn longer.

Basic Attack — Aqua Bolt

- **Cooldown:** None
- Sends a condensed water sphere from the grimoire.
- **Effects:** Single Target Damage; 15% chance to apply **Slow** (1 turn).

Skill 1 — Tidal Sigil

- **Cooldown:** 2 turns | **Multiplier:** 1.8× AoE
- Grimoire pages flutter outward forming ocean sigils beneath enemies.
- **Effects:** AoE Damage; applies **Slow** (2 turns); 25% chance per enemy for **Accuracy Down**.

Skill 2 — Forbidden Tides

- **Cooldown:** 3 turns | **Multiplier:** 2.2× single-target
- Massive arcane water construct slams a single target.
- **Effects:** Heavy Single Target Damage; bonus +30% damage if target is Slowed OR Frostbite-d; generates 2 Tide stacks.

Ultimate — Leviathan Archive

- **Cooldown:** 5 turns | **Multiplier:** 2.6× AoE
- Neria opens the grimoire fully, summoning the Leviathan's recorded memories.
- **Effects:**
 - Massive AoE Damage
 - Removes all enemy buffs
 - Applies **DEF Down** + **Slow** to all enemies
 - Bonus +20% damage per Tide stack at cast time

Character Identity

Strengths: Powerful AoE damage; strong debuff control; pairs naturally with Glacier/Auron; scales well in long fights.

Weaknesses: Low HP/DEF; requires Tide stack setup; vulnerable to assassins; relies on debuffs.

Chibi Animations

- **Idle:** floating grimoire beside her, water sigil glows under feet, bubbles drift
- **Attack:** grimoire flips open → water bolt fires forward
- **Skill 1:** pages flutter outward → ocean sigil appears under enemies → water bursts
- **Skill 2:** grimoire glows blue → massive water construct rises → slams target
- **Ultimate:** grimoire opens fully → Leviathan silhouette emerges in spectral form → wave engulfs battlefield → enemy buffs scatter
- **Death:** grimoire snaps shut → falls to her side → fades into ocean mist

Team Role

Primary AoE damage + control. Excellent in Tower of Trials due to high Tide-scaling longevity.

Recommended Partners

- **Glacier / Auron** — Frostbite + Slow synergy
- **Marina** — keeps her safe; cleanses debuffs from Neria when needed
- **Mei Lin** (future banner) — back-line protection via taunt-shield
- **Kael** — burst that finishes targets she debuffed

Auto-AI Defaults

- **Skill 1:** Cast on cooldown when 2+ enemies alive
- **Skill 2:** Cast on cooldown if any enemy is Slowed or Frostbite-d (otherwise hold)
- **Ultimate:** Cast on cooldown when 3+ enemies alive AND ≥ 3 Tide stacks active

Quote

“The ocean remembers every name. I’m just reading them aloud.”

Lyria — The Tempest Huntress

Field	Value
Element	 Wind
Class	 Archer
Role	Single-Target DPS / Critical Hit Specialist
Weapon	Zephyrvine Longbow
Rarity	★★★★★
Banner Pool	Standard (available at launch)

Appearance

- Athletic huntress with deep green-and-white hunter’s leathers
- Long flowing emerald hair tied with wind ribbons
- Bright green eyes with golden flecks
- Zephyrvine Longbow — ornate bow of living vine and wind crystal
- Wind feathers drift around her shoulders
- Color palette: emerald green, white, gold accents, light tan leather

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
D	S	D	A	S	A	S

Passive — Huntress’s Instinct

Whenever Lyria lands a Critical Hit, she gains 1 **Focus** stack (max 5).

Each stack grants: - +4% Crit Damage - +4% Accuracy

At max stacks: ignore 25% enemy DEF.

Basic Attack — Zephyr Shot

- **Cooldown:** None
- Lyria fires a wind-infused arrow with deadly precision.
- **Effects:** Single Target Damage; +15% Critical Rate on this attack.

Skill 1 — Gale Piercer

- **Cooldown:** 2 turns | **Multiplier:** 1.8× single-target
- Draws back her bow and releases a concentrated wind arrow.
- **Effects:** Heavy Single Target Damage; **ignores 30% DEF**; +20% Critical Chance on this attack.

Skill 2 — Tempest Volley

- **Cooldown:** 3 turns | **Multiplier:** 0.55× per hit × 5 hits = 2.75× total
- Leaps backward while unleashing a flurry of enchanted arrows.
- **Effects:** Hits 5 random enemies (or same target if only 1 alive); each hit may crit independently; **grants 1 Focus stack** (one total per cast, not per crit).

Ultimate — Skyfall Barrage

- **Cooldown:** 5 turns | **Multiplier:** 3.5× single-target (plus splash)
- Calls upon the spirits of the wind to rain countless arrows.
- **Effects:**
 - Massive Single Target Damage
 - Additional 0.5× Splash Damage to adjacent enemies
 - **Consumes Focus Stacks** for bonus damage (+30% per stack consumed)
 - **Guaranteed Critical Hit**

Character Identity

Strengths: Highest Crit Rate at launch; excellent boss killer; exceptional accuracy; strong burst damage; reliable DPS.

Weaknesses: Fragile; limited crowd control; requires protection; focuses on single targets.

Chibi Animations

- **Idle:** holds ornate bow, hair sways in the wind, small feathers drift around her
- **Attack:** draws bow → fires wind arrow → returns to stance
- **Skill 1:** wind gathers around arrow → releases piercing shot → emerald trail
- **Skill 2:** backstep → fires multiple arrows rapidly → wind feathers scatter
- **Ultimate:** leaps into the air → massive wind vortex forms → hundreds of glowing arrows descend
- **Death:** bow slips from her hands → falls to one knee → wind energy disperses

Team Role

Primary single-target DPS. Best-in-class boss killer. Especially strong against Water-element bosses (Wind > Water).

Recommended Partners

- **Sylphie** — keeps her topped up; speed boost
- **Valdrik** — frontline that keeps Lyria's back row safe
- **Marina** — sustains in long boss fights
- **Mei Lin** (future) — Wind synergy taunt-shield setup

Auto-AI Defaults

- **Skill 1:** Cast on cooldown
- **Skill 2:** Cast on cooldown when 1-2 enemies alive (high-yield against single target via multi-hit)
- **Ultimate:** Cast on cooldown when boss-tagged enemy alive AND ≥ 3 Focus stacks active

Quote

"The prey never hears the wind that carries the final arrow."

Sylphie — The Wind Spirit

Field	Value
Element	 Wind
Class	 Support
Role	Primary Healer / Team Buffer
Weapon	Verdant Spirit Staff
Rarity	★★★★★
Banner Pool	Standard (available at launch)

Appearance

- Petite adult woman with graceful, delicate build
- Long mint-green hair flowing to her thighs
- Bright emerald eyes
- Soft, gentle features
- White and green healer robes with gold filigree, leaf motifs
- Open shoulders, flowing sleeves, layered skirt
- Floral hair ornaments
- Carries the Verdant Spirit Staff
- Color palette: mint green, white, gold, soft emerald

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
B	D	C	A	D	D	S

Passive — Spirit’s Embrace

Whenever an ally falls below 50% HP: - Restore a small amount of HP (10% of Sylphie’s ATK) - Remove 1 debuff - Sylphie gains 1 **Nature Blessing** stack (max 5)

Each stack grants: - +3% Healing Done

Basic Attack — Verdant Bolt

- **Cooldown:** None
- Sylphie launches a concentrated wind blossom at an enemy.
- **Effects:** Single Target Damage; small chance to reduce enemy Accuracy (-15%, 1 turn).

Skill 1 — Nature's Embrace

- **Cooldown:** 2 turns | **Heal multiplier:** 0.5× ATK
- Sylphie summons healing winds around her allies.
- **Effects:** **Heal All Allies**; removes 1 debuff; small **Heal Over Time** (2 turns).

Skill 2 — Zephyr's Renewal

- **Cooldown:** 3 turns | **Multiplier:** 1.0× ATK (heal + buff)
- A powerful breeze strengthens the entire team.
- **Effects:** **ATK Up** (+30%); **Speed Up** (+30%); **Healing Received Up** (+25%); lasts 2 turns.

Ultimate — Sanctuary of the World Tree

- **Cooldown:** 5 turns | **Heal multiplier:** 1.3× ATK
- Calls forth the spirit of the World Tree.
- **Effects:**
 - Massive **Team Heal**
 - **Removes all debuffs**
 - Grants **Regeneration** (2 turns)
 - Grants **Damage Reduction** (+30%, 2 turns)

Character Identity

Strengths: Strongest healer at launch; team-wide cleansing; excellent sustain; powerful buffs; easy to fit into any team.

Weaknesses: Very low damage; fragile defense; vulnerable to burst; relies on allies for kills.

Chibi Animations

- **Idle:** staff gently floating beside her, spirit butterflies circling, leaves drifting around
- **Attack:** staff points forward → wind blossom projectile → returns to support stance
- **Skill 1:** green healing circle appears → butterflies spread outward → healing particles rain
- **Skill 2:** wind spirals around allies → emerald leaves burst → buff symbols appear briefly
- **Ultimate:** massive World Tree manifests → healing winds sweep battlefield → light blossoms bloom everywhere
- **Death:** staff slips from hands → kneels softly → falls beside a bed of leaves

Team Role

Primary Support across nearly every Wind team. Strong universal sustainer — fits even mono-Fire or mono-Water teams when needed.

Recommended Partners

- **Lyria / Kael** — Wind core team
- **Valdrik / Glacier** — sustains them through tough fights
- **Emberia** — back-line healing while she nukes
- **Kaida** — keeps Kaida alive long enough to stack momentum

Auto-AI Defaults

- **Skill 1:** Cast when 2+ allies below 80% HP OR any ally has 1+ debuff
- **Skill 2:** Cast on cooldown when team has no active buffs (pre-engagement) OR boss-tagged enemy appears
- **Ultimate:** Cast when 2+ allies below 50% HP OR team has 2+ debuffs

Quote

“The wind always returns. So do I.”

Kael — The Tempest Shadow

Field	Value
Element	 Wind
Class	 Assassin
Role	Burst DPS / Ambush Specialist
Weapon	Tempest Fangs (twin shortswords with spectral tiger spirit)
Rarity	★★★★★
Banner Pool	Standard (available at launch)

Appearance

- Lean, agile silhouette with hooded cape that flows in the wind
- Twin blades drawn at low guard
- Dark forest-green hood + tunic with white-silver trim
- Glowing emerald eyes when in combat (spectral tiger eye effect)
- Wind aura swirls around him
- Color palette: forest green, silver, black, white accents

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
D	S	D	S	A	S	B

Passive — Stormfang’s Eye

Whenever Kael lands a Critical Hit, gain 1 **Predator** stack (max 5).

Each stack grants: - +4% Critical Damage - +4% Speed

At max stacks: ignore 25% enemy DEF.

Basic Attack — Tiger Fang

- **Cooldown:** None
- Quick twin-blade slash with wind claw projectile.
- **Effects:** Single Target Damage; +15% Critical Rate on this attack.

Skill 1 — Emerald Pounce

- **Cooldown:** 2 turns | **Multiplier:** 1.5× single-target
- Spectral Stormfang Tiger lunges alongside Kael.
- **Effects:** Heavy Single Target Damage; +25% Critical Chance; ignores 20% DEF.

Skill 2 — Phantom Hunt

- **Cooldown:** 3 turns | **Multiplier:** 0.55× per hit × 4 hits = 2.2× total
- Kael disappears into the wind and strikes from multiple angles.
- **Effects:** Hits 4 times on selected target; each hit can crit; generates 1 Predator stack per crit.

Ultimate — Stormfang Execution

- **Cooldown:** 5 turns | **Multiplier:** 3.5× single-target
- Massive spectral wind tiger emerges from a cyclone and tears through.
- **Effects:**
 - Massive Single Target Damage
 - **Guaranteed Critical Hit**
 - Consumes Predator stacks for bonus damage (+30% per stack consumed)
 - Ignores 30% enemy DEF

Character Identity

Strengths: Extremely high burst damage; exceptional mobility; strong boss killer; excellent crit scaling; fastest Wind assassin.

Weaknesses: Fragile; requires setup; limited team utility; vulnerable if focused.

Chibi Animations

- **Idle:** twin blades drawn, wind swirls around him, spectral tiger eyes appear briefly
- **Attack:** quick slash → wind claw launches rightward
- **Skill 1:** Stormfang Tiger appears → tiger lunges rightward
- **Skill 2:** dash afterimages → multiple strikes rightward
- **Ultimate:** massive cyclone forms → Stormfang Tiger emerges → tiger charges → wind explosion
- **Death:** blades remain visible briefly → falls to one knee → collapses as wind disperses

Team Role

Wind-element burst DPS. Mirror of Kaida (Fire Assassin) — slightly more crit-scaling, slightly less Burn-dependent. Picks targets, dies, picks more targets.

Recommended Partners

- **Sylphie** — keeps him alive long enough to stack
- **Lyria** — Wind-team boss DPS synergy
- **Marina** — sustain when paired with Glacier
- **Mei Lin** (future) — Wind synergy + taunt protection

Auto-AI Defaults

- **Skill 1:** Cast on cooldown
- **Skill 2:** Cast on cooldown when target has >50% HP (maximize hit count value)
- **Ultimate:** Cast on highest-HP enemy at the moment of activation; prefer boss-tagged

Quote

“By the time you hear the wind... I’ve already struck.”

Auron — The Frostbound Prodigy

Field	Value
Element	 Water
Class	 Archer
Role	Single-Target DPS / Freeze Specialist
Weapon	Frostbound Bracer (cryogenic projectile bracer — no traditional bow)
Rarity	★★★★★
Banner Pool	 LAUNCH LIMITED BANNER — first paid banner, live at launch

Appearance

- Cool, composed young hunter with silver-white hair
- Light blue cryogenic bracer on dominant arm — projects ice crystals as ammunition
- Pale blue tunic with frost-edge embroidery
- Calm, ice-blue eyes
- Subtle frost particles drift around him
- Color palette: ice blue, silver, white frost, navy accents

Visual identity note: Auron’s design uses ice/frost imagery (Frostbound Bracer, Frost Charges, Freeze status). Mechanically he is **Water-element** — there is no separate Ice element. Frost/Freeze are Water-flavored status effects.

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
C	A	D	A	B	A	S

Passive — Cryogenic Resonance

Whenever Auron lands a Critical Hit, gain 1 **Frost Charge** stack (max 5).

Each stack grants: - +5% Critical Damage

At max stacks: Auron's abilities ignore 20% enemy DEF.

Basic Attack — Frost Shard

- **Cooldown:** None
- The Frostbound Bracer creates a razor-sharp ice crystal and launches it at the target.
- **Effects:** Single Target Damage; 10% chance to inflict **Freeze** (1 turn).

Skill 1 — Glacial Lance

- **Cooldown:** 2 turns | **Multiplier:** 1.8× single-target
- Condenses cryogenic energy into a massive ice spear.
- **Effects:** High Single Target Damage; guaranteed hit (Accuracy auto-passes); +25% Critical Chance.

Skill 2 — Crystal Bloom

- **Cooldown:** 3 turns | **Multiplier:** 0.55× per hit × 4 hits = 2.2× total
- The bracer summons multiple floating frost crystals that fire rapidly.
- **Effects:** Hits 4 times on selected target; each hit can crit; 20% chance per hit to **Freeze**; generates 1 Frost Charge per crit.

Ultimate — Absolute Zero Domain

- **Cooldown:** 5 turns | **Multiplier:** 3.5× single-target
- Auron releases the full power of the Frostbound Bracer.
- **Effects:**
 - Massive Single Target Damage
 - Ignores 30% enemy DEF
 - **Guaranteed Freeze** (1 turn)
 - Bonus damage against Frozen enemies (+30%)
 - Consumes Frost Charges for bonus damage (+25% per stack consumed)

Character Identity

Strengths: Highest Accuracy in Water roster; strong single-target damage; freeze specialist; excellent boss killer; reliable crits; strong scaling through Frost Charges.

Weaknesses: Low survivability; limited AoE; requires setup; no team utility.

Chibi Animations

- **Idle:** Frostbound Bracer glowing softly, snowflakes drifting, floating ice crystals orbit nearby
- **Attack:** raises bracer → ice shard forms instantly → frost projectile launches rightward
- **Skill 1:** large ice spear forms → cryogenic energy gathers → Glacial Lance fires rightward
- **Skill 2:** floating frost crystals appear → multiple ice projectiles fire rightward → crystal fragments scatter
- **Ultimate:** frozen magic circle appears → ice crystals surround battlefield → massive convergence attack rightward → frozen explosion
- **Death:** bracer loses its glow → falls to one knee → collapses as ice particles fade

Team Role

Launch banner star. Primary single-target Water DPS. Synergizes brilliantly with Glacier — Glacier Frostbites the field, Auron executes Frozen targets for massive bonus damage.

Recommended Partners

- **Glacier** — primary synergy partner (Frost ecosystem)
- **Marina** — sustains his fragile back line
- **Neria** — debuffs targets that Auron then executes
- **Sylphie** — pure healing safety net

Auto-AI Defaults

- **Skill 1:** Cast on cooldown
- **Skill 2:** Cast on cooldown when target HP > 50%
- **Ultimate:** Cast on cooldown — prefer Frozen / Frostbite-d targets; otherwise highest-HP boss-tagged enemy

Banner Notes

- Launch Banner duration: 21 days
- Banner pity carries to next Limited banner
- Featured Hero Starter Bundle (\$14.99) and Power Bundle (\$29.99) available

Quote

“Perfection isn’t luck. It’s precision.”

Ragnar — The Flamehunter

Field	Value
Element	 Fire
Class	 Archer
Role	AoE DPS / Burn Specialist
Weapon	Inferno Recurve Bow (volcanic-steel bow with phoenix-feather fletching)
Rarity	★★★★★
Banner Pool	 Future Limited Banner (post-launch release)
Status	Design draft — kit from source PDF, polish pass pending

Appearance

- Rugged hunter aesthetic with wild crimson hair
- Golden eyes glowing with phoenix-flame light
- Black and red leather armor, exposed shoulders + collarbone
- Burning phoenix-feather cloak that smolders
- Large infernal recurve bow forged from volcanic steel
- Color palette: red, orange, black, gold accents

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
C	A	D	A	B	A	A

Passive — Kindling Arrow

Critical Hits apply **Burn** (2 turns).

This makes Ragnar a strong Burn spreader even without active skill use — every basic crit ticks the Burn count up.

Basic Attack — Blazing Shot

- **Cooldown:** None
- Fires a flaming arrow.
- **Effects:** Single Target Damage; 10% chance to apply **Burn** independently (separate from crit-Burn).

Skill 1 — Explosive Arrow

- **Cooldown:** 2 turns | **Multiplier:** 1.8× AoE
- Ragnar fires an arrow that explodes on impact.
- **Effects:** AoE Damage; applies **Burn** to all hit enemies.

Skill 2 — Phoenix Volley

- **Cooldown:** 3 turns | **Multiplier:** 0.55× per hit × 5 hits = 2.75× total
- Rapid flaming arrows into the sky that rain down.
- **Effects:** Hits 5 random enemies (or same target); each hit can crit; each hit may apply Burn (40% chance per hit).

Ultimate — Solar Flare Barrage

- **Cooldown:** 5 turns | **Multiplier:** 2.6× AoE
- Ragnar channels the power of the sun. A massive flaming arrow ascends, then explodes into a rain of fire.
- **Effects:**
 - Massive AoE Damage
 - **Burn All Enemies** (3 turns — longer than standard 2)
 - +20% bonus damage to already-Burned enemies (stacking)

Character Identity

Strengths: Strong AoE damage; reliable Burn spreader; excellent wave clear; high crit damage; Wind-element bosses melt under his fire (off-element penalty for them).

Weaknesses: Fragile back-line; limited crowd control; needs protection from front-line.

Chibi Animations

- **Idle:** bow resting on shoulder, small embers drifting around him
- **Attack:** quick draw → fire arrow launches rightward
- **Skill 1:** arrow glows → explosive shot fires → fire explosion erupts at target
- **Skill 2:** rapid firing stance → multiple flaming arrows rain down across enemy formation
- **Ultimate:** massive solar arrow forms → launches skyward → meteor-like firestorm crashes down
- **Death:** drops bow → falls to one knee → collapses

Team Role

Second AoE option in the Fire roster. Where Emberia is sustained Burn-stacker, Ragnar is **immediate explosive burst**. Different feel; both viable.

Recommended Partners

- **Valdrik** — Taunts pull aggro off the fragile Ragnar
- **Akari** — buffs his ATK; revives if he dies first
- **Emberia** — Burn synergy (chain explosions)
- **Kaida** — single-target finisher behind Ragnar's AoE

Auto-AI Defaults

- **Skill 1:** Cast on cooldown when 2+ enemies alive
- **Skill 2:** Cast on cooldown (random multi-hit always useful)
- **Ultimate:** Cast on cooldown when 3+ enemies alive AND at least 1 is Burned

Quote

"Even the sun rises only to fall. And I am the fall."

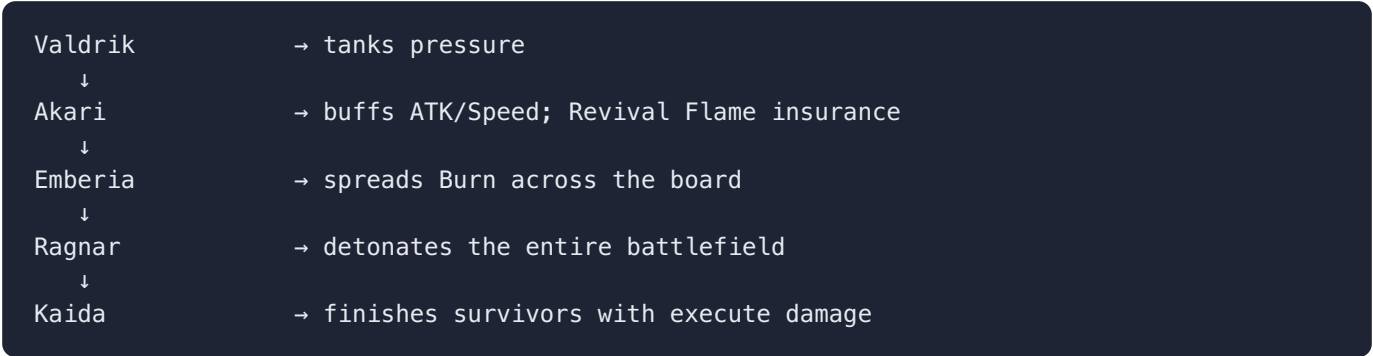
Fire Roster Comparison

(Per canonical source:)

Hero	Role	Specialty
Valdrik	Tank	Frontline protection + Burn application
Akari	Support	Bufs + Revival Flame
Emberia	Mage	AoE Burn stacker (Crimson Flame)
Kaida	Assassin	Single-target executioner
Ragnar	Archer	AoE Burn DPS

Fire Team Synergy Flow

The canonical Fire team loop with Ragnar:



Ragnar becomes the Fire faction’s dedicated ranged AoE damage dealer — the first true AoE Archer in Project I.

Status notes (design polish pending): - Skill 2 vs Lyria’s Tempest Volley — similar mechanic, may need differentiation - Confirm Burn-on-crit triggers per-hit or once per attack roll - Cloak VFX cost may be high — placeholder

Akari — The Phoenix Chronicler

Field	Value
Element	 Fire
Class	 Support
Role	Buffer / Revival Support
Weapon	Phoenix Lantern (floating fire lantern blessed by the Phoenix Spirit)
Rarity	★★★★★
Banner Pool	 Future Limited Banner (post-launch release)
Status	Design draft — kit from source PDF, polish pass pending

Appearance

- Elegant eastern-inspired shrine maiden
- Long crimson hair tied with golden phoenix ornament
- Amber eyes that glow softly
- White and red ceremonial robes with phoenix-feather trim
- Golden phoenix ornaments at wrists and waist
- Floating fire lantern beside her with a small Phoenix Spirit inside
- Color palette: red, gold, white, ember orange

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
B	D	B	A	D	D	S

Passive — Phoenix Ember

Whenever an ally falls below 50% HP: - Restore a small amount of HP (10% of Akari's ATK) - Can trigger **once per ally per battle**

Basic Attack — Lantern Flame

- **Cooldown:** None
- Akari sends a small phoenix flame toward an enemy.
- **Effects:** Single Target Damage.

Skill 1 — Blessing of Ashes

- **Cooldown:** 2 turns | **Heal multiplier:** 0.5× ATK
- Akari's lantern releases restorative embers.
- **Effects:** **Heal All Allies**; applies **ATK Up** (+30%, 2 turns).

Skill 2 — Crimson Rebirth

- **Cooldown:** 3 turns
- Akari surrounds allies with phoenix flames.
- **Effects:** Apply **Revival Flame** to all allies (2 turns): if an ally is defeated during this window, they revive with 20% HP. Each ally can be revived only once per battle.

Ultimate — Phoenix Festival

- **Cooldown:** 5 turns | **Heal multiplier:** 1.2× ATK
- A massive phoenix appears from Akari's lantern and circles the battlefield.
- **Effects:**
 - Large **Team Heal**
 - **ATK Up** (+30%, 2 turns)
 - **Speed Up** (+30%, 2 turns)
 - Apply **Revival Flame** to all allies (2 turns)

Character Identity

Strengths: Strong buffs; emergency recovery; unique revival utility (only hero with proactive revive); excellent team support; high accuracy.

Weaknesses: Low damage; limited crowd control; relies on team to win.

Chibi Animations

- **Idle:** lantern floating beside her, small phoenix embers orbiting, warm glow beneath feet
- **Attack:** lantern swings → phoenix spark fires forward
- **Skill 1:** lantern opens → healing embers scatter → warm light spreads outward
- **Skill 2:** phoenix flames envelop each ally → revival sigils appear briefly
- **Ultimate:** large phoenix spirit emerges → flames spiral around allies → golden feathers rain down
- **Death:** lantern falls → flame extinguishes → Akari kneels and fades

Team Role

The Fire-faction Support counterpart to Sylphie. Where Sylphie is pure sustain, Akari trades a bit of healing for **revival insurance** — extremely powerful in hard content where one bad turn would wipe.

Recommended Partners

- **Valdrik** — durable frontline that Akari's revives protect
- **Emberia** — gets her ATK Up + Speed Up buffs to maximum AoE burn
- **Kaida** — gets revived after she dives too deep
- **Auron** — fragile Water Archer she keeps alive through Frostbound bursts

Fire Team Synergy Flow

The intended Fire-faction loop with Akari at the heart of it:

```

Valdrik      → tanks pressure
  ↓
Akari        → buffs ATK/Speed + applies Revival Flame
  ↓
Emberia      → burns the entire battlefield with stacked Crimson Flame
  ↓
Kaida        → executes the weakened survivors
  
```

Akari gives the Fire faction what it otherwise lacks: **sustain, recovery, and second chances** — while keeping the aggressive fire identity through phoenix-themed support magic.

Auto-AI Defaults

- **Skill 1:** Cast on cooldown when 2+ allies below 80% HP
- **Skill 2:** Cast on cooldown when entering boss waves OR when team has 1+ ally below 40% HP
- **Ultimate:** Cast when 2+ allies below 50% HP OR boss-tagged enemy appears

Quote

“Fall — and rise again. The Phoenix remembers.”

Status notes (design polish pending): - Heal multipliers may need tuning vs Sylphie's - Revival Flame should NOT trigger if Akari herself dies — confirm in implementation - Phoenix Spirit visual should be subtle, not overpower other VFX

Seraphina — The Abyssal Reaper

Field	Value
Element	 Water
Class	 Assassin
Role	Burst DPS / Executioner
Weapon	Tidal Scythe (large crescent water-blade scythe)
Rarity	★★★★★
Banner Pool	 Future Limited Banner (post-launch release)
Status	Design draft — kit from source PDF, polish pass pending

Appearance

- Elegant assassin with long silver-blue hair flowing
- Ocean-blue eyes with a faint serpentine glow
- Light armored dress in white and teal with flowing water ribbons
- Massive crescent-shaped Tidal Scythe — water-blade form
- Bare midriff, subtle scaling armor accents
- Color palette: deep blue, white, silver, teal

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
D	S	D	A	A	S	B

Passive — Death Tide

Whenever Seraphina lands a Critical Hit, gain 1 **Tide** stack (max 5).

Each stack grants: - +5% Critical Damage

(Note: this is Seraphina’s “Tide” stack — separate from Neria’s “Tide” stacks. Each hero’s stacks are isolated.)

Basic Attack — Tidal Reap

- **Cooldown:** None
- Swift scythe slash.
- **Effects:** Single Target Damage.

Skill 1 — Undertow Slash

- **Cooldown:** 2 turns | **Multiplier:** 1.5× single-target
- Wave surges beneath target as Seraphina swings her scythe.
- **Effects:** High Single Target Damage; applies **Slow** (2 turns).

Skill 2 — Whirlpool Reaping

- **Cooldown:** 3 turns | **Multiplier:** 0.55× per hit × 4 hits = 2.2× total
- Spins her scythe, creating a violent whirlpool around target.
- **Effects:** Hits 4 times on selected target; each hit can crit; +30% bonus damage against Slowed enemies.

Ultimate — Leviathan's Judgment

- **Cooldown:** 5 turns | **Multiplier:** 3.5× single-target + execute bonus
- A gigantic spectral sea serpent appears behind Seraphina as she performs a devastating execution strike.
- **Effects:**
 - Massive Single Target Damage
 - **Guaranteed Critical Hit**
 - +50% bonus damage based on target's missing HP%
 - **If target is below 30% HP:** deal additional execute damage (+2.0× multiplier extra)

Character Identity

Strengths: Extremely high single-target burst; excellent boss killer; strong execute potential; high crit scaling. Slightly different feel from Kael — Seraphina has the **execute window** as her signature.

Weaknesses: Fragile; limited AoE; requires setup; no team utility.

Chibi Animations

- **Idle:** scythe resting on shoulder, water droplets drifting upward, small currents around feet
- **Attack:** quick scythe swing → water crescent releases
- **Skill 1:** wave rises → scythe cleaves downward → water eruption
- **Skill 2:** spinning reaper attack → whirlpool forms around her
- **Ultimate:** Leviathan spirit appears → massive overhead scythe swing → ocean explodes outward
- **Death:** drops scythe → falls to one knee → collapses

Team Role

Execute-focused Water Assassin. Best against bosses with HP phase windows. Pair with debuffers (Glacier, Neria) for setup.

Recommended Partners

(Per canonical source — implied via Water Roster Comparison context:)

- **Glacier** — Frostbite-d / Slowed targets eat extra damage; pair with Whirlpool Reaping
- **Marina** — keeps her fragile self alive (heal + cleanse)
- **Neria** — debuffs that Seraphina exploits
- **Auron** — both single-target Water DPS; together they shred bosses

Water Roster Comparison

Hero	Role	Specialty
Glacier	Tank	Frontline + Frost ecosystem
Marina	Support	Heal / cleanse
Neria	Mage	Magic DPS / control
Auron	Archer	Freeze specialist / single-target
Seraphina	Assassin	Critical Execution Damage

Auto-AI Defaults

- **Skill 1:** Cast on cooldown when target is not already Slowed
- **Skill 2:** Cast on cooldown when target is Slowed (max value)
- **Ultimate:** Cast when target HP < 30% (execute window) OR when target is at full HP and boss-tagged

Quote

“The tide does not ask permission. It just arrives.”

Status notes (design polish pending): - Execute threshold (30%) may scale with difficulty - Tide stack vs Neria's Tide — clarify naming in implementation (consider renaming Seraphina's to “Abyss” stacks) - Scythe weapon archetype confirmed (Assassin-class allowed)

Mei Lin — The Jade Guardian

Field	Value
Element	 Wind
Class	 Tank
Role	Protector / Shield Specialist
Weapon	Jade Tower Shield (massive jade shield engraved with protective runes)
Rarity	★★★★★
Banner Pool	 Future Limited Banner (post-launch release)
Status	Design draft — kit from source PDF, polish pass pending

Appearance

- Elegant eastern-inspired guardian, tall and composed
- Long black hair with jade ornaments
- Emerald-green eyes
- White and jade ceremonial armor with golden trim
- Flowing silk ribbons infused with wind magic (drift around her)
- Massive jade tower shield engraved with ancient protective runes
- Color palette: jade green, white, gold, black

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
S	C	S	B	D	D	B

Passive — Guardian’s Resolve

Whenever Mei Lin receives damage, gain 1 **Resolve** stack (max 5).

Each stack grants: - +4% DEF - +4% Shield Strength

(Note: same name as Valdrik's Resolve — each hero's stacks are isolated.)

Basic Attack — Jade Strike

- **Cooldown:** None
- Strikes with her shield, releasing compressed wind.
- **Effects:** Single Target Damage.

Skill 1 — Emerald Barrier

- **Cooldown:** 2 turns
- Mei Lin creates a protective wall of jade wind around her allies.
- **Effects:** Grants **Shield** to all allies (15% of Mei Lin's max HP, 2 turns).

Skill 2 — Guardian's Gale

- **Cooldown:** 3 turns | **Multiplier:** 1.8× AoE
- Mei Lin spins her shield and creates a powerful wind vortex.
- **Effects:** AoE Damage; **Taunt All Enemies** (2 turns); **DEF Up** to self (+30%, 2 turns).

Ultimate — Heavenly Jade Sanctuary

- **Cooldown:** 5 turns
- Mei Lin plants her shield into the ground and summons an enormous jade barrier.
- **Effects:**
 - Massive **Team Shield** (40% of Mei Lin's max HP, 2 turns)
 - **DEF Up** to all allies (+50%, 2 turns)
 - **Damage Reduction** to all allies (+30%, 2 turns)

Character Identity

Strengths: Highest defensive utility in Wind roster; strong team shields; excellent survivability; reliable protection; good turn cycling for a tank.

Weaknesses: Low damage; limited offensive pressure; relies on teammates for kills.

Chibi Animations

- **Idle:** shield planted beside her, jade talismans flutter in the wind, soft emerald particles drift
- **Attack:** shield bash → wind burst releases → returns to defensive stance
- **Skill 1:** raises shield → emerald barrier projects across allies → wind ribbons swirl
- **Skill 2:** spins shield → jade wind vortex forms → enemies are pulled toward her
- **Ultimate:** plants shield → massive jade barrier rises → protective winds surround team → ancient runes glow beneath her feet
- **Death:** shield falls beside her → drops to one knee → wind particles fade → collapses gracefully

Team Role

The Wind-faction Tank counterpart to Valdrik / Glacier. Shield-focused (vs Valdrik’s Taunt-focused, vs Glacier’s CC-focused). Best for protecting fragile back lines.

Recommended Partners

(Per canonical source sheet:)

- **Lyria** — back-line Crit Archer protected by Mei Lin’s shields
- **Kael** — assassin gets shield insurance for his dive
- **Ayame** — fragile AoE Mage protected by Mei Lin
- **Sylphie** — sustain stacks with Mei Lin’s shields

Wind Roster Comparison

Hero	Role	Specialty
Lyria	Archer	Precision DPS
Kael	Assassin	Burst single-target
Ayame	Mage	AoE Wind Sigil stacking
Sylphie	Support	Healer / cleanser
Mei Lin	Tank	Shields & Damage Reduction

Auto-AI Defaults

- **Skill 1:** Cast on cooldown when team has no active shields
- **Skill 2:** Cast on cooldown when 2+ enemies alive AND any ally HP < 70%
- **Ultimate:** Cast when boss-tagged enemy appears OR 2+ allies below 50% HP

Quote

“The jade does not break. Neither does my word.”

Status notes (design polish pending): - Shield value (15%/40% of max HP) may need scaling with rarity/level - Wind ribbons VFX should be performant in compact mode - Taunt duration on Skill 2 vs Valdrik's 40% chance — Mei Lin is guaranteed but only on AoE

Ayame — The Storm Dancer

Field	Value
Element	 Wind
Class	 Mage
Role	AoE DPS / Debuffer
Weapon	Tempest Ritual Fan (ceremonial fan with kanji talismans + wind sigils)
Rarity	★★★★★
Banner Pool	 Future Limited Banner (post-launch release)
Status	Design draft — kit from source PDF, polish pass pending

Appearance

- Eastern shrine-maiden / kagura-dancer aesthetic
- Long flowing black hair tied with green-and-white ribbons
- Bright green eyes with subtle wind-glow
- White and green ceremonial robes with kanji paper talismans pinned at sleeves
- Tempest Ritual Fan carried in dominant hand — talismans float around her
- Color palette: white, jade green, gold-paper, black accents

Stat Grades

HP	ATK	DEF	SPD	CR	CD	ACC
D	S	D	A	B	A	A

Passive — Dancing Winds

Whenever Ayame uses a skill, gain 1 **Wind Sigil** stack (max 5).

Each stack grants: - +5% Skill Damage - +3% Speed

At max stacks: Ayame ignores 20% enemy DEF.

Basic Attack — Wind Charm

- **Cooldown:** None
- Releases an enchanted talisman carried by the wind.
- **Effects:** Single Target Damage; 20% chance to inflict **Speed Down** (1 turn).

Skill 1 — Whirling Talisman

- **Cooldown:** 2 turns | **Multiplier:** 1.8× AoE
- Ayame scatters sacred talismans across the battlefield.
- **Effects:** AoE Damage; applies **Speed Down** (2 turns).

Skill 2 — Sacred Gale Dance

- **Cooldown:** 3 turns | **Multiplier:** 1.8× AoE
- Ayame performs an ancient shrine ritual summoning slicing winds.
- **Effects:** AoE Damage; applies **Accuracy Down** (-30%, 2 turns); generates 2 Wind Sigil stacks.

Ultimate — Heavenly Tempest Rite

- **Cooldown:** 5 turns | **Multiplier:** 2.6× AoE
- Ayame opens a sacred storm circle and commands the wind spirits.
- **Effects:**
 - Massive AoE Damage
 - **Removes all enemy buffs**
 - Applies **Wind Vulnerability** (-25% Wind resistance, all enemies, 2 turns)
 - Generates maximum (5) Wind Sigil stacks immediately

Character Identity

Strengths: Highest AoE damage in Wind team; excellent wave clear; strong debuff utility; fast skill cycling; scales well in long battles.

Weaknesses: Low HP / DEF; vulnerable to assassins; relies on skills for damage; weak basic attacks.

Chibi Animations

- **Idle:** holds ritual fan, paper talismans float around her, sleeves sway in wind
- **Attack:** throws talisman → fan slash effect → returns to ritual stance
- **Skill 1:** scatters talismans outward → wind cuts across enemies → talismans flutter away
- **Skill 2:** ritual dance pose → spiraling winds form → slicing wind blades hit enemies
- **Ultimate:** sacred shrine circle appears → storm clouds gather overhead → talismans swirl around enemies → massive tornado erupts
- **Death:** drops ritual fan → talismans scatter away → falls gracefully

Team Role

Wind AoE Mage. Mirror of Emberia (Fire) and Neria (Water). Pairs with Wind-team for max element synergy. Excellent in chapters with Water-element enemies (Wind > Water).

Recommended Partners

(Per canonical source sheet:)

- **Mei Lin** — shields Ayame's fragile back line (Wind synergy)
- **Sylphie** — heals + buffs Ayame's AoE
- **Kael** — assassin finisher for surviving enemies
- **Marina** — alternate sustain across mixed Water/Wind teams
- **Valdrik** — alternate frontline tank with Burn synergy on AoE-cleared boards

Auto-AI Defaults

- **Skill 1:** Cast on cooldown when 2+ enemies alive
- **Skill 2:** Cast on cooldown when 2+ enemies alive AND target has no Accuracy Down already
- **Ultimate:** Cast on cooldown when 3+ enemies alive OR boss-tagged enemy alive AND enemies have buffs to strip

Quote

"The fan turns, the wind answers."

Status notes (design polish pending): - Wind Sigil vs Lyria's Focus — both 5-stack passives; clarify uniqueness - Skill 1 and Skill 2 are both AoE — consider differentiating one as Mid-AoE (3 random) vs full AoE - Sacred Gale Dance Accuracy Down value (-30%) may be too punishing; tune

4★ Roster — Designs TBD

Status: User is producing 4★ hero designs separately. This file is a placeholder + design constraints reference.

Quantity

15 4★ heroes planned for launch.

Suggested element + class distribution

To mirror the 5★ roster cleanly, the recommended split is **one 4★ per element + class combination**:

Element	Tank	Assassin	Archer	Mage	Support
 Fire	4★ TBD	4★ TBD	4★ TBD	4★ TBD	4★ TBD
 Water	4★ TBD	4★ TBD	4★ TBD	4★ TBD	4★ TBD
 Wind	4★ TBD	4★ TBD	4★ TBD	4★ TBD	4★ TBD

= 15 4★ heroes, one per cell.

(User may choose a different split — this is a recommendation only.)

Mechanical role of 4★ heroes

Per `gdd/03-stats-and-progression.md` and `gdd/05-gacha-and-currencies.md`:

- 4★ heroes are obtainable from **Epic-tier gacha drops** (7.2% gacha rate).
- 4★ heroes drop from **Rare Egg** hatches (small chance).
- 4★ heroes drop from **Expedition rewards** at higher mission tiers.
- 4★ heroes are **valid combat heroes** — not just fodder. Use them in early/mid-game teams.
- 4★ heroes can be **promoted to 5★** via the standard fusion ladder (5 × 4★ + 50,000 Gold + Sacred Crystal × 30 + Pristine Element Crystal).
- 4★ → 5★ promotion via fusion produces a “generic” 5★ — not a *named* hero. (To get a named 5★, you must summon them.)

Design constraints (when authoring 4★ sheets)

When the user provides 4★ designs, each sheet should follow the same template as 5★ but with these differences:

Field	5★ approach	4★ approach
Stat grades	Mix of S / A / B / C / D	Mostly A / B / C — no S grades except in their core stat
Passive	5-stack passive with synergy	Simpler passive — single-condition trigger, fewer stacks
Skill multipliers	1.5× / 2.2× / 3.5×	1.3× / 1.9× / 3.0× (15-20% lower)
Ultimate complexity	Multi-effect bundles	Simpler, 1-2 effects
Chibi anims	Full unique set	Can share base attack archetype within class
Splash art	Unique per hero	Unique per hero

Naming considerations

- 4★ names should feel distinctive, not “lesser versions” of 5★ names.
- Each 4★ should have personality — they’re not background characters.
- Consider: a 4★ might foreshadow a future 5★ via shared lore (e.g. “Apprentice of the Phoenix” → eventual 5★ “Master of the Phoenix”).

When to author full 4★ sheets

After user provides 4★ designs: 1. Create one `.md` file per 4★ hero in `design/heroes/4-star/` 2. Follow the same template as 5★ sheets (see `README.md` § Design Conventions) 3. Update master plan to reflect the named 4★ roster 4. Re-render PDF

Open questions for 4★

When user provides designs, also confirm:

- Do 4★ heroes have splash art OR just hero cards?
- Do 4★ heroes appear in Hero Trials / Boss Rush?
- Can 4★ heroes go to Tower of Trials?

- Are 4★ heroes featured on Limited banners (as 4★ rate-up) or only on Standard?

Project I — Production & Business Plan

*Sits alongside the design docs (`design/gdd/*`) but covers business, legal, marketing, engineering process, QA, customer support, analytics, community, and post-launch ops. Not a game-design doc — a **company / process** doc.*

1. Engineering / Tech ADRs

(ADR = Architecture Decision Record. Lock these before writing serious code.)

1.1 Engine

- **Unity 2023 LTS** (or 6.0 LTS once stable)
- Built-in render pipeline for compact mode (lightweight)
- URP for expanded mode if richer visuals are needed
- Skip HDRP — overkill for chibi 2D game

1.2 C# / coding standards

- C# 10+
- Nullable reference types enabled
- `dotnet format` enforced in CI
- Folder structure: `Assets/Project/Scripts/<system>/`
- Unit tests in `Tests/<system>Tests/`
- One file per class. Internal classes for collaborators.

1.3 Save format

- JSON serialized via `Newtonsoft.Json` (or `System.Text.Json` on .NET 7+)
- Obfuscation: XOR cipher with player-derived key
- Checksum: SHA-256
- See `design/gdd/17-save-system.md` for full spec

1.4 Asset management

- Unity Addressables (NOT Resources)

- Remote content delivery NOT required at launch (single-build app)
- Lazy-load splash arts, chibi sheets per hero
- Preload tutorial assets

1.5 Networking

- Steamworks.NET for Steam integration
- No game-server architecture (single-player, server-validated state only)
- Custom config server: Firebase Remote Config OR AWS Lambda + S3
- IAP validation: server-side via Steam Web API

1.6 Audio

- Wwise OR FMOD (Wwise free for ≤ 200 SFX assets, FMOD free for indie under \$500k revenue)
- OGG Vorbis for music, WAV/OGG for SFX

1.7 UI

- Unity UI Toolkit (UI Builder + USS) for expanded mode — modern, performant
- Compact mode: hand-laid UI via canvas + sprite (small surface area, doesn't need toolkit complexity)

1.8 CI/CD

- GitHub Actions for build pipeline
- Builds: Windows x64 (launch), macOS (post-launch)
- Automated unit tests on PR
- Asset bundle builds nightly
- Steam Build Tools for upload

1.9 Crash & error tracking

- **Sentry** (or Unity Cloud Diagnostics) integrated from Day 1
- Stack-trace deobfuscation enabled
- PII-stripped reports

1.10 Analytics

- **PostHog** (open-source, GDPR-friendly) OR **GameAnalytics** (gacha-friendly free tier)
- Event taxonomy from `design/gdd/21-liveops-and-telemetry.md` § 3.4

2. Legal & Compliance

2.1 Required documents (lawyer-drafted before launch)

Document	Status
End-User License Agreement (EULA)	 Need draft
Privacy Policy	 Need draft (GDPR + CCPA + PIPEDA compliant)
Terms of Service	 Need draft
Refund Policy	 Need draft (Steam policy compliant)
Cookie Policy (if web companion)	 If applicable
Open Source Licenses page	 Auto-generate from package.json equivalents

2.2 Content rating

- **ESRB** (North America): Submit for “E10+” or “Teen” rating depending on combat tone
- **PEGI** (Europe): Submit for “PEGI 7” or “PEGI 12”
- **USK** (Germany): Submit
- **CERO** (Japan): Submit if Japanese localization planned

Rating submissions take 2-6 weeks. Plan ahead.

2.3 Compliance frameworks

- **GDPR** (EU): Privacy policy, data export, right-to-be-forgotten — addressed in `design/gdd/17-save-system.md § 3.8` + `gdd/21-liveops-and-telemetry.md`
- **CCPA** (California): Similar requirements
- **COPPA**: If under-13 users — Project I targets 13+ minimum (set Steam age gate accordingly)
- **PCI-DSS**: Payment data — Steam handles this; we never see credit card numbers
- **DSA (EU Digital Services Act)**: If user-generated content emerges — N/A at launch

2.4 Gacha disclosure laws (per region)

Region	Requirement
Japan	Disclose all gacha rates publicly (we do — <code>gdd/05-gacha-and-currencies.md</code>)
South Korea	Same
China	If launching there, requires version + government approval (separate process)
Belgium	Loot boxes banned — region-restrict purchases OR exclude from Belgian sales
Netherlands	Loot boxes restricted — same handling

Decision: Geo-block Crystal purchases in Belgium and Netherlands. Players can still play F2P.

2.5 Trademark

- “Project I” is a working title. Final game name needs:
 - Trademark search (USPTO, EUIPO, JPO)
 - Domain registration (.com, .gg, .net)
 - Steam URL slug reservation

3. Marketing & Store Page

3.1 Steam page assets needed (~6 weeks before launch)

- **Header capsule** (1920×620)
- **Library capsule** (600×900)
- **Hero capsule** (1840×930)
- **Main capsule** (616×353)
- **Small capsule** (296×175)
- **Page background** (1438×810)
- **Screenshots** (1920×1080) — minimum 5, recommended 8-12
- **Trailer** (30s-2min, 1920×1080, MP4)
- **Animated GIFs** (optional, increases conversion)
- **Game logo** (transparent PNG)
- **Page description** (short + long)

- **Tags** (e.g., Idle, Gacha, RPG, Anime, Free to Play, Singleplayer, Turn-Based, 2D)

3.2 Trailer outline (30-60 second cut)

1. **0-3s** — Compact window appears on desktop (TBH-style hook — visual differentiation)
2. **4-15s** — Quick montage: Lyria firing arrows, Emberia AoE flames, gacha pull animation, Tower climb, character roster
3. **16-25s** — Voice-over / text: “Idle in your taskbar. Battle when you want. Build the team you dream of.”
4. **26-30s** — Logo + “Launching [date] on Steam” CTA

3.3 Soft launch strategy

Option A — Open Steam Early Access in 2-3 small English-speaking regions (e.g., Canada, Australia, NZ) - Pros: Real feedback before global launch; preserves day-1 cleanliness for major markets - Cons: Splits attention; piracy / region-shopping risk

Option B — Closed Beta via Steam Playtest (~500-2,000 players) - Pros: Free, low-risk, signed NDAs possible - Cons: Self-selected hardcore players, not representative

Option C — Direct global launch with rapid post-launch hotfix capability - Pros: Single moment of attention, easier marketing - Cons: Bugs visible to everyone

Recommendation: Option B (closed beta 4-6 weeks pre-launch) then Option C (global launch). Use hot-config to tune balance based on beta data.

3.4 Content creator strategy

- Outreach to YouTubers / Twitch streamers in the gacha + idle game niche 4-6 weeks before launch
- Provide early access keys + ~\$50 of in-game gem packs for content creation
- Offer banner exclusives at affiliate / creator events
- Target list: ~20-50 mid-tier creators (10k-200k followers) over 3-5 mega creators

3.5 Community channels

Channel	Launch Status	Purpose
Discord	Live 4 weeks pre-launch	Closed beta coordination, community building, support channel
Reddit (r/ProjectI or similar)	Live 2 weeks pre-launch	Long-form discussions, memes, guides
Twitter/X	Live 2 weeks pre-launch	Patch notes, hype posts, banner announcements
TikTok	Optional	Short-form gameplay clips
YouTube (official channel)	Live with trailer	Trailers, dev diaries, tutorials
Website / Landing page	Live 4 weeks pre-launch	Newsletter signup, Steam wishlist CTA

3.6 Pricing strategy

- Base game: **Free to play** (F2P)
- Monetization per `gdd/14-monetization.md`
- **Wishlist target before launch:** 30,000-50,000 (industry rule of thumb for indie F2P)
- **Regional pricing:** Use Steam’s recommended tiers — don’t undercut, don’t overprice
- **Launch discount:** none (F2P has no base price to discount); but a “Launch Founder Pack” (\$14.99-\$24.99) is a viable one-week-only purchase

4. QA & Test Process

4.1 Test environments

- **Dev** — daily builds, local testing
- **Staging** — bi-weekly QA builds, automated test pass
- **Beta** (closed) — externally distributed via Steam Playtest
- **Production** — live Steam build

4.2 Test types

Test type	Owned by	Cadence
Unit tests (combat formulas, gacha rates, save serialization)	Engineering	Every commit (CI)
Integration tests (state transitions, IAP, cloud save)	Engineering	Every PR
Smoke tests (game launches, tutorial completable)	QA	Daily on dev builds
Regression suite (known-bug list)	QA	Pre-release builds
Manual playtest (full gameplay session)	QA + designers	Weekly
Localization QA	Localization team	Per-locale release
Platform certification (Steam Deck Verified)	Engineering + QA	Pre-launch + per-version
Compliance check (rate disclosures, refund flow)	Legal + QA	Pre-launch

4.3 Bug tracking

- **Linear** or **Jira** or **GitHub Issues** — pick one
- Severity levels:
 - **P0 (Blocker)** — Crash / data loss / IAP broken → fix within 24h
 - **P1 (Critical)** — Major feature broken → fix within 1 week
 - **P2 (Major)** — Important issue, no workaround → fix within sprint
 - **P3 (Minor)** — Polish / cosmetic → fix when convenient
 - **P4 (Trivial)** — Backlog

4.4 Test plan template

For each launch feature (BP, Gacha, Tower, etc.): 1. Happy path test cases 2. Edge cases (boundary values, empty states, max values) 3. Error states (no internet, save corrupted, etc.) 4. Performance tests (memory, FPS in compact mode) 5. Compatibility tests (Win 10 / Win 11, integrated graphics, low-RAM PC)

5. Customer Support

5.1 Tooling

- **Helpshift** (specialized for games) OR **Zendesk** (general SaaS) — pick before launch
- Integrated SDK in-game for ticket submission
- Email fallback: `support@<projectname>.gg`

5.2 Response time SLA

- **Billing issues:** 48 hours
- **Account / data loss:** 72 hours
- **General gameplay:** 7 days
- **Bug reports:** acknowledged within 7 days

5.3 FAQ articles (launch-ready)

20 articles ready Day 1: - How does gacha work? - What is pity? - How do I get Apex Catalyst? - How does the Battle Pass work? - How do I refund a purchase? (links to Steam policy) - Why is the game in my taskbar? - Can I play on Mac? - How do I report a bug? - I lost my account / I want to delete my account - Cross-platform save? - (10 more covering common questions)

5.4 Response templates

Pre-written responses for common scenarios: - Refund denied (item already consumed) - Account recovery - Compensation gift - Server downtime explanation - Cheating accusation handling

5.5 Escalation path

Tier 1 (support agent) → Tier 2 (senior support) → Tier 3 (engineering / community manager) → Tier 4 (legal / executive)

6. Analytics & Data

6.1 Stack

Layer	Tool
Event collection	PostHog SDK / GameAnalytics SDK
Crash reporting	Sentry
Server logs	CloudWatch / Datadog
BI / dashboard	PostHog dashboards OR Metabase + Postgres
A/B testing	PostHog feature flags OR custom

6.2 Key dashboards (build pre-launch)

1. **Funnel:** Install → Tutorial Complete → First Pull → First Purchase
2. **Retention:** D1, D7, D14, D30 cohort curves
3. **Monetization:** ARPU, ARPPU, conversion rate, top-spender distribution
4. **Engagement:** DAU, session length, sessions per day, idle session %
5. **Content:** Stage completion rates, Tower floor reach distribution, BP tier reach
6. **Performance:** Crash-free session %, average FPS by mode
7. **Gacha health:** Pity hit rate, Legendary distribution by source

6.3 KPI targets (launch + 30 days)

Metric	Target
D1 retention	35-45%
D7 retention	20-25%
D30 retention	8-12%
D1 conversion (any spend)	2-3%
Tutorial completion	75-85%
Avg session length	8-15 minutes
ARPPU (paying users)	\$30-60 lifetime
Crash-free session %	99.5%+

If hitting **below 50% of any** target after 14 days → emergency review meeting.

7. Team Structure (Required Roles)

(These are the suggested roles needed. Sizes vary based on budget.)

Role	Min Headcount	Responsibilities
Game Director / Producer	1	Vision, priorities, schedule
Game Designer	1-2	Mechanics, balance, content
Lead Engineer	1	Architecture, tech direction
Engineers (Unity / C#)	2-4	Implementation
Tools Engineer	0-1	Internal tools, build pipeline
Lead Artist / Art Director	1	Style guide, art bible
Artists (chibi, splash, UI)	2-4	Hero / boss / UI art
Animator	1-2	Chibi animations, VFX motion
VFX Artist	0-1	Particle systems, skill VFX
Audio Designer	0-1	SFX library, music direction
UI/UX Designer	1	Layouts, wireframes
QA Lead	1	Test strategy, regression suite
QA Testers	1-2	Manual testing
Community Manager	0-1	Discord, social, content creator outreach
Customer Support	0-2	Ticket handling (post-launch grows)
Localization Lead	0-1	When adding locales
Marketing	0-1	Steam page, ads, partnerships
Legal	Contract	EULA, privacy, compliance

Solo / two-person team: combine roles. A two-person studio (one engineering + one design/art) can absolutely ship Project I — many indie gachas have shipped this way.

8. Release Timeline & Milestones

(Template — adjust to actual schedule.)

Phase	Duration	Goal
Pre-production	3 months	All GDDs locked (✓ this phase complete), engine setup, art bible, tech ADRs
Vertical slice	2 months	One playable chapter + tutorial + basic gacha + idle. Validates core loop.
Production Alpha	6 months	All 12 chapters playable, 10 heroes implemented, BP/Tower/Expedition functional, but rough
Beta / Closed Playtest	2 months	Polished, missing some content. ~500 external testers via Steam Playtest
Pre-launch hardening	1 month	Final QA, certification, store page finalization, marketing push
Launch	1 week	Day 1 + first week monitoring + hotfix readiness
Post-launch ops	Ongoing	Banners every ~3-4 weeks, BP seasons every 60 days, balance patches

Total pre-launch: ~14-16 months from end of design phase.

Critical milestones (with owners)

- Vertical slice playable internally (validates “the game is fun”)
- First playtest external (validates “players get it”)
- All 10 launch heroes implemented + balanced
- Tower of Trials Season 1 fully populated (100 floors)
- BP Season 1 fully populated (50 tiers)
- Localization-ready (string extraction complete, even if EN only)
- Compliance & legal approved

-  Steam page approved + wishlist drive started
-  Content creator outreach complete
-  Soft launch / closed beta
-  Launch day

9. Risk Management

Risk	Impact	Probability	Mitigation
Cheaters edit save → bypass IAP	Lost revenue	Medium	Tamper detection (<code>gdd/17-save-system.md</code>); IAP validation server-side
Compact window concept doesn't resonate	Bad reviews	Medium	Beta playtest validates ASAP; have expanded-mode as fallback
Apex Catalyst supply tuning wrong	Player frustration OR power inflation	High	Hot-config; adjust within 2 weeks of telemetry signal
Battle Pass season can't be authored in time	Live-ops gap	Medium	First 3 BP seasons authored before launch
Steam algorithm doesn't favor F2P	Low organic traffic	High	Strong wishlist push + content creator partnerships
Cloud save conflict bugs	Player data loss perception	Medium	Conflict resolution UI; preserve discarded saves for 30 days
Server downtime breaks gacha	Players furious	Low	Local fallback for config; queue purchases offline
Localization missing market	Reduced global reach	Medium	EN-first; CJK + EU as post-launch wave
Belgium/NL gacha ban affects sales	Minimal revenue impact	Confirmed	Geo-restrict IAP, allow F2P

10. Post-Launch Live-Ops Model

10.1 Patch cadence

- **Hotfix:** as needed (P0/P1 issues, hot-config tweaks) — same-day to 1-week
- **Minor patch:** every 2 weeks (balance changes, small QoL, bug fixes)
- **Major content patch:** every 4-6 weeks (new banner, new BP season transitions, new chapter)

10.2 First-year content roadmap (recommendation)

Month	New Content
Month 1 (launch)	Day-1 content per <code>gdd/24-day-1-launch-content.md</code>
Month 2	Banner #2 (e.g. Ragnar or Mei Lin), first limited event (e.g. “Phoenix Rite” mini-stage)
Month 3	BP Season 2, Banner #3, first chapter content drop (Chapter 13 if ready)
Month 4	Banner #4, “Tower Echoes” (Tower variant mode), first cosmetic shop rotation
Month 5	BP Season 3, Banner #5, first 4★ designs released
Month 6	Half-year anniversary event, free 5★ pick, Banner #6
Month 7-12	Continue cadence; introduce new endgame (Hero Trials new tier, Boss Rush Hell unlock)

10.3 Community ops

- Weekly Discord dev update (every Friday)
- Monthly newsletter (mid-month)
- Quarterly producer’s letter (state of the game)

10.4 Hotfix decision tree

When a P0 bug is detected: 1. Acknowledge in Discord within 1 hour 2. Hot-config workaround if possible (no app patch) 3. If client patch required: target same-day deployment 4. Post-mortem and compensation mail within 48 hours 5. Public patch notes published

11. Budget (rough planning template)

(Numbers are placeholders — fill in based on actual budget.)

Category	Pre-launch	Year 1 Live-Ops
Team salaries	\$300k-\$1M (small team)	\$400k-\$1.5M
Art outsourcing (chibis, splashes, BP skins)	\$50k-\$150k	\$100k-\$300k
Audio (SFX library + music)	\$10k-\$50k	\$20k-\$50k
Tooling (Unity Pro, Sentry, Helpshift, etc.)	\$10k-\$30k	\$10k-\$30k
Marketing (ads, content creators, events)	\$20k-\$100k	\$50k-\$300k
Legal	\$10k-\$30k	\$5k-\$15k
Steam fees	\$100 (one-time per app)	30% revenue share
Servers (config, telemetry)	\$5k-\$15k	\$10k-\$50k
QA outsourcing (LQA, certification)	\$10k-\$30k	\$5k-\$20k
Contingency	15-20%	15-20%

12. Acceptance Criteria (this doc)

1. ☒ All engineering ADRs documented (§ 1)
2. ☒ Legal requirements listed (§ 2)
3. ☒ Marketing checklist provided (§ 3)
4. ☒ QA process defined (§ 4)
5. ☒ Customer support model defined (§ 5)
6. ☒ Analytics stack picked (§ 6)
7. ☒ Team roles enumerated (§ 7)
8. ☒ Release timeline template ready (§ 8)
9. ☒ Risk register populated (§ 9)
10. ☒ Live-ops model defined (§ 10)

11.  Budget template provided (§ 11)

Project I — Unity Setup Guide

*Step-by-step setup for starting Project I in Unity. Run this top-to-bottom on your Mac (or future PC).
Total time ~30-60 minutes for steps 1-7. Steps 8+ are ongoing.*

0. Prerequisites

Before you start:

- ☐ macOS Sonoma+ (or Windows 10+)
 - ☐ 16 GB RAM minimum (32 GB recommended)
 - ☐ 50 GB free disk
 - ☐ Stable internet connection
 - ☐ Apple ID or Microsoft account (for Unity Hub)
 - ☐ GitHub account (you have `im2basic`)
 - ☐ Free Steam account (for Steamworks integration later)
-

1. Install Unity Hub

1. Download from <https://unity.com/download>
 2. Open the downloaded `.dmg` and drag Unity Hub to Applications
 3. Launch Unity Hub
 4. Sign in (or create a free Unity account)
 5. Choose a license: **Personal** is free for revenue under \$200k. **Plus** (\$399/year) when you cross that threshold.
-

2. Install Unity Editor

1. In Unity Hub, click **Installs** → **Install Editor**
 2. Choose **Unity 2023.3 LTS** (or the latest LTS version available)
 3. In the modules dialog, check these:
 - ☒ **Mac Build Support (IL2CPP)** (you're on Mac)
 - ☒ **Windows Build Support (IL2CPP)** ← critical, this is the launch target
 - ☒ **Visual Studio for Mac OR JetBrains Rider** (your code editor — Rider preferred for C# dev)
 - ☒ **Documentation**
 4. Click **Install**. Wait ~20-30 minutes.
-

3. Install C# IDE

Choose ONE:

- **JetBrains Rider** (recommended for serious development) — <https://www.jetbrains.com/rider/> — free non-commercial license available
- **Visual Studio Code** (free) — <https://code.visualstudio.com/> + install the C# extension
- **Visual Studio for Mac** (free, deprecated by MS but functional) — bundled in Unity install option

In Unity → Preferences → External Tools → set **External Script Editor** to your choice.

4. Install Git & Configure GitHub

```
# Check if git is installed
git --version

# If not installed, install via Homebrew
brew install git

# Configure your identity (one-time setup)
git config --global user.name "Song"
git config --global user.email "your-github-email@example.com"

# Set default branch to main
git config --global init.defaultBranch main
```

Then authenticate with GitHub:

```
# Install GitHub CLI (easiest auth path)
brew install gh

# Login
gh auth login
```

Choose “GitHub.com” → “HTTPS” → “Login with a web browser” → follow prompts.

5. Clone the Planning Repo

```
# Pick where you want the repo on disk
cd ~/Documents
mkdir -p ProjectI
cd ProjectI

# Clone the planning repo I created for you
gh repo clone im2basic/Project-I-Planning planning

# OR if you don't use gh CLI:
# git clone https://github.com/im2basic/Project-I-Planning.git planning

cd planning
ls -la
```

You should see all the design docs.

6. Create a Separate Unity Project Repo

Planning lives in `Project-I-Planning`. The actual Unity project should live in a **separate repo** (it has very different content + large binary assets).

Option A — Create on GitHub via web

1. Go to <https://github.com/new>
2. Repository name: `Project-I-Game` (or similar — avoid copyrighted name yet)
3. Visibility: **Private** (until launch)
4. Initialize with `.gitignore` template: **Unity**
5. Add a README

6. Click **Create repository**

7. Clone locally:

```
cd ~/Documents/ProjectI
gh repo clone im2basic/Project-I-Game game
```

Option B — Create via CLI

```
cd ~/Documents/ProjectI
gh repo create Project-I-Game --private --description "Project I – Unity game source" --
    gitignore Unity --clone
mv Project-I-Game game
cd game
```

7. Create the Unity Project

1. Open Unity Hub
2. Click **New Project**
3. Template: **2D (URP)** (we want Universal Render Pipeline for expanded mode polish)
4. Project name: `ProjectI`
5. Location: `~/Documents/ProjectI/game/` (the repo you just made)
6. Click **Create project**

Unity creates the folder structure. **Wait** ~5 minutes for the first import.

7a. Verify .gitignore

Open the `game/` repo directory in Finder. You should see a `.gitignore` (Unity template). If not, copy this content into `.gitignore` :

```

# Unity generated
/Library/
/Temp/
/Obj/
/Build/
/Builds/
/Logs/
/UserSettings/
/MemoryCaptures/

# Visual Studio / Rider
.idea/
.vs/
*.csproj
*.unityproj
*.sln
*.suo
*.tmp
*.user
*.userprefs
*.pidb
*.booproj
*.svd
*.pdb
*.mdb
*.opendb
*.VC.db

# Mac / Windows OS
.DS_Store
Thumbs.db

# Builds
*.apk
*.aab
*.unitypackage
*.app

# Crashlytics
crashlytics-build.properties

```

7b. Initial commit

```

cd ~/Documents/ProjectI/game
git add .
git commit -m "Initial Unity project - Project I"
git push origin main

```

8. Recommended Folder Structure

In `Assets/`, create this folder structure (in Unity Project window — right-click → Create → Folder):

```
Assets/
├── Project/
│   ├── Scripts/
│   │   ├── Core/           # Game loop, save system, config
│   │   ├── Combat/        # Combat engine, AI
│   │   ├── Heroes/        # Hero data + behavior
│   │   ├── Gacha/         # Gacha system
│   │   ├── Idle/          # Idle simulation
│   │   ├── UI/            # UI screens, widgets
│   │   ├── Window/        # Desktop window control (compact / expanded / tray)
│   │   ├── Audio/         # Audio manager
│   │   ├── Localization/  # String table system
│   │   ├── Analytics/     # Telemetry hooks
│   │   └── Tests/         # Unit / integration tests
│   ├── Art/
│   │   ├── Heroes/        # Splash, chibi, hero cards (per hero subfolder)
│   │   ├── Enemies/       # Enemy art
│   │   ├── UI/            # UI sprites
│   │   ├── Icons/         # Element / class / status icons
│   │   ├── VFX/           # Particle effects
│   │   └── Backgrounds/   # Stage backgrounds (incl. compact strips)
│   ├── Audio/
│   │   ├── Music/
│   │   ├── SFX/
│   │   └── Voice/         # Reserved for future VO
│   ├── Data/              # ScriptableObjects: hero stats, enemy stats, drop tables, etc.
│   ├── Scenes/
│   │   ├── Bootstrap.unity # First scene that loads on launch
│   │   ├── Compact.unity  # Compact mode UI
│   │   ├── Expanded.unity # Expanded mode UI
│   │   ├── Tutorial.unity # Tutorial flow
│   │   └── Combat.unity   # Combat scene (can be additive)
│   ├── Prefabs/           # Reusable game objects
│   └── ThirdParty/        # Imported packages (Steamworks, etc.)
```

9. Install Required Unity Packages

In Unity → Window → Package Manager:

Package	Why
Addressables	Asset loading (mandatory)
Input System	Modern input (replaces old Input class)
2D Sprite	If 2D template didn't include
Universal RP	Render pipeline
TextMeshPro	UI text (mandatory)
Newtonsoft Json	Save serialization
Unity Test Framework	Unit tests

In Package Manager, also import these from the Asset Store / Git later:

- **Steamworks.NET** — Steam integration. From: <https://steamworks.github.io/>
- **DOTween** — Tweening library (free). From Asset Store.
- **Sentry SDK for Unity** — Crash reporting. From <https://docs.sentry.io/platforms/unity/>

10. First Code: Set Up the Game Bootstrap

In Unity → `Assets/Project/Scripts/Core/` , create a new C# script: `GameBootstrap.cs`


```

using UnityEngine;

namespace ProjectI.Core
{
    public class GameBootstrap : MonoBehaviour
    {
        [SerializeField] private bool startInCompactMode = true;

        private void Start()
        {
            Debug.Log("Project I bootstrap started.");

            // TODO: load save data
            // TODO: initialize config (hot-config fetch)
            // TODO: route to Tutorial or main menu based on save state
            // TODO: switch to compact or expanded mode based on settings

            if (startInCompactMode)
            {
                // Placeholder – actual compact mode switch comes later
                Debug.Log("Starting in compact mode.");
            }
        }
    }
}

```

Create an empty GameObject in the `Bootstrap.unity` scene → name it `GameBootstrap` → attach the script.

Press **Play** in the editor. You should see “Project I bootstrap started.” in the Console.

✓ **Hello-world milestone hit.**

11. Set Up Reference to Planning Repo

The Unity project (`game/`) doesn't include the planning docs. Reference them like this:

Option A — Submodule (clean, but more git complexity)

```

cd ~/Documents/ProjectI/game
git submodule add https://github.com/im2basic/Project-I-Planning.git docs/planning
git submodule update --init --recursive
git add .
git commit -m "Add planning repo as submodule"
git push

```

Now `docs/planning/` inside the Unity project links to the planning repo. When the planning repo updates, run:

```
git submodule update --remote
```

Option B — Just keep both repos side-by-side (simpler)

Keep the planning repo at `~/Documents/ProjectI/planning/` and the Unity project at `~/Documents/ProjectI/game/`. Open both in your file browser. Reference docs in `planning/` whenever you need them.

Recommended: Option B for simplicity. Move to A once you have CI / multiple devs.

12. Working Across Mac + PC

Since you're planning on Mac + want to work on Windows PC too:

On your Mac

```
# Daily: pull latest before starting work
cd ~/Documents/ProjectI/planning
git pull

cd ~/Documents/ProjectI/game
git pull

# Edit / develop ...

# Push when done
cd ~/Documents/ProjectI/game
git add .
git commit -m "Describe what you did"
git push
```

On your PC (one-time setup)

1. Install Git for Windows: <https://git-scm.com/download/win>
2. Install GitHub CLI: <https://cli.github.com/>
3. Open Command Prompt or PowerShell:

```
gh auth login
# follow same auth flow
```

4. Install Unity Hub for Windows: <https://unity.com/download>

5. Install the same Unity 2023.3 LTS version

6. Clone the repos:

```
mkdir C:\dev\ProjectI
cd C:\dev\ProjectI
gh repo clone im2basic/Project-I-Planning planning
gh repo clone im2basic/Project-I-Game game
```

7. Open Unity Hub → Add → select `C:\dev\ProjectI\game\` → open



You can now work on either machine. Push when done; pull before starting.

13. Daily Workflow Cheat Sheet

```
# Morning – start of work session
cd ~/Documents/ProjectI/game # or whichever repo you're in
git pull

# Work in Unity ...

# Mid-day – commit often (good habit)
git add .
git commit -m "WIP: implementing hero stats system"

# End of day – push to GitHub so it's safe + available on your other PC
git push
```

Rules: - **Never** commit while Unity is open and unsaved. - **Always** save scenes and assets before committing. - **Pull first** if working on multiple machines. - **Push often** — at least daily.

14. Recommended Next Tasks (after setup)

In rough order, the first ~3 months of implementation work:

1. **Save system** (Scripts/Core/SaveSystem) — JSON serialization, obfuscation, checksums, atomic writes

2. **Hot-config fetcher** (Scripts/Core/Config) — fetches gacha rates, BP schedule, etc.
3. **Hero data system** (Scripts/Heroes/HeroData.cs) — ScriptableObjects for the 10 launch heroes
4. **Combat engine — turn order** (Scripts/Combat/ActionGauge.cs) — see `design/gdd/01-combat.md § 3.2`
5. **Combat engine — damage formula** (Scripts/Combat/DamageCalculator.cs) — see `gdd/01-combat.md § 3.5`
6. **Stat system** (Scripts/Heroes/HeroStats.cs) — implements the 7 core stats per `gdd/03`
7. **Enemy AI** (Scripts/Combat/EnemyAI.cs) — see `gdd/25-enemy-ai.md`
8. **Compact window controller** (Scripts/Window/CompactWindowController.cs) — borderless window, always-on-top, click-through
9. **Auto-AI for player heroes** (Scripts/Combat/AutoAI.cs) — see `gdd/01 § 3.14`
10. **First test battle scene** — combine 1-9 into one playable Lyria vs. enemy battle

This is “vertical slice” work. Validates the engine before you build all 15 heroes.

15. Troubleshooting

Unity is slow / crashes on first import - Wait 30 minutes. First-time imports are heavy. - If still slow: check Activity Monitor — Unity needs RAM headroom.

Git rejects push: “non-fast-forward” - Someone (or another machine) pushed before you. Pull first: `bash git pull --rebase` - Then push again.

“package.json” conflicts during pull - Unity package manifest changed on both sides. Resolve by accepting the newer version OR merging manually.

Unity won’t open the project - Check Unity version matches. If you opened a project made in 2023.3 in 2022.x, it won’t work.

16. Resources & References

- **Unity Manual:** <https://docs.unity3d.com/Manual/index.html>
 - **C# style guide:** <https://learn.microsoft.com/en-us/dotnet/csharp/fundamentals/coding-style/coding-conventions>
 - **Steamworks.NET docs:** <https://steamworks.github.io/>
 - **GitHub CLI docs:** <https://cli.github.com/manual/>
 - **Project I planning docs:** local copy at `~/Documents/ProjectI/planning/`
-

17. Done with setup?

✔ Unity Hub installed ✔ Unity 2023.3 LTS installed with Windows + Mac build support ✔ Git + GitHub CLI configured ✔ Both repos cloned to `~/Documents/ProjectI/` ✔ Unity project created and opens ✔ Folder structure created ✔ First bootstrap script runs in editor ✔ Mac + PC workflows verified

You're ready to start implementing the Vertical Slice. Begin with the save system (it underpins everything else).

Refer to `design/gdd/01-combat.md` and onward for spec details.

Project I — Backlog (Deferred Features)

Features kept on file for post-launch consideration. NOT in v1 scope. NOT abandoned — preserved so they can be revisited at the right time.

Player-Facing Systems

B-01 — Pet / Companion System

- **TBH-like pet that follows the party.** Inactive in v1 (user decision). Could be reintroduced as a cosmetic-only “follower” or as a stat-bonus auxiliary. If reintroduced, design carefully to avoid power-creep.
- **Effort:** Medium (new entity, new UI, new gacha-adjacent pool)
- **Earliest candidate window:** Post-launch +3 months

B-02 — Hero Skins / Costumes

- Alternate visual identities for existing heroes. Pure cosmetic, no stat impact. Monetization-friendly.
- **Effort:** Per-hero costume = ~2 weeks of art + integration
- **Earliest candidate window:** Post-launch +1 month (first event season)

B-03 — Hero Affinity / Friendship

- Heroes used in battle gain “affinity” with the player. At thresholds, small flavor lines unlock and small stat bonuses apply.
- **Effort:** Medium (affinity tracking, dialogue strings, UI surface)
- **Earliest candidate window:** Post-launch +3 months

B-04 — Hero Synergy Bonuses

- Specific team compositions grant party-wide buffs (e.g., “All-Wind team: +10% Speed”, “Lyria + Sylphie + Mei Lin: +15% Crit”).
- **Effort:** Low-Medium (data table + buff plumbing)
- **Earliest candidate window:** Post-launch +2 months
- **Caveat:** Risk of pigeonholing meta; needs careful tuning to avoid “always bring these 3” syndrome

B-05 — Hero Trial Mastery → Talent Tree

- Beyond Gold-tier Hero Trials, future-state could unlock a per-hero talent tree (small per-hero choices like “+5% CritDamage on Skill 2” or “Skill 1 hits 2 enemies”).
- **Effort:** Large (UI, balance, save schema)
- **Earliest candidate window:** Post-launch +6 months

B-06 — DPS Meter / Combat Log (toggleable)

- Detailed per-hero damage tracking and historical combat log. Toggleable in settings.
- **Effort:** Low-Medium (logging is already there per `gdd/01-combat.md` ; just surface in UI)
- **Earliest candidate window:** Post-launch +1 month (QoL update)

B-07 — Replay Video / Step-Through Playback

- Full battle replay with scrubbable timeline. Currently only the damage log is in v1.
- **Effort:** Large (replay storage, deserialization, playback)
- **Earliest candidate window:** Post-launch +6+ months

Social & Community

B-08 — Friends List & Referral

- Player friends, referral rewards, friend “high-five” daily bonus.
- **Effort:** Medium (single-player → social changes architecture)
- **Note:** User said single-player only design locked. Revisit only if user changes mind.

B-09 — Co-op Boss Battles

- Multi-player raid-style boss fights. Architecturally requires real-time multiplayer.
- **Effort:** Very Large (multiplayer netcode, matchmaking, real-time sync)
- **Note:** Out of scope per user direction.

B-10 — Guild System

- Guilds, guild chat, guild boss raids, guild shop.
- **Effort:** Very Large
- **Note:** Out of scope per user direction.

B-11 — PvP Arena

- Async ladder PvP. Out of scope per user.
-

Accessibility (Phase 2)

B-12 — Full Screen Reader Support

- v1 includes hooks; v2 adds full NVDA / JAWS / VoiceOver navigation through all menus, combat, gacha. Includes ARIA-equivalent labels on all UI.
- **Effort:** Medium-Large (audit pass + a lot of integration)
- **Earliest candidate window:** Post-launch +3-6 months

B-13 — Hold-to-Skip Animations

- For accessibility (motion sensitivity) and for fast play, allow hold-to-skip on all gacha pulls, ultimate animations.
- **Effort:** Low
- **Earliest candidate window:** Post-launch +1 month

B-14 — Localization VO Recording

- Voice acting for major Asian markets (JP, KR, CN). Currently no VO at all.
 - **Effort:** Very Large (recording, lip-sync, files)
 - **Earliest candidate window:** Post-launch +6+ months, market-driven
-

QoL & Convenience

B-15 — Sweep / Skip Tickets

- Instant-clear cleared stages 10x at once. User explicitly excluded from v1.
- **Effort:** Low
- **Note:** Revisit only if user changes mind.

B-16 — Cross-Platform Save Sync (beyond Steam)

- EGS, GOG, mobile cross-save. Steam Cloud is v1.

- **Effort:** Medium per platform
- **Earliest candidate window:** When a new platform launch is planned.

B-17 — Steam Market Trading (TBH-style)

- Sell in-game items on the Steam Market. Explicitly excluded by user.
- **Effort:** Large (server-side, anti-cheat, IRS-related complications)
- **Note:** Permanently parked unless monetization strategy changes radically.

B-18 — Hero Equipment Sets (More than 6)

- Launch ships with 6 sets. Future expansions can introduce new sets (e.g., Phoenix Set, Tide Set, Tempest Set).
- **Effort:** Low (data-driven once architecture is in)
- **Earliest candidate window:** Post-launch +1 month (first BP season)

B-19 — Custom Compact-Mode Skin Pack

- Multiple compact-mode visual frames (TBH-style themes). Already proposed as Premium Shop item in monetization GDD.
- **Effort:** Low (graphics only)
- **Earliest candidate window:** Post-launch +1 month

B-20 — Mobile Port

- Touchscreen UI, mobile-specific battery / data considerations.
- **Effort:** Very Large
- **Note:** PC-first decision locked. Revisit only after PC succeeds.

B-21 — Console Port (Steam Deck verified, then console)

- Steam Deck “Verified” badge is achievable quickly (controller support + 720p compact mode). Full console ports (PS5 / Xbox / Switch) are large.
 - **Effort:** Steam Deck = small; console = Large
 - **Earliest candidate window:** Steam Deck within 2-4 months; console post-launch +12 months
-

Live-Ops & Events (post-launch content)

B-22 — Seasonal Events (limited stages)

- Limited-time stages with unique rewards, mini-bosses, banners.
- **Effort:** Per-event = 2-4 weeks design + content
- **Earliest candidate window:** Post-launch +1 month (first event season)

B-23 — Crossover Events

- Branded crossovers with other gacha games. Marketing lever.
- **Effort:** Large (legal, content production)
- **Earliest candidate window:** Post-launch +6+ months, market-driven

B-24 — Endless / Roguelike Mode

- Procedural endless climb beyond Tower of Trials. Different from Tower in that floors are procedural, not curated.
- **Effort:** Medium-Large
- **Earliest candidate window:** Post-launch +6 months

B-25 — Guild Wars / Mass Boss Events

- Out of scope (single-player); see B-10.

Technical / Infrastructure

B-26 — Server-Side Combat Validation (full)

- v1 has server-side gacha + IAP + leaderboard validation. v2 could validate every combat for ladder integrity.
- **Effort:** Large
- **Earliest candidate window:** When ladder cheating becomes a problem.

B-27 — Public Game API

- Allow third-party tools (e.g., damage calculators, build planners) to read public game data.
- **Effort:** Medium
- **Earliest candidate window:** Post-launch +6 months, community-driven

B-28 — Player Profile Page (public/shareable)

- Public-facing profile with stats, hero collection, achievements. Sharable URL.
- **Effort:** Medium (web frontend + auth + privacy controls)
- **Earliest candidate window:** Post-launch +3 months

B-29 — Tournament Mode / Speedrun Leaderboards

- Curated competitive modes. Out of v1 scope.
- **Effort:** Medium
- **Earliest candidate window:** Post-launch +6+ months

Tracking

This backlog is **not a roadmap**. It's a list of options that have been considered and parked. Promote items to the active roadmap only when:

1. Player-data justifies the feature (telemetry shows demand or correlation with retention)
2. Capacity exists in the team
3. The feature aligns with current goals